Multispecies Stock Assessment: Mathematical Methods, Data, and Software

A Mathematical Review Towards Dynamic Size-Based Stock Assessment in mizer

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Introduction

From Single-Species to Multispecies Tactical Assessment

Traditional fisheries stock assessment relies on the single-species paradigm (Beverton and Holt 1957; Cadima 2003; Lart 2022), in which each exploited population is modeled in isolation from its biological community. In single-species Virtual Population Analysis (VPA), Statistical Catch-at-Age (SCAA), or integrated assessment models such as Stock Synthesis (Methot and Wetzel 2013) and SAM (Nielsen and Berg 2014), all non-fishing mortality is aggregated into a single natural mortality rate $M_{s,a,t}$ that is frequently assumed constant across time and often across ages (or scaled empirically with body size and life-history parameters; (Gislason et al. 2010)). Somatic growth (weight-at-age $w_{s,a,t}$) and maturity ogives ($\text{Mat}_{s,a,t}$) are typically entered as empirical covariates rather than state variables coupled to prey availability.

Decades of empirical and modeling work in the North Sea, Baltic Sea, Barents Sea, Iceland, Bering Sea, and Georges Bank have demonstrated that this single-species simplification breaks down in strongly interacting fish communities:

1. **Non-Stationary Predation Mortality ($M2$)**: Predation mortality inflicted by piscivorous fish, marine mammals, and seabirds typically exceeds fishing mortality F on juvenile age classes and forage fish, and varies by orders of magnitude as predator biomasses fluctuate (Lewy and Vinther 2004; Trijoulet et al. 2019). Ignoring time-varying $M2$ biases historical stock-recruitment relationships, reference points (F_{MSY}, B_{lim}), and short-term catch forecasts (ICES 2023).
2. **Prey-Driven Somatic Growth and Starvation**: Predators are food-limited across parts of their ontogeny. Depletion of forage fish (e.g., capelin *Mallotus villosus* in the Barents Sea and Icelandic waters, or sprat *Sprattus sprattus* in the Baltic Sea) directly reduces cod (*Gadus morhua*) feeding levels, somatic growth rates, condition factors, and reproductive output (Stefánsson and Pálsson 1998; Taylor and Stefánsson 2004; Hartvig et al. 2011).
3. **Ontogenetic Size-Structured Omnivory and Cannibalism**: In marine ecosystems, "who eats whom" is governed primarily by body size rather than taxonomic identity alone (Ursin 1973; Andersen 2020). Large piscivores grow through five to six orders of magnitude in body weight from egg to adult, transitioning from planktivory to benthivory/piscivory and cannibalism, which creates strong internal regulatory feedback loops (Canales et al. 2020).
4. **Incompatibility of Single-Species F_{MSY} Targets**: Attempting to simultaneously harvest all predator and prey stocks at their single-species F_{MSY} targets is frequently

biologically impossible because rebuilding predator stocks erodes prey yields and vice versa (Pope et al. 2006, 2019; Andersen 2020).

Chapter Structure

This monograph provides a six-chapter mathematical and statistical review of multispecies stock assessment models, length-structured integrated assessments, stomach content observation models, and dynamic statistical calibration methods for size-spectrum models, accompanied by a comprehensive References section.

Chapter	Title & Core Scope	Key Models & Methodologies Covered
Chapter 1	**Taxonomy and Theoretical Foundations of Multispecies Stock Assessment Models**	Strategic vs. Tactical MICE "sweet spot"; State-space coordinate systems (Age, Age-Length, Length, Continuous Mass, Biomass Pools); One-Way ($M2$-only) vs. Two-Way (Bioenergetic) trophic feedback; Andersen & Ursin (1977) differential equations and Livingston (1983) nonlinear sensitivity analysis, and how `SMS` vs. `mizer` diverged; Jacobian community matrices (**J**) and quadratic multispecies yield approximations.

Chapter	Title & Core Scope	Key Models & Methodologies Covered
Chapter 2	**Age-Structured Multispecies Stock Assessment Models**	`MSVPA/MSFOR` iterative cohort equations; `SMS / smsR` seasonal survival, separable F, Andersen–Ursin size preference, evacuation rations, exact $M2$, and Lognormal + Dirichlet likelihoods; `MSASA / MSCAA / Hydra` and 7 predator functional responses including Pre-emption and Ecosim Foraging Arena; `CEATTLE / Rceattle` temperature-dependent generalized VBGF, Wisconsin bioenergetics, 12 MBRP harvest control rules, and hybrid predator scaling.
Chapter 3	**Age-Length and Length-Structured Assessment Models**	`Gadget2` & `gadget3` 5D state tensor $N_{s,a,l,r,t}, W_{s,a,l,r,t}$, MULTSPEC/Jones consumption-coupled growth, **Beta-Binomial** discrete bin transitions (`g3a_grow_impl_bbinom`), Type II/III predation with `dif_pmin` overconsumption cap, Stefánsson's two-stage iterative reweighting, and Baltic cod–herring–sprat–*Saduria* case study; `MULTIFAN-CL` robustified normal length likelihood; `Stock Synthesis (SS3)` growth morphs/platoons & Rosa Lee's phenomenon; **Size-Transition Matrices** $\mathbf{X}_{l,l'}$; `LIME` length-based mixed effects; `LeMans / LeMaRns` and `CSMS / CSM`.

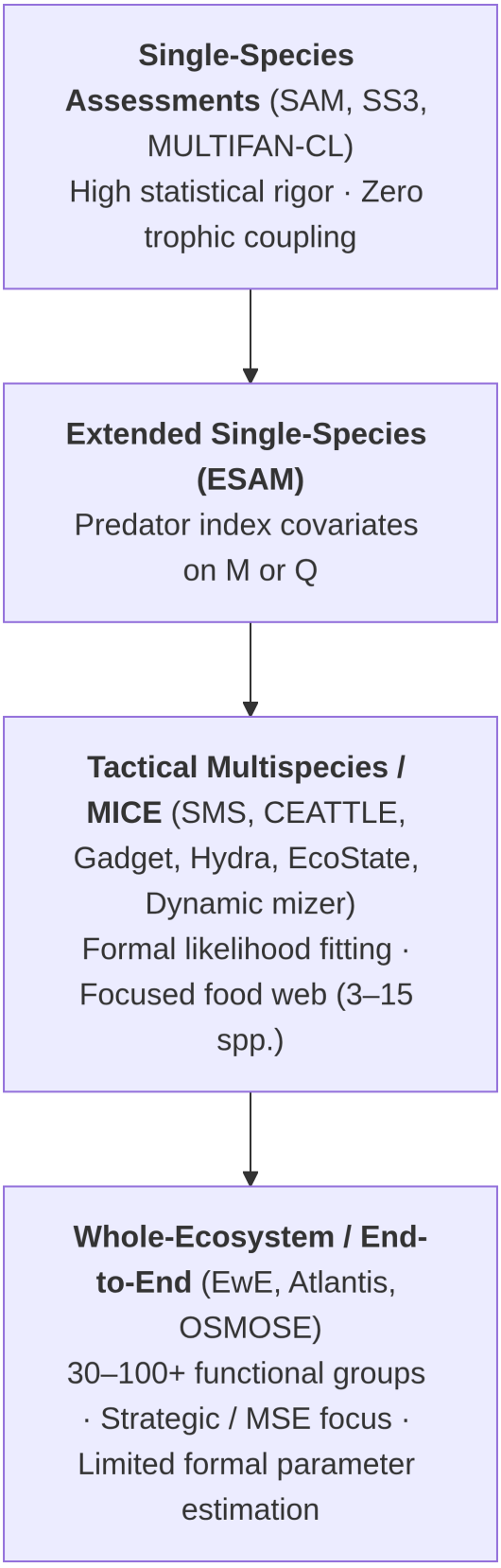
Chapter	Title & Core Scope	Key Models & Methodologies Covered
Chapter 4	**Continuous Size-Spectrum (`mizer`) and State-Space Food-Web Models**	**`mizer`** complete PDEs, bioenergetics, reproduction, starvation, and semi-implicit upwind discretization; **Jump-Growth Fokker–Planck PDE** and second-order growth diffusion $d_i(w,t)$; **Regulation without SRR**; **`Ecopath`, `Ecosim`, `Ecospace`, `Rpath`**, Walters' **Foraging Arena Theory** (v_{ij} exchange), and **`EcoState`** TMB state-space ODE model.
Chapter 5	**Stomach Content Data, Gastric Evacuation Physics, and Diet Likelihood Formulations**	Biophysical **cylinder model of gastric evacuation** ($\frac{dS}{dt} = -\rho L^{\lambda} e^{\delta T} E^{-\mu} S^{0.5}$); **Jensen's inequality pooling bias** and Taylor correction; Prey condition & **`%F`** vs. **`%W`**; **ICES & `DAPSTOM`** databases, **`FishStomachs`** CPUE raising, and **`sdmTMB`** Tweedie spatiotemporal diet standardization; Multinomial, Dirichlet, **Dirichlet-Multinomial**, Logistic-Normal, and **Exponential Family / Gamma log-PPMR likelihoods**.

Chapter	Title & Core Scope	Key Models & Methodologies Covered
Chapter 6	**Statistical Estimation, Data Weighting, Uncertainty Quantification, and `mizer` Roadmap**	Master comparative table across 8 model families; Francis `TA1.8` vs. McAllister–Ianelli vs. Dirichlet-Multinomial data weighting; **TMB Laplace approximation**; GA + Parallel Tempering MCMC and History Matching + SMC + Multi-Model Ensembles; **Five-Pillar Architectural Blueprint for Dynamic Statistical Stock Assessment in `mizer`.**
References	**Bibliography**	Complete bibliography with DOIs for all cited monographs, primary research articles, ICES reports, and software packages.

1 Taxonomy and Theoretical Foundations of Multispecies Stock Assessment Models

1.1 The Complexity Spectrum: Strategic Models, Tactical Assessments, and MICE

Fisheries ecosystem models span a wide continuum of structural complexity, taxonomic resolution, and statistical rigor (Plagányi et al. 2014; Collie et al. 2016; Karp et al. 2023):


Single-Species Assessments (SAM, SS3, MULTIFAN-CL)
High statistical rigor · Zero trophic coupling
Extended Single-Species (ESAM)
Predator index covariates on M or Q
Tactical Multispecies / MICE (SMS, CEATTLE, Gadget, Hydra, EcoState, Dynamic mizer)
Formal likelihood fitting · Focused food web (3–15 spp.)
Whole-Ecosystem / End-to-End (EwE, Atlantis, OSMOSE)
30–100+ functional groups · Strategic / MSE focus · Limited formal parameter estimation

1.1.1 Models of Intermediate Complexity for Ecosystem Assessments (MICE)

(Plagányi et al. 2014) and (Collie et al. 2016) formalized the concept of **MICE** (Models of Intermediate Complexity for Ecosystem assessments) as the "sweet spot" for tactical fisheries management. As model complexity increases, **structural bias** (from omitting trophic interactions) decreases, whereas **parameter estimation uncertainty** (from fitting hundreds of poorly identified interaction rates to noisy data) increases rapidly (Collie et al. 2016, fig. 1).

A model qualifies as a tactical multispecies assessment or MICE when it satisfies four operational criteria (Plagányi et al. 2014; Karp et al. 2023):

1. **Context-Driven Taxonomic Scope**: It models only the core interacting species (typically 3–12 commercial predator and prey stocks plus 1–3 aggregated "other food" or resource pools) responsible for the majority of predation mortality and fishery yield.
2. **Formal Statistical Estimation**: Parameters are estimated by maximizing a formal likelihood (or posterior density) against empirical time series of commercial catches, scientific survey indices, age/size compositions, and stomach contents, rather than relying purely on literature plug-in values or manual tuning.
3. **Diagnostic and Uncertainty Quantification**: It provides standard stock assessment diagnostics (residual patterns, retrospective Mohn's ρ, profile likelihoods, Hessian/Laplace standard errors, or MCMC/SMC credible intervals).
4. **Operational Cadence**: It can be updated on the 1–5 year assessment cycle used by management bodies (e.g., ICES Working Group on Multispecies Assessment Methods [WGSAM] key runs supplying time-varying $M = M1 + M2$ matrices to single-species SAM assessments, or NOAA NPFMC Bering Sea CEATTLE runs).

1.2 Fundamental Axes of Structural Classification

Every multispecies stock assessment model can be positioned along three fundamental mathematical axes (Stefánsson and Pálsson 1998; Andersen et al. 2016; Pennino et al. 2025) (see Table 1.1):

1.2.1 Axis 1: Individual State Space (Age vs. Length vs. Continuous Body Mass)

Table 1.1: Primary State-Space Representations in Multispecies Fisheries Assessment

State Space	State Variable	Representative Models
1. Biomass Pool / Multi-Stanza	$B_i(t)$	`Ecopath with Ecosim` (Walters et al. 1997), `Rpath` (Lucey et al. 2020), `EcoState` (Thorson et al. 2024)
2. Discrete Age-Structured	$N_{i,a,t}$	`MSVPA`/`MSFOR`, `SMS` (Lewy and Vinther 2004), `MSCAA`/`Hydra`, `CEATTLE`
3. Joint Age-Length Structured	$N_{i,a,l,t}$, $W_{i,a,l,t}$	`MULTSPEC`, `BORMICON`, `Gadget2`, `gadget3` (Stefánsson and Pálsson 1998; Begley and Howell 2004)
4. Discrete Length-Structured (Age-Free)	$N_{i,l,t}$	`LeMans` (Hall et al. 2006), `LeMaRns` (Spence et al. 2020), `CSMS` (Pope et al. 2006)
5. Continuous Body-Mass Spectrum	$N_i(w,t)$ (numbers $\cdot$ g^{-1})	`mizer` (Hartvig et al. 2011; Scott et al. 2014; Datta et al. 2010)

1. **Biomass Pool / Multi-Stanza ($B_i(t)$)**:
   - *Governing Dynamical Equation*: Coupled system of ordinary differential equations (ODEs):

$$\frac{dB_i}{dt} = g_i \sum_j Q_{ij} - \sum_k Q_{ki} - (M0_i + F_i)B_i \tag{1.1}$$

   - *Strengths & Limitations*: Fast; low state dimension; captures whole-food-web energy flow; cannot directly fit length or age compositions without multi-stanza delay-difference approximations.

2. **Discrete Age-Structured ($N_{i,a,t}$)**:
   - *Governing Dynamical Equation*: Difference equation along cohort diagonals with empirical weight-at-age $w_{i,a,t}$:

$$N_{i,a+1,t+1} = N_{i,a,t} \exp(-Z_{i,a,t}) \tag{1.2}$$

   - *Strengths & Limitations*: Directly compatible with single-species age-based assessments; simple cohort tracking; requires aging data (otoliths); size-based predation must be mapped through fixed $w_{i,a,t}$ or age-length keys.

3. **Joint Age-Length Structured ($N_{i,a,l,t}$ and $W_{i,a,l,t}$)**:

- *Governing Dynamical Equation*: Advection-dispersion across length bins within each discrete age class:

$$\mathbf{N}_{i,a,t+\Delta t} = \mathbf{G}_{i,a,t}\mathbf{S}_{i,a,t}\mathbf{N}_{i,a,t} \tag{1.3}$$

- *Strengths & Limitations*: Separates chronological age from physical size; resolves size-selective mortality effects on mean length-at-age; computationally intensive due to the $(s \times a \times l \times r)$ state tensor.

4. **Discrete Length-Structured / Age-Free ($N_{i,l,t}$)**:
   - *Governing Dynamical Equation*: Matrix projection across length bins:

$$\mathbf{N}_{i,t+1} = \mathbf{X}_i\mathbf{S}_{i,t}\mathbf{N}_{i,t} + \mathbf{R}_{i,t} \tag{1.4}$$

   - *Strengths & Limitations*: Requires no otolith aging; matches survey length-frequency data directly; deterministic fractional bin transition ($\phi_{i,l}$) introduces numerical cohort diffusion unless a stochastic transition matrix $\mathbf{X}_{l,l'}$ is used.
5. **Continuous Body-Mass Spectrum ($N_i(w,t)$, numbers $\cdot\, \mathbf{g}^{-1}$)**:
   - *Governing Dynamical Equation*: McKendrick–von Foerster PDE (or Fokker–Planck Jump-Growth PDE):

$$\frac{\partial N_i}{\partial t} + \frac{\partial (g_i N_i)}{\partial w} = \frac{1}{2}\frac{\partial^2 (d_i N_i)}{\partial w^2} - \mu_i N_i \tag{1.5}$$

   - *Strengths & Limitations*: Mechanistic mass and energy conservation from plankton to top predators; emergent growth, maturity, and mortality; requires PDE discretization and diffusion treatment to track cohorts dynamically.

1.2.2 Axis 2: Direction of Trophic Feedback (One-Way vs. Two-Way Coupling)

A critical distinction in multispecies stock assessment—emphasized by (Stefánsson and Pálsson 1998) and (Andersen et al. 2016)—is whether trophic interactions operate **one-way (top-down only)** or **two-way (top-down predation mortality + bottom-up bioenergetic growth/reproduction)**:

1.2.2.1 One-Way Trophic Feedback ("Predation-Only" / Fixed-Ration Models)

Used in `MSVPA`, `SMS`, `LeMans/LeMaRns`, and standard modes of `MSCAA/Hydra`. Predator abundance determines the predation mortality rate $M2$ experienced by prey:

$$\text{Predator Abundance } N_{\text{pred}} \implies \text{Prey Mortality } M2_{\text{prey}} \tag{1.6}$$

However, **predator somatic growth rate, weight-at-age $w_{a,t}$, and per-capita total food consumption rate $R_{a,t}$ are independent of prey abundance**. Even if all modeled fish prey collapse, predators maintain their prescribed ration $R_{a,t}$ (implicitly compensating by consuming an inexhaustible "other food" pool B_{other}) and follow fixed growth curves.

⚠ Management Consequence of One-Way Coupling

In forward projections under high forage-fish fishing mortality, one-way (predation-only) models predict that piscivores suffer **zero loss of somatic growth or reproductive output** when their prey are fished down (Jacobsen et al. 2016; Pope et al. 2019).

1.2.2.2 Two-Way Trophic Feedback ("Bioenergetic / Size-Spectrum" Models)

- Used in `mizer`, `MULTSPEC/Gadget` (when configured with `g3a_grow_length_multspec` or `g3a_grow_weight_weightjones`), bioenergetic `CEATTLE`, and `Ecosim/EcoState`.
- Every predation event simultaneously removes biomass from the prey cohort ($M2_{\text{prey}}$) **and** supplies assimilable energy to the predator cohort, driving somatic growth g_{pred}, energy reserves/condition W_{pred}, egg production $R_{p,\text{pred}}$, and (when energy intake falls below metabolic maintenance) starvation mortality $\mu_{s,\text{pred}}$:

$$\text{Prey Abundance } N_{\text{prey}} \Longleftrightarrow \begin{cases} \text{Top-Down: Predation Mortality } M2_{\text{prey}} \\ \text{Bottom-Up: Predator Growth } g_{\text{pred}},\ \text{Reproduction } R_{p,\text{pred}},\ \text{Starvation } \mu \end{cases} \tag{1.7}$$

1.2.3 Axis 3: Calibration Paradigm (Deterministic VPA vs. Pattern-Oriented/History Matching vs. Integrated Statistical Likelihood)

1. **Deterministic / Iterative Backward VPA (`MSVPA`)**: Assumes catch-at-age and stomach contents are known without error; solves for historical $N_{a,t}$, $F_{a,t}$, and $M2_{a,t}$ via fixed-point iteration.
2. **Equilibrium / Pattern-Oriented / History Matching (`mizer` steady-state, `LeMaRns` SMC, `EwE`)**: Calibrates time-averaged catch or biomass against a steady-state reference period, or uses rejection/Sequential Monte Carlo sampling with implausibility metrics (Spence, Thorpe, et al. 2021) to identify ensembles of plausible parameter vectors.
3. **Integrated Maximum Likelihood / State-Space Assessment (`SMS`, `Gadget/gadget3`, `CEATTLE`, `MSCAA/Hydra`, `EcoState`)**: Formulates a differentiable forward process model coupled to explicit statistical observation models for each raw data stream (commercial catch-at-age/length, survey CPUE indices, stomach content proportions), optimizing the joint log-likelihood (or marginal Laplace likelihood in TMB) via automatic differentiation.

1.3 The Andersen & Ursin (1977) Theoretical Foundation and Livingston's (1983) Evaluation

Virtually all modern size- and age-structured multispecies models—including **MSVPA**, **SMS**, **Gadget**, **LeMans/LeMaRns**, **CSMS**, and `mizer`—trace their mathematical lineage to the landmark monograph by K. P. Andersen and Erik Ursin (1977), *"A Multispecies Extension to the Beverton and Holt Theory of Fishing, with Accounts of Phosphorus Circulation and Primary Production"*, Ursin's (1973) empirical analysis of prey size preferences in cod and dab, and Patricia Livingston's (1983) mathematical evaluation and sensitivity analysis of the Andersen–Ursin equations for North Pacific fisheries.

Because (Andersen and Ursin 1977) is the shared ancestor of both statistical multispecies assessments (`SMS`, `MSCAA`, `CEATTLE`) and continuous size-spectrum models (`mizer`), examining its exact differential equations and parameter sensitivities—as documented by (Livingston 1983, 2–11, Table 1)—clarifies how these two branches diverged.

1.3.1 Core Coupled Differential Equations of Andersen & Ursin (1977)

(Andersen and Ursin 1977) extended the three basic (Beverton and Holt 1957) differential equations for individual body weight $w_i(t)$, cohort abundance $N_i(t)$, and fishery yield $Y_i(t)$ of computational entity i (a species-age group) (Livingston 1983, Eqs. 1–8):

1. **Bioenergetic Pütter / Generalized von Bertalanffy Somatic Growth**: The rate of food consumption $\frac{dR_i}{dt}$ by an individual of entity i with body mass w_i is proportional to its allometric absorbing surface w_i^m and its dimensionless **feeding level** $f_i \in [0, 1]$ (capped in the full model at a critical feeding level $f_c = 0.44$):

$$\frac{dR_i}{dt} = f_i \, h_i \, w_i^m \qquad (0 \le f_i \le f_c \le 1) \tag{1.8}$$

   Only a fraction $v_i = (1 - \alpha)\beta$ of consumed food is net-assimilated into tissue, where $\beta = 0.80$ is the digested fraction of food and $\alpha = 0.40$ is the fraction of digested energy lost in specific dynamic action / energy transfer costs (so $v_i = 0.48$ universally; (Livingston 1983, Table 1, p. 11)). Subtracting fasting metabolic catabolism $k_i w_i^n$ gives the individual weight ODE:

$$\frac{dw_i}{dt} = v_i \frac{dR_i}{dt} - k_i w_i^n = (1 - \alpha)\beta \, f_i \, h_i \, w_i^m - k_i w_i^n \tag{1.9}$$

   where (Andersen and Ursin 1977) estimated universal exponents $m = 0.56$ (or $m = 2/3$ in the simplified skeleton model) and $n = 0.83$ (or $n = 1$ in the skeleton model) across 81 fish species (Ursin 1967).

2. **Ursin's (1973) Lognormal Size-Preference Kernel and Suitability (G_{ij})**: Analyzing thousands of stomach records of cod (*Gadus morhua*) and dab (*Limanda limanda*), (Ursin 1973, Eq. 2) demonstrated that predator i of weight w_i evaluates a

potential prey item j of weight w_j according to a symmetric Gaussian distribution on the logarithmic predator-prey mass ratio $\ln(w_i/w_j)$:

$$g_{ij}(w_i, w_j) = \exp\left(-\frac{\left(\ln \frac{w_i}{w_j} - \eta_i\right)^2}{2\sigma_i^2} \right) \tag{1.10}$$

where η_i is the preferred log predator-prey weight ratio ("a prey twice the preferred size is as acceptable as a prey half the preferred size"; (Livingston 1983, 3, Fig. 1)) and σ_i is the standard deviation of diet breadth. Multiplying by species-specific behavioral/habitat vulnerability $\rho_{ij} \in (0, 1]$ gives total suitability:

$$G_{ij} = \rho_{ij}\, g_{ij}(w_i, w_j) \tag{1.11}$$

3. **Available Food (Ψ_i), Search Volume ($q_i w_i^r$), and Feeding Level (f_i)**: Total suitable food biomass encountered by predator i is $\Psi_i = \sum_j \Psi_{ij} = \sum_j G_{ij} w_j N_j$. The feeding level f_i follows a Holling Type II saturation curve (Livingston 1983, Eqs. 5–6):

$$f_i = \frac{\Psi_i}{\Psi_i + Q_i}, \qquad Q_i = \frac{V \text{ (or } A)}{q_i\, w_i^r} \tag{1.12}$$

where V (pelagic water volume) or A (demersal seabed area) scales the half-saturation food biomass Q_i, $r = 0.15$ is the universal search rate exponent, and q_i is the search rate coefficient (23 $\text{m}^3\ \text{g}^{-(1+r)}$ for pelagic habitat or 0.53 $\text{m}^2\ \text{g}^{-(1+r)}$ for demersal habitat; (Livingston 1983, Table 1, p. 10)).

4. **Cohort Mortality Partitioning (F_i, $\hat{M}_i$, $M2_i$) and Discrete Spawning**: Cohort numbers $N_i(t)$ decay according to (Livingston 1983, Eqs. 2A, 8):

$$\frac{dN_i}{dt} = -\left(F_i + \hat{M}_i + M2_i\right) N_i, \qquad \frac{dY_i}{dt} = F_i\, N_i\, w_i \tag{1.13}$$

where predation mortality $M2_i$ inflicted on prey i by all predator groups j is the ratio of total biomass of i consumed per unit time to the standing biomass $N_i w_i$ of i:

$$M2_i = \frac{1}{N_i w_i} \sum_j \frac{dR_j}{dt}\, N_j\, \frac{\Psi_{ji}}{\Psi_j} \tag{1.14}$$

Residual natural mortality $\hat{M}_i = M1_i + M4_i + M5_i + M6_i$ comprises density-dependent larval mortality $M1_i$ plus background disease ($M4_i \approx 0.03\ \text{yr}^{-1}$), senescence ($M5_i$), and spawning stress ($M6_i$). At discrete spawning dates, mature age groups lose a weight fraction π_i to release $E = \sum \pi_i (1 - \pi_{\text{male}}) N_i w_i / \omega_i$ eggs of weight ω_i, which hatch into age-0 recruits via a Beverton–Holt density-dependent survival function (Livingston 1983, Eq. 9, p. 13):

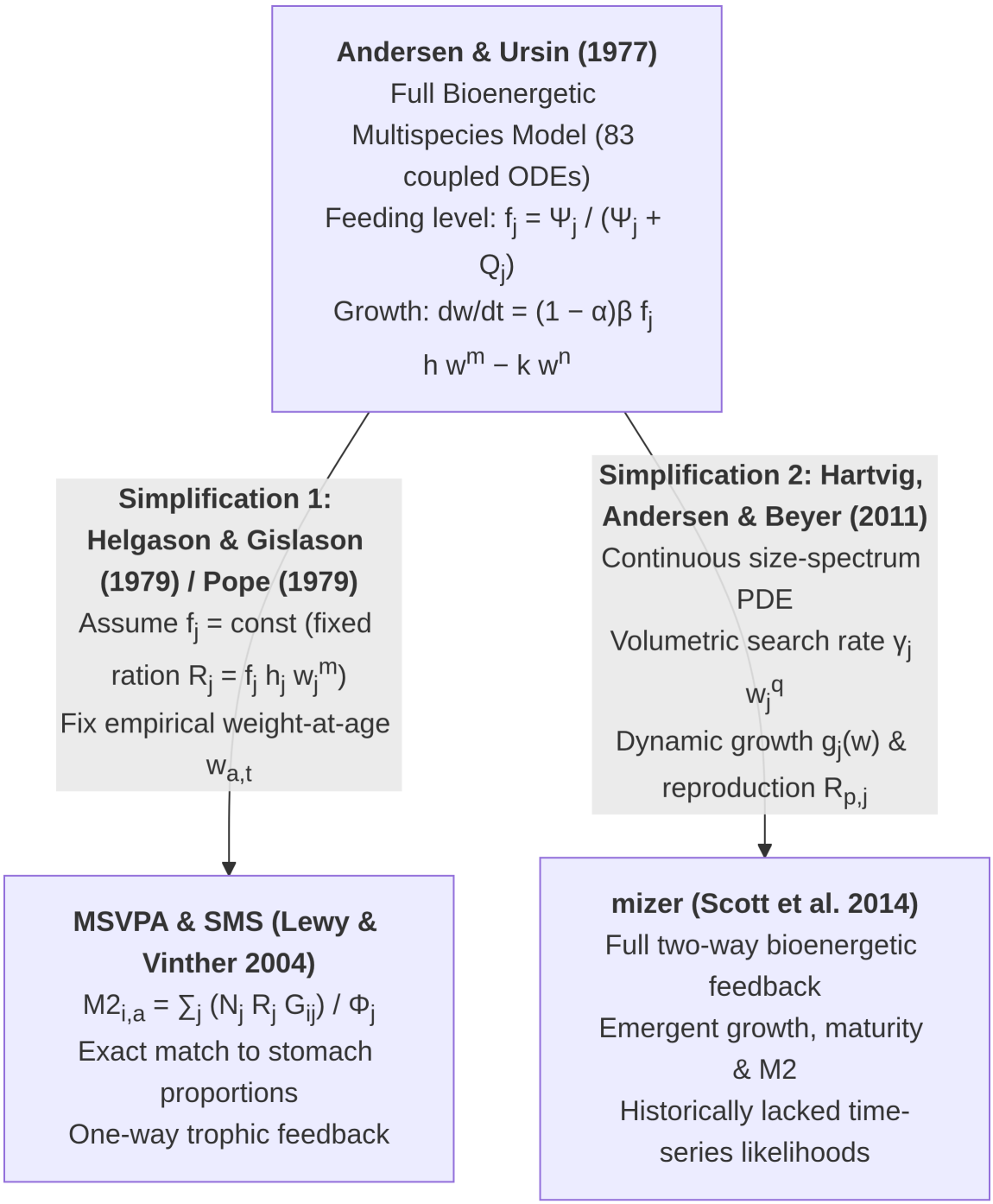
$$E1 = E\left(\frac{C}{C + E}\right) \tag{1.15}$$

1.3.2 Nonlinear Parameter Sensitivity of the Andersen–Ursin System (Livingston 1983)

By perturbing each parameter of the 14-entity Andersen–Ursin skeleton model (phytoplankton, 3 zooplankton age classes, 4 herring age classes, 6 cod age classes) by $\pm 10\%$ and $\pm 20\%$ and fitting the nonlinear power-law response $y = \alpha x^{\beta}$ (Livingston 1983, 13–27, Tables 4–6), Livingston discovered a striking hierarchy of parameter sensitivities:

- **Low Sensitivity ($|\beta| \leq 0.5$, Table 4)**: Spawning fraction π, residual natural mortality $M1$, fish egg weights ω_2, ω_3, and herring fishing mortalities F_6–F_8 had weak, nearly linear effects on equilibrium numbers and body weights.
- **Moderate Sensitivity ($0.5 < |\beta| \leq 1.5$, Table 5)**: Predator search rate scalar Q, density-dependent larval recruitment caps C_2, C_3, and adult cod fishing mortality F_{14} had moderate effects; increasing Q (i.e., reducing search efficiency q_i) lowered individual body weights W_i ($\beta = -0.34$ to -0.66) while increasing prey numbers N_i ($\beta = +0.12$ to $+0.53$) via reduced predation.
- **High, Strongly Nonlinear Sensitivity ($|\beta| > 1.5$, Table 6)**: The **anabolic growth and assimilation parameters** (v, H_1, H_2, H_3, m), catabolic exponent (n), and **mean prey size preference** (η) produced massive, strongly nonlinear responses across the food web (e.g., $\beta = +7.17$ for total weight with respect to anabolic exponent m, and $\beta = +2.39$ to $+3.37$ for cod weight with respect to assimilation efficiency v).
- **Mathematical Rationale**: In a two-way size-structured model, increasing anabolic growth (H_i, m, v) simultaneously (1) increases predator body mass w_j, (2) increases per-capita predator ration $h_j w_j^m$, and (3) shifts the predator's preferred prey size window $w_j e^{-\eta_j}$ across the prey size spectrum, creating a multiplicative trophic cascade across cohorts (Livingston 1983, 26–27).

1.3.3 The Fundamental Divergence: How `SMS/MSVPA` vs. `mizer` Simplified Andersen & Ursin (1977)



1.3.3.1 The MSVPA / SMS Simplification (Helgason and Gislason 1979; Pope 1979; Lewy and Vinther 2004)

In the late 1970s, the ICES Multispecies Working Group realized that Andersen & Ursin's full 83-ODE system had too many unknown bioenergetic parameters to estimate directly from fisheries data, **unless** one assumed that feeding level f_j (and thus total consumption rate $R_j(w_j) = f_j h_j w_j^n$ and somatic growth $w_{a,t}$) was known externally from gastric evacuation studies and commercial weight-at-age sampling.

Under this assumption, the fraction of predator j's diet consisting of prey (i, w_i) is simply the ratio of suitable prey biomass to total suitable food:

$$\text{Diet Proportion } D_{ij} = \frac{G_{ij} B_i}{\Phi_j} \tag{1.16}$$

Consequently, the instantaneous predation mortality rate $M2_i$ on prey i simplifies to:

$$M2_i = \sum_j \frac{\bar{N}_j \, R_j \, G_{ij}}{\Phi_j} \tag{1.17}$$

> **Key Historical Insight: The MSVPA Simplification**
>
> This simplification made G_{ij}/Φ_j **directly observable in stomach content data** (since stomach content weight proportions after evacuation correction estimate D_{ij}), launching the entire MSVPA and SMS assessment tradition. However, the price paid was the loss of dynamic somatic growth feedback (dw/dt).

1.3.3.2 The `mizer` Reformulation (Hartvig et al. 2011; Scott et al. 2014)

Thirty-four years after (Andersen and Ursin 1977), (Hartvig et al. 2011) restored the full bioenergetic growth and reproduction equations of Andersen & Ursin within a continuous McKendrick–von Foerster PDE framework, replacing the arbitrary species-specific half-saturation constants $Q_j(w_j)$ with a mechanistic **volumetric search rate** $\gamma_j(w_j) = \gamma_j w_j^q$.

In `mizer`, the encounter rate of food is $E_j(w_j) = \gamma_j(w_j)\Phi_j(w_j)$, which implies an emergent half-saturation food density $Q_j(w_j) = \frac{h_j w_j^n}{\gamma_j w_j^q}$. All life-history parameters scale allometrically with individual weight w and asymptotic species size $W_{\infty,j}$, reducing the parameter space from hundreds of cohort-specific constants to a compact set of scaling exponents and species traits.

1.4 Community-Level Dynamics and the Multispecies Jacobian Matrix

When comparing multispecies stock assessment models across these structural paradigms (Tsou and Collie 2001; Aydin et al. 2005; Jacobsen et al. 2016; Spence et al. 2018; Pope et al. 2019), (Pope et al. 2006), (Pope et al. 2019), and (Jacobsen et al. 2016) introduced two unifying analytical diagnostics:

1.4.1 The Yield Vector and Jacobian Interaction Matrix (J)

Let $\mathbf{F} = (F_1, F_2, \dots, F_S)^\top$ be the vector of long-term fishing mortality rates across S species, and let $\mathbf{Y}(\mathbf{F}) = (Y_1, Y_2, \dots, Y_S)^\top$ and $\mathbf{SSB}(\mathbf{F}) = (\mathrm{SSB}_1, \dots, \mathrm{SSB}_S)^\top$ be the resulting equilibrium yield and spawning stock biomass vectors. Near a reference fishing vector $\mathbf{F}_0$, the response of the multispecies community is characterized by the **Jacobian matrix** (Pope et al. 2019, 5):

$$J_{ik} = \left.\frac{\partial \ln \mathrm{SSB}_i}{\partial F_k}\right|_{\mathbf{F}=\mathbf{F}_0} \quad \text{or} \quad J^{(Y)}_{ik} = \left.\frac{\partial Y_i}{\partial F_k}\right|_{\mathbf{F}=\mathbf{F}_0} \tag{1.18}$$

- **Diagonal elements ($J_{ii} < 0$)**: Measure the direct single-species depletion + intraspecific cannibalism recovery response of species i to its own fishing mortality F_i. Strong cannibalism (e.g., cod, whiting, hake) dampens $|J_{ii}|$ because fishing down large adults releases juveniles from cannibalistic $M2$.
- **Off-diagonal elements (J_{ik} for $i \neq k$)**:
  - In **one-way models** (`SMS`, `LeMans`), $J_{\mathrm{prey,pred}} > 0$ (fishing down a predator increases prey SSB by reducing $M2$), while $J_{\mathrm{pred,prey}} \approx 0$ (fishing down a prey has zero direct effect on predator SSB, or only an indirect "prey-switching/shielding" effect if total suitable food Φ_{pred} changes cannibalism rates).
  - In **two-way bioenergetic models** (`mizer`, `Gadget`, `Ecosim`), $J_{\mathrm{pred,prey}} < 0$ because fishing down forage fish reduces predator feeding level f_{pred}, slowing predator somatic growth g_{pred} and lowering reproductive output $R_{p,\mathrm{pred}}$.

1.4.2 Quadratic Approximation of Total Multispecies Yield

(Pope et al. 2006) and (Pope et al. 2019) showed that across widely different multispecies models of the North Sea (`SMS`, `Gadget`, `EwE`, `LeMans`, `mizer`, `CSMS`, `CSM`), the equilibrium yield of species i as a function of the fishing mortality vector $\mathbf{F}$ is well approximated by a coupled quadratic form (a multispecies generalization of the Schaefer surplus production parabola):

$$Y_i(\mathbf{F}) \approx a_i F_i + \sum_{k=1}^{S} b_{ik} F_i F_k \tag{1.19}$$

In matrix notation, total community yield $Y_{\text{tot}}(\mathbf{F}) = \sum_{i=1}^{S} Y_i(\mathbf{F}) = \mathbf{a}^\top \mathbf{F} + \mathbf{F}^\top \mathbf{B}\mathbf{F}$. An unconstrained community-wide MSY exists at $\mathbf{F}_{\text{MSY}} = -\frac{1}{2}(\mathbf{B}+\mathbf{B}^\top)^{-1}\mathbf{a}$ only if the symmetric part of the interaction matrix $\frac{1}{2}(\mathbf{B} + \mathbf{B}^\top)$ is negative definite; in practice, strong predation terms $b_{ik} > 0$ often make the unconstrained optimum corner-valued ("fish out the predators to maximize forage fish yield"), necessitating conservation constraints ($B_i \geq B_{\text{lim},i}$) or value-weighted objective functions (Pope et al. 2019).

2 Age-Structured Multispecies Stock Assessment Models (`MSVPA`, `SMS/smsR`, `MSCAA/Hydra`, `CEATTLE`)

2.1 Overview of the Age-Structured Multispecies Family

Age-structured multispecies models are direct extensions of single-species Virtual Population Analysis (VPA) and Statistical Catch-at-Age (SCAA) models. They partition total instantaneous mortality $Z_{s,a,y,q}$ of species s, age class a, year y, and season/quarter q into three additive components (first formalized by (Helgason and Gislason 1979) and (Pope 1979)):

$$Z_{s,a,y,q} = F_{s,a,y,q} + M1_{s,a,q} + M2_{s,a,y,q} \tag{2.1}$$

where:

- $F_{s,a,y,q}$ is the instantaneous fishing mortality rate,
- $M1_{s,a,q}$ is the **residual (background) natural mortality rate** due to disease, senescence, abiotic stress, and predation by unmodeled predators,
- $M2_{s,a,y,q}$ is the **endogenous predation mortality rate** inflicted by the modeled predators in the ecosystem.

Because these models track discrete age cohorts (s, a) rather than continuous body size, individual body weight $w_{s,a,y,q}$ and length $l_{s,a,y,q}$ are typically treated as known empirical inputs (measured from port and survey sampling), or mapped via pre-computed age-length keys ($\text{ALK}_{s,a,l,y,q}$).

Four major model lineages define this family:

- **Foundations and `MSVPA`**: (Ursin 1967), (Ursin 1973), and (Andersen and Ursin 1977) established the size-based bioenergetic and prey-suitability equations, evaluated for North Pacific stocks by (Livingston 1983) and simplified into operational `MSVPA` by (Helgason and Gislason 1979), (Pope 1979), and (Sparre 1980).
- **Stochastic Multispecies Model (`SMS` / `smsR`)**: (Lewy and Vinther 2004) formulated the seasonal age-length statistical multispecies model and Dirichlet stomach-content likelihood used in ICES key runs (ICES 2021, 2023).

- **North American Statistical Catch-at-Age (`MSASA` / `MSCAA` / `Hydra`)**: (Jurado-Molina et al. 2005), (Kinzey and Punt 2009), (Van Kirk et al. 2010), (Curti et al. 2013), and (Trijoulet et al. 2019) developed and tested forward-projecting multispecies catch-at-age models across alternative predator functional responses and diet observation error regimes.
- **Climate-Enhanced Multispecies Assessment (`CEATTLE` / `Rceattle`)**: (Holsman and Aydin 2015), (Holsman et al. 2016), and (Adams et al. 2022) coupled temperature-dependent von Bertalanffy growth (Essington et al. 2001) and Wisconsin bioenergetics (Kitchell et al. 1977; Thornton and Lessem 1978) with age-structured stock assessment and multispecies biological reference points.

2.2 Multispecies Virtual Population Analysis (`MSVPA`) and `MSFOR`

2.2.1 Historical Development

Developed simultaneously by (Helgason and Gislason 1979) and (Pope 1979) following the (Andersen and Ursin 1977) monograph, **MSVPA** was the standard operational multispecies model of the ICES Multispecies Working Group (MSWG) throughout the 1980s and 1990s, motivating the massive 1981 and 1991 ICES "Years of the Stomach" sampling programs in the North Sea (Stefánsson and Pálsson 1998; Lewy and Vinther 2004). **MSFOR** is its forward-projection counterpart used to evaluate management scenarios given recruitment assumptions.

2.2.2 Mathematical Formulation of MSVPA

Within each quarter $q \in \{1, 2, 3, 4\}$ of year y, cohort abundance $N_{i,a,y,q}$ of prey species i, age a decays exponentially:

$$N_{i,a,y,q+1} = N_{i,a,y,q} \exp\!\left(-Z_{i,a,y,q}\right), \qquad \bar{N}_{i,a,y,q} = N_{i,a,y,q} \frac{1 - \exp(-Z_{i,a,y,q})}{Z_{i,a,y,q}} \tag{2.2}$$

Observed catch-at-age in numbers $C_{i,a,y,q}$ is assumed exact (zero observation error), obeying the Baranov catch equation:

$$C_{i,a,y,q} = \frac{F_{i,a,y,q}}{Z_{i,a,y,q}} N_{i,a,y,q} \left(1 - \exp(-Z_{i,a,y,q})\right) = F_{i,a,y,q} \bar{N}_{i,a,y,q} \tag{2.3}$$

The predation mortality $M2_{i,a,y,q}$ on prey (i, a) by all predator species j and predator ages b is:

$$M2_{i,a,y,q} = \sum_j \sum_b \frac{\bar{N}_{j,b,y,q}\, R_{j,b,q}\, S_{i,a,j,b,q}}{\Phi_{j,b,y,q}} \tag{2.4}$$

where:

- $R_{j,b,q}$ is the prescribed per-capita quarterly food ration (in mass) consumed by predator (j, b) in quarter q,
- $S_{i,a,j,b,q} \in [0, 1]$ is the **suitability coefficient** of prey (i, a) for predator (j, b) in quarter q,
- $\Phi_{j,b,y,q}$ is the **total available (suitable) food biomass** for predator (j, b):

$$\Phi_{j,b,y,q} = \sum_k \sum_c S_{k,c,j,b,q}\, \bar{N}_{k,c,y,q}\, w_{k,c,y,q} + S_{\text{other},j,b,q}\, B_{\text{other},q} \tag{2.5}$$

2.2.3 Iterative Fixed-Point Solution Algorithm

In classic MSVPA, suitability $S_{i,a,j,b,q}$ is not parameterized as a smooth function of predator-prey size ratio; instead, it is estimated directly from observed average stomach content weight proportions $\bar{W}_{i,a,j,b,y,q}$ via fixed-point iteration:

1. **Initialize**: Set $M2^{(0)}_{i,a,y,q} = 0$ (or a prior guess) and run standard backward VPA on each species to obtain initial cohort abundances $\bar{N}^{(0)}_{i,a,y,q}$ and fishing mortalities $F^{(0)}_{i,a,y,q}$.
2. **Update Suitabilities**: For years $y \in \mathcal{Y}_{\text{stom}}$ with stomach content observations $\bar{W}_{i,a,j,b,y,q}$, compute empirical suitabilities:

$$\tilde{S}_{i,a,j,b,y,q} = \frac{\bar{W}_{i,a,j,b,y,q}}{\bar{N}_{i,a,y,q}\, w_{i,a,y,q}} \tag{2.6}$$

   Normalize so $\sum_{i,a} \tilde{S}_{i,a,j,b,y,q} = 1$, and average across stomach-sampling years $y \in \mathcal{Y}_{\text{stom}}$ (weighted by predator stomach sample size) to obtain time-invariant quarter-specific suitabilities $S^{(m)}_{i,a,j,b,q}$.
3. **Update** $M2$: Substitute $S^{(m)}_{i,a,j,b,q}$ and $\bar{N}^{(m-1)}$ into the $M2$ equation to compute $M2^{(m)}_{i,a,y,q}$.
4. **Update VPA**: Re-run backward VPA for all species using $Z^{(m)}_{i,a,y,q} = F^{(m)}_{i,a,y,q} + M1_{i,a,q} + M2^{(m)}_{i,a,y,q}$ to obtain $\bar{N}^{(m)}_{i,a,y,q}$.
5. **Iterate**: Repeat Steps 2–4 until $\max |M2^{(m)} - M2^{(m-1)}| < \epsilon$.

2.2.4 Limitations of MSVPA

As analyzed by (Stefánsson and Pálsson 1998) and (Lewy and Vinther 2004):

- **No Observation Error**: MSVPA treats commercial catch-at-age and stomach content proportions as exact deterministic quantities. Sampling noise in rare stomach items propagates directly into suitability $S_{i,a,j,b,q}$.
- **Over-Parameterization of Suitability**: Estimating a free suitability parameter for every (i, a, j, b, q) combination requires thousands of parameters and leaves zero degrees of freedom for prey age classes not observed in a particular predator age's stomachs.
- **No Uncertainty Estimates**: Because there is no statistical likelihood function, MSVPA cannot produce confidence intervals on $M2$, SSB, or recruitment, nor can it perform formal hypothesis testing (e.g., AIC/LRT between functional responses).

2.3 The Stochastic Multispecies Model (`SMS` and `smsR`)

To overcome the deterministic limitations of MSVPA, Peter Lewy and Morten Vinther at DTU Aqua developed **SMS** (Stochastic Multispecies Model; (Lewy and Vinther 2004)), implemented in **ADMB** (Auto Differentiation Model Builder) and recently ported to **TMB/R** as `smsR` (Nis Sand Jacobsen et al.; (ICES 2021, 2023)).

SMS is the flagship operational multispecies stock assessment model in ICES, providing the official time-varying natural mortality matrices ($M = M1 + M2$) used in single-species stock assessments (e.g., `SAM` and `SS3`) for the North Sea and Baltic Sea (ICES 2023).

2.3.1 State-Space Indexing and Population Dynamics

SMS is a forward-projecting, seasonal (typically quarterly, $q \in \{1, 2, 3, 4\}$), age-structured statistical catch-at-age model. In addition to tracked dynamic VPA species (which can act as both predators and prey), SMS includes **"other predators"** (e.g., seabirds, grey seals, harbour porpoises, horse mackerel, hake) whose abundance time series $N_{A,a,y,q}$ are entered as known external covariates, and an **"other food"** biomass pool B_{OF} (Lewy and Vinther 2004, 7).

Following the notation of (Lewy and Vinther 2004):

- Let (A, a) denote a predator species A of age (or size class) a, with mean weight $w_{A,a,y,q}$.
- Let (B, b) denote a prey species B of age (or size class) b, with mean weight $w_{B,b,y,q}$.

Stock survival across seasons q within year y follows:

$$N_{s,a,y,q+1} = N_{s,a,y,q} \exp\left(-Z_{s,a,y,q}\right), \qquad q = 1, 2, 3 \tag{2.7}$$

At the transition from the final quarter $q = 4$ of year y to $q = 1$ of year $y + 1$:

$$N_{s,a+1,y+1,1} = N_{s,a,y,4} \exp\left(-Z_{s,a,y,4}\right), \qquad a < a_{\max} - 1 \tag{2.8}$$

and for the plus-group $a_{\max}$:

$$N_{s,a_{\max},y+1,1} = N_{s,a_{\max}-1,y,4} \exp\bigl(-Z_{s,a_{\max}-1,y,4}\bigr) + N_{s,a_{\max},y,4} \exp\bigl(-Z_{s,a_{\max},y,4}\bigr) \tag{2.9}$$

Recruitment $N_{s,a_{\mathrm{rec}},y,q_{\mathrm{rec}}}$ is parameterized either as free log-parameters $\ln R_{s,y}$ or via a stochastic stock-recruitment relationship (Ricker, Beverton-Holt, or hockey-stick) with lognormal process error:

$$N_{s,a_{\mathrm{rec}},y,q_{\mathrm{rec}}} = f_{\mathrm{SRR}}\bigl(\mathrm{SSB}_{s,y-a_{\mathrm{rec}}}\bigr) \exp\bigl(\epsilon_{R,s,y}\bigr), \qquad \epsilon_{R,s,y} \sim \mathcal{N}\bigl(0, \sigma^2_{R,s}\bigr) \tag{2.10}$$

2.3.2 Separable Fishing Mortality Model

To prevent overfitting catch-at-age noise, SMS models instantaneous fishing mortality $F_{s,a,y,q}$ using a partially separable structure across age a, year y, and season q within multi-year regime blocks Y_j (Lewy and Vinther 2004, Eq. 2):

$$F_{s,a,y,q} = v_{s,a,Y_j}\, F^{(y)}_{s,y}\, F^{(q)}_{s,a,Y_j,q} \tag{2.11}$$

where:

- v_{s,a,Y_j} is the age effect (fishery selectivity at age a in year-block Y_j),
- $F^{(y)}_{s,y}$ is the annual fishing effort/mortality multiplier in year y,
- $F^{(q)}_{s,a,Y_j,q}$ is the seasonal allocation fraction of fishing mortality in quarter q (constrained so the mean across seasons equals 0.25).

Expected commercial catch-at-age $\hat{C}_{s,a,y,q}$ is given by the Baranov equation:

$$\hat{C}_{s,a,y,q} = \frac{F_{s,a,y,q}}{Z_{s,a,y,q}} N_{s,a,y,q} \left(1 - \exp(-Z_{s,a,y,q})\right) = F_{s,a,y,q} \bar{N}_{s,a,y,q} \tag{2.12}$$

2.3.3 Parametric Suitability and the Andersen–Ursin Size-Selection Submodel

Instead of MSVPA's non-parametric empirical suitabilities, SMS parameterizes the suitability $S(B, b, A, a, q)$ of prey (B, b) for predator (A, a) in season q as a multiplicative product of three biological components (Lewy and Vinther 2004, Eq. 5):

$$S(B, b, A, a, y, q) = \mathrm{vul}(B, A) \times \mathrm{size}(B, b, A, a, y, q) \times \mathrm{overlap}(B, A, q) \tag{2.13}$$

1. **Species Vulnerability (vul(B, A))**: Time-invariant parameter measuring the behavioral/morphological vulnerability of prey species B to predator species A (with $\mathrm{vul}(\text{Other Food}, A) \equiv 1$ as the reference baseline).
2. **Spatial/Seasonal Overlap (overlap(B, A, q))**: Parameter representing the spatial co-occurrence of prey B and predator A in season q (fixed to 1 in seasons with full overlap, estimated in seasons with migratory separation).

3. **Asymmetric Lognormal Size Preference (size(B, b, A, a, y, q))**: Following (Andersen and Ursin 1977), SMS models size preference as a Gaussian (or asymmetric split-Gaussian) function of the log predator-to-prey weight ratio $\xi = \ln\left(\frac{w_{A,a,y,q}}{w_{B,b,y,q}}\right)$, restricted to the physiologically feasible range $[\xi_{\min,A}, \xi_{\max,A}]$ (Lewy and Vinther 2004, Eq. 6):

$$\text{size}(B,b,A,a,y,q) = \begin{cases} \exp\left(-\dfrac{\left(\ln\frac{w_{A,a}}{w_{B,b}} - \eta_A\right)^2}{2\sigma_{1,A}^2}\right) & \text{if } \xi_{\min,A} \le \ln\frac{w_{A,a}}{w_{B,b}} \le \eta_A \\ \exp\left(-\dfrac{\left(\ln\frac{w_{A,a}}{w_{B,b}} - \eta_A\right)^2}{2\sigma_{2,A}^2}\right) & \text{if } \eta_A < \ln\frac{w_{A,a}}{w_{B,b}} \le \xi_{\max,A} \\ 0 & \text{otherwise} \end{cases} \tag{2.14}$$

When SMS is run in **length-class stomach mode** (because ICES stomach databases record predator and prey *lengths* l, not ages a), SMS converts age abundances $N_{s,a,y,q}$ into length classes l using seasonal empirical Age-Length Keys ($\text{ALK}_{s,a,l,y,q}$) before evaluating $\text{size}(B, l_B, A, l_A)$ using constant length-weight relationships $w_s(l) = \alpha_s l^{\beta_s}$, and then maps $M2$ back from length classes to age classes (Lewy and Vinther 2004, 8–9).

2.3.4 Gastric-Evacuation-Based Consumption Rate ($R_{A,a,y,q}$)

Total quarterly food intake (ration) $R_{A,a,y,q}$ of predator (A, a) is input as an empirical covariate derived from average observed stomach fullness $\bar{S}_{A,a,q}$ and temperature-dependent gastric evacuation models (Andersen and Beyer 2005; Lewy and Vinther 2004, Eqs. 7–8):

$$R_{A,a,y,q} = \alpha_{R,A}\, w_{A,a,y,q}^{\beta_{R,A}} \quad \text{or} \quad R_{A,a,y,q} = \rho_A\, \bar{S}_{A,a,q}^{\gamma_A} \exp(\delta_A T_{y,q})\, L_{A,a}^{\lambda_A}\, D_q \tag{2.15}$$

where $T_{y,q}$ is ambient water temperature and D_q is the number of hours in quarter q.

2.3.5 Predation Mortality ($M2$) and Exact Integration Over the Quarter

Let $\bar{B}_{B,b,y,q} = \bar{N}_{B,b,y,q} w_{B,b,y,q}$ be the mean quarterly biomass of prey (B, b), and let total available food for predator (A, a) be:

$$\Phi(A,a,y,q) = \sum_B \sum_b \bar{B}_{B,b,y,q}\, S(B,b,A,a,y,q) + B_{\text{OF},A}\, S(\text{OF},A,a,y,q) \tag{2.16}$$

In differential form at instant $t \in [q, q+1]$, the instantaneous predation rate is $M2_{B,b}(t) = \sum_{A,a} \frac{N_{A,a}(t) r_{A,a} S(B,b,A,a)}{\Phi(A,a,t)}$. Assuming $\Phi(A, a, t)$ is approximately constant within a single quarter (since B_{OF} is large and individual prey decline is offset across cohorts), integrating

over the quarter yields the exact discrete-time SMS $M2$ equation (Lewy and Vinther 2004, Eq. 11):

$$M2_{B,b,y,q} = \sum_{A} \sum_{a} \frac{\bar{N}_{A,a,y,q}\, R_{A,a,y,q}\, S(B,b,A,a,y,q)}{\Phi(A,a,y,q)} \tag{2.17}$$

Since $\bar{N}_{B,b,y,q} = N_{B,b,y,q} \frac{1-e^{-Z_{B,b,y,q}}}{Z_{B,b,y,q}}$ depends on $M2_{B,b,y,q}$ (and predator survival also depends on $M2$ for intermediate predators like whiting or young cod), SMS performs 2–4 internal Newton/fixed-point iterations per quarter evaluation within the ADMB/TMB objective function, or uses the explicit beginning-of-quarter biomass approximation when quarterly time steps are short.

2.3.6 Expected Stomach Content Proportions

Under the SMS equations, the expected proportion by weight $\hat{W}(B,b,A,a,y,q)$ of prey (B,b) in the stomach of predator (A,a) in year y, quarter q is (Lewy and Vinther 2004, Eq. 12):

$$\hat{W}(B,b,A,a,y,q) = \frac{\bar{B}_{B,b,y,q}\, S(B,b,A,a,y,q)}{\Phi(A,a,y,q)} \tag{2.18}$$

Note that $\sum_{B,b} \hat{W}(B,b,A,a,y,q) + \hat{W}(\mathrm{OF},A,a,y,q) = 1$.

2.3.7 Statistical Likelihood Function of SMS

SMS estimates all parameters $\theta = (\ln N_{\text{init}}, \ln R_{s,y}, v_{s,a}, F^{(y)}_{s,y}, F^{(q)}_{s,q}, q_{\text{survey}}, \text{vul}(B,A), \eta_A, \sigma_A, \text{overlap}, \dots)$ by minimizing the negative joint log-likelihood across four data types (Lewy and Vinther 2004, 10–16):

$$\mathcal{L}_{\text{total}}(\theta) = \mathcal{L}_{\text{catch}}(\theta) + \mathcal{L}_{\text{CPUE}}(\theta) + \mathcal{L}_{\text{stomach}}(\theta) + \mathcal{L}_{\text{SRR}}(\theta) \tag{2.19}$$

2.3.7.1 1. Commercial Catch-at-Age Likelihood (Lognormal)

Observed catch-at-age $C_{s,a,y,q}$ is assumed lognormally distributed around expected catch $\hat{C}_{s,a,y,q}$:

$$\ln C_{s,a,y,q} \sim \mathcal{N}\left(\ln \hat{C}_{s,a,y,q},\ \sigma^2_{\text{catch},s,a,q}\right) \tag{2.20}$$

contributing to the negative log-likelihood:

$$-\ln L_{\text{catch}} = \frac{1}{2} \sum_{s,a,y,q} \left[\ln\left(2\pi\sigma^2_{\text{catch},s,a,q}\right) + \frac{\left(\ln C_{s,a,y,q} - \ln \hat{C}_{s,a,y,q}\right)^2}{\sigma^2_{\text{catch},s,a,q}} \right] \tag{2.21}$$

2.3.7.2 2. Survey Abundance Index (CPUE) Likelihood (Lognormal)

For survey fleet f operating from season fraction α_f to β_f in quarter q, the expected survey index $\hat{I}_{f,s,a,y}$ is:

$$\hat{I}_{f,s,a,y} = Q_{f,s,a}\, N_{s,a,y,q} \frac{\exp(-\alpha_f Z_{s,a,y,q}) - \exp(-\beta_f Z_{s,a,y,q})}{(\beta_f - \alpha_f) Z_{s,a,y,q}} \tag{2.22}$$

with lognormal negative log-likelihood:

$$-\ln L_{\text{CPUE}} = \frac{1}{2} \sum_{f,s,a,y} \left[\ln\!\left(2\pi\sigma^2_{\text{CPUE},f,s,a}\right) + \frac{\left(\ln I_{f,s,a,y} - \ln \hat{I}_{f,s,a,y}\right)^2}{\sigma^2_{\text{CPUE},f,s,a}} \right] \tag{2.23}$$

2.3.7.3 3. Stomach Content Composition Likelihood (Dirichlet vs. Multinomial)

Let $U = (A, a, y, q)$ index a specific predator-age-year-quarter sampling stratum, and let $k \in \{1, \ldots, K_U\}$ index the K_U prey entities (B, b) (plus "other food") observed in that predator's diet, with observed weight proportions $\mathbf{W}_U = (W_{1,U}, \ldots, W_{K_U,U})$ satisfying $\sum_{k=1}^{K_U} W_{k,U} = 1$, and model expectations $\hat{p}_{k,U} = \hat{W}(k, U)$.

(Lewy and Vinther 2004, 11–14) derived two alternative likelihood formulations for continuous stomach weight proportions:

- **Option A: Multinomial Approximation on Effective Prey Counts** Defining an effective number of independent prey observations $N_{\text{stom},U}$ (e.g., number of stomachs or hauls sampled), $N_{\text{stom},U}\mathbf{W}_U$ is treated as a multinomial vector with log-likelihood kernel:

$$\ln L_{\text{stom, Mult}} = \sum_U N_{\text{stom},U} \sum_{k=1}^{K_U} W_{k,U} \ln \hat{p}_{k,U} \tag{2.24}$$

- **Option B: Exact Dirichlet Distribution (Default in SMS)** Because $W_{k,U} \in (0, 1)$ is a continuous biomass proportion rather than a discrete integer count, (Lewy and Vinther 2004, Eqs. 19–24) modeled $\mathbf{W}_U$ using a **Dirichlet distribution** with concentration parameters $\alpha_{k,U} = v_U \hat{p}_{k,U}$ and total precision sum $v_U = \sum_{k=1}^{K_U} \alpha_{k,U}$:

$$f_{\text{Dir}}(\mathbf{W}_U \mid v_U \hat{\mathbf{p}}_U) = \frac{\Gamma(v_U)}{\prod_{k=1}^{K_U} \Gamma(v_U \hat{p}_{k,U})} \prod_{k=1}^{K_U} W_{k,U}^{v_U \hat{p}_{k,U} - 1} \tag{2.25}$$

  Under the Dirichlet distribution, the expectation and variance of each prey share $W_{k,U}$ are:

$$\mathbb{E}[W_{k,U}] = \hat{p}_{k,U}, \qquad \text{Var}(W_{k,U}) = \frac{\hat{p}_{k,U}(1 - \hat{p}_{k,U})}{v_U + 1} \tag{2.26}$$

Defining the variance-scaling factor $V_U = \frac{1}{v_U+1} \iff v_U = \frac{1}{V_U} - 1$, SMS relates V_U to the predator species' overall diet dispersion parameter $\theta_{\text{stom},A}$ and the number of stomachs/hauls sampled $N_{\text{haul},U}$ via:

$$V_U = \frac{\theta_{\text{stom},A}}{N_{\text{haul},U}} \implies v_U = \frac{N_{\text{haul},U}}{\theta_{\text{stom},A}} - 1 \tag{2.27}$$

This allows $\theta_{\text{stom},A}$ to be estimated directly inside the maximum likelihood optimization via the exact Gamma-function log-likelihood! (Wait: to handle observed zeros $W_{k,U} = 0$, SMS pools unobserved interior age classes with adjacent age classes or applies a small threshold; see Chapter 5 for full details).

2.4 Statistical Multispecies Catch-at-Age (`MSASA`, `MSCAA`, and `Hydra`)

In North America (particularly NOAA AFSC for the Aleutian Islands, Gulf of Alaska, and Bering Sea, and NOAA NEFSC for Georges Bank and the Gulf of Maine), a family of **Multispecies Statistical Catch-at-Age (`MSASA` / `MSCAA`)** models was developed in ADMB and TMB by (Kinzey and Punt 2009), (Van Kirk et al. 2010), (Curti et al. 2013), Gaichas et al. (`Hydra` MSE operating model), and evaluated in simulation studies by (Trijoulet et al. 2019).

2.4.1 Kinzey & Punt (2009): Aleutian Islands Age–Length Hybrid `MSCAA` and the Seven Predator Functional Responses

(Kinzey and Punt 2009) developed a foundational integrated multispecies statistical catch-at-age framework in ADMB and applied it to the Aleutian Islands (1960–2003) for walleye pollock (*Theragra chalcogramma*, ages 0–15+, 1–80 cm in 5 cm bins), Atka mackerel (*Pleurogrammus monopterygius*, ages 0–15+, 1–65 cm), and Pacific cod (*Gadus macrocephalus*, ages 0–12+, 1–120 cm, harvested by 3 fleets: trawl, pot, and longline).

1. **Predation Mortality ($P_{k,a,y}$), Gamma Length-Ratio Selectivity ($\gamma^{r,u}_{k,a}$), and Age–Length Transition Mapping ($\chi^{r,u,l}$)**: Total mortality on prey species k, age a in year y is $Z_{k,a,y} = \sum_f F^f_{k,a,y} + P_{k,a,y} + M_k$, where predation mortality $P_{k,a,y}$ inflicted by all predator species r and predator ages u is (Kinzey and Punt 2009, Eqs. 6–7, p. 77):

$$P_{k,a,y} = \sum_{r,u} V^{r,u}_{k,a,y}, \qquad V^{r,u}_{k,a,y} = N_{r,u,y}\, \phi^{r,u}_{k,a,y}\, \gamma^{r,u}_{k,a} \tag{2.28}$$

   Instead of a symmetric lognormal weight-ratio kernel, (Kinzey and Punt 2009, Eq. 18, p. 80) parameterize predator (r,u) size selectivity $\gamma^{r,u}_{k,a} \in (0,1]$ for prey (k,a) as a

skewed Gamma function of the log predator-to-prey expected length ratio $G^{r,u}_{k,a} = \ln(\bar{L}_{r,u}/\bar{L}_{k,a})$, normalized to equal 1 at its mode $\tilde{G}^r = (\alpha^r - 1)\beta^r$:

$$\gamma^{r,u}_{k,a} = \left(\frac{G^{r,u}_{k,a}}{\tilde{G}^r}\right)^{\alpha^r - 1} \exp\left[-\frac{G^{r,u}_{k,a} - \tilde{G}^r}{\beta^r}\right], \qquad \tilde{G}^r = (\alpha^r - 1)\beta^r \tag{2.29}$$

The total number of prey (k, a) eaten by predator (r, u) in year y follows the Baranov predation equation:

$$E^{r,u}_{k,a,y} = \frac{V^{r,u}_{k,a,y}}{Z_{k,a,y}} N_{k,a,y}\left(1 - e^{-Z_{k,a,y}}\right) \tag{2.30}$$

Because stomach samples ($1,987$ pollock, $1,033$ mackerel, and $3,278$ cod stomachs) are recorded by 5 **cm predator length bin** l and 5 **cm prey length bin** m, Kinzey & Punt map age-specific predation into length space using prespecified empirical age-to-length transition matrices $\chi^{r,u,l}$ (Kinzey and Punt 2009, Eqs. 9–14 & Table 3, pp. 77–84):

- **Prey numbers eaten by predator length class** (r, l): $E^{r,l}_{k,a,y} = \sum_u E^{r,u}_{k,a,y}\chi^{r,u,l}$.
- **Predicted prey length composition** $\tilde{T}^{r,l}_{k,m}$ **inside predator** (r, l) **stomachs** (fitted in multinomial likelihood L_6):

$$\tilde{T}^{r,l}_{k,m} = \frac{\sum_y n^{r,l}_y \sum_a \chi^{k,a,m} E^{r,l}_{k,a,y}}{\sum_y n^{r,l}_y \sum_{a'} E^{r,l}_{k,a',y}} \tag{2.31}$$

- **Predicted prey species mass** $Q^{r,u}_{k,y}$ **and diet weight proportions** $\tilde{Q}^{r,l}_{k,y}$ (fitted in multinomial likelihood L_7):

$$Q^{r,u}_{k,y} = \begin{cases} \sum_{a=0}^{A_{L,k}} E^{r,u}_{k,a,y} w_{k,a} & \text{modeled prey } k \\ Q^r_0\left(1 - e^{-\phi^{r,u}_o N_{r,u,y}}\right) & \text{"other" prey} \end{cases}, \qquad Q^{r,l}_{k,y} = \sum_u Q^{r,u}_{k,y}\chi^{r,u,l}, \qquad \tilde{Q}^{r,l}_{k,y} = \frac{Q^{r,l}_{k,y}}{\sum_{k'} Q^{r,l}_{k',y}} \tag{2.32}$$

- **Predicted vs. Bioenergetic Daily Ration** ($\hat{\Omega}^{r,u}_y$**, fitted in likelihood** L_5):

$$\hat{\Omega}^{r,u}_y = \frac{\sum_k Q^{r,u}_{k,y}}{365\, N_{r,u,y} e^{-Z_{r,u,y}/2}}, \qquad L_5 = \sum_{r,u} \frac{\left(\Omega^{r,u} - \bar{\hat{\Omega}}^{r,u}\right)^2}{2\sigma^2_d} \tag{2.33}$$

  where $\Omega^{r,u} = \frac{1}{365}\frac{K^r}{A^r}(w^{r,u})^{d^r}(w^{r,\infty})^{d^r - 1}$ is the (Essington et al. 2001) bioenergetic daily ration!

2. **Unified Equilibrium-Normalized Parameterization of Seven Functional Responses (Table 1, p. 79)**: To compare prey-dependent and predator-dependent functional responses on an equal footing, (Kinzey and Punt 2009, Eqs. 15–17 & Table 1) define the predator age-effect $\theta^{r,u} = 1 + \frac{\omega^r \tilde{\omega}^r}{u + \tilde{\omega}^r}$ and normalize available prey

abundance $\Phi_y^{r,u}$ and competing predator abundance $\Psi_y^{k,a}$ **relative to their unfished equilibrium values in year** $y_P = 1960$:

$$\Phi_y^{r,u} = \frac{\sum_{k,a} N_{k,a,y} \gamma_{k,a}^{r,u}}{\sum_{k,a} N_{k,a,y_P} \gamma_{k,a}^{r,u}}, \qquad \Psi_y^{k,a} = \frac{\sum_{r,u} N_{r,u,y} \gamma_{k,a}^{r,u}}{\sum_{r,u} N_{r,u,y_P} \gamma_{k,a}^{r,u}} \tag{2.34}$$

Because $\Phi_{y_P}^{r,u} = \Psi_{y_P}^{k,a} = 1$ at unfished equilibrium, all seven functional responses for $\phi_{k,a,y}^{r,u}$ share the exact same unfished baseline $\phi_{k,a,y_P}^{r,u} = v_k^r \theta^{r,u}$ (see Table 2.1):

Table 2.1: Comparison and Empirical AIC of Seven Functional Responses in Aleutian Islands Walleye Pollock and Atka Mackerel (Kinzey and Punt 2009)

Type	Functional Response Name	Mathematical Formula for $\phi_{k,a,y}^{r,u}$	Params	Aleutian AIC (ΔAIC)
I	**Linear (Lotka–Volterra)**	$\theta^{r,u} v_k^r$	1	7411 (+716)
II	**Asymptotic (Holling Type II)**	$\frac{v_k^r \theta^{r,u} \left[1 + \tilde{v}_k^r\right]}{1 + \tilde{v}_k^r \Phi_y^{r,u}}$	2	7080 (+384)
III	**Sigmoid (Holling Type III)**	$\frac{v_k^r \theta^{r,u} \left(1 + \tilde{v}_k^r\right) \left(\Phi_y^{r,u}\right)^{\tilde{\tilde{v}}_k^r - 1}}{1 + \tilde{v}_k^r \left(\Phi_y^{r,u}\right)^{\tilde{\tilde{v}}_k^r}}$	3	7306 (+611)
IV	**Predator Interference (Beddington–DeAngelis)**	$\frac{v_k^r \theta^{r,u} \left[1 + \tilde{v}_k^r\right]}{1 + \tilde{v}_k^r \Phi_y^{r,u} + \tilde{\tilde{v}}_k^r \left(\Psi_y^{k,a} - 1\right)}$	3	7098 (+403)
V	**Predator Pre-emption (Multiplicative Interference)**	$\frac{v_k^r \theta^{r,u} \left[1 + \tilde{v}_k^r\right]}{\left(1 + \tilde{v}_k^r \Phi_y^{r,u}\right) \left[1 + \tilde{\tilde{v}}_k^r \left(\Psi_y^{k,a} - 1\right)\right]}$	3	6696 (0)
VI	**Hassell–Varley**	$\frac{v_k^r \theta^{r,u} \left[1 + \tilde{v}_k^r\right]}{\tilde{v}_k^r \Phi_y^{r,u} + \left(\Phi_y^{r,u}\right)^{\tilde{\tilde{v}}_k^r}}$	3	7169 (+473)
VII	**Ecosim (Foraging Arena Vulnerability)**	$\frac{v_k^r \theta^{r,u}}{1 + \tilde{v}_k^r \left(\Psi_y^{k,a} - 1\right)}$	2	7046 (+351)

3. **Key Empirical Findings (Kinzey and Punt 2009, 89–101)**:

   - **Dominance of Predator-Dependent Functional Responses (Types V & VII)**: The two best-fitting functional responses by AIC were **Type V (Predator Pre-emption, ΔAIC = 0)** and **Type VII (Ecosim Foraging Arena, ΔAIC = 351)**, both of which outperformed standard prey-only Holling Type II (ΔAIC = 384) and Type III (ΔAIC = 611). In Type V, predator interference $\Psi_y^{k,a}$ multiplicatively dampens per-capita attack rates at low and intermediate prey densities, preventing unrealistic over-depletion when predator stocks are large.
   - **Divergence Between Single-Species and Multispecies Trajectories**: The multispecies model estimated that Aleutian pollock spawning biomass declined **over 3× more steeply** from 1964 to 2003 than estimated by the single-species model (with non-overlapping 95% confidence bands in the 1960s–1980s, Fig. 9), and estimated age-0 pollock predation mortality $P_{1,0,y} \approx 5.0\ \mathrm{yr}^{-1}$ (only $\sim 0.5\%$ survival from age 0 to age 1).

2.4.2 Van Kirk et al. (2010): Gulf of Alaska `MSASA` and Simultaneity vs. Identifiability

(Van Kirk et al. 2010) constructed an ADMB-based **Multispecies Age-Structured Assessment (`MSASA`)** model for three strongly interacting species in the Gulf of Alaska (1981–2001): walleye pollock (*Theragra chalcogramma*, ages 1–10+), Pacific cod (*Gadus macrocephalus*, ages 1–12+), and arrowtooth flounder (*Atheresthes stomias*, ages 1–15+), fitted to commercial catch, NMFS triennial bottom-trawl survey data, and 2,704 predator stomachs from the AFSC REEM database.

1. **Predation Mortality ($P_{i,a,t}$) and Beginning-of-Year Approximation**: Total mortality is $Z_{i,a,t} = M^r_{i,a,t} + F_{i,a,t} + P_{i,a,t}$. Because true instantaneous predation mortality $P^{\text{inst}}_{i,a,t} = \frac{1}{\bar{B}_{i,a,t}} \sum_j \sum_b I_{j,b} \bar{N}_{j,b,t} \frac{\varphi_{i,a,j,b,t}}{\varphi_{j,b,t}}$ is implicitly recursive in $\bar{N}_{i,a,t}$ and $\bar{B}_{i,a,t}$, `MSASA` uses the explicit beginning-of-year abundance approximation (Van Kirk et al. 2010, Eqs. 1–6):

$$P_{i,a,t} = \frac{1}{B_{i,a,t}} \sum_j \sum_b I_{j,b}\, N_{j,b,t}\, \frac{\varphi_{i,a,j,b,t}}{\varphi_{j,b,t}} \tag{2.35}$$

   where $I_{j,b}$ ($\mathrm{kg} \cdot \mathrm{yr}^{-1}$) is the annual predator ration derived from (Essington et al. 2001) / (Aydin et al. 2005) von Bertalanffy bioenergetics, and available prey biomass is:

$$\varphi_{i,a,j,b,t} = \nu_{i,a,j,b}\, B_{i,a,t}, \qquad \varphi_{j,b,t} = B_{\text{oth}} + \sum_i \sum_a \nu_{i,a,j,b,t}\, B_{i,a,t} \tag{2.36}$$

with constant non-modeled prey biomass $B_{\text{oth}} = e^{15}$ ($\nu_{\text{oth}} \equiv 1$) and Andersen–Ursin suitability:

$$\nu_{i,a,j,b} = \rho_{i,j}\, g_{i,a,j,b}, \qquad g_{i,a,j,b} = \exp\left[-\frac{1}{2\sigma_j^2}\left(\ln\frac{w_{j,b}}{w_{i,a}} - \eta_j\right)^2\right] \tag{2.37}$$

2. **Joint Objective Function and $\sqrt{n}$-Weighted Stomach Least Squares**: Unlike `SMS` (which uses a Dirichlet stomach likelihood), `MSASA` combines lognormal total catch and survey biomass, multinomial catch-at-age and survey-at-age/length (with fixed effective sample size $w = 100$), and a **weighted least-squares (normal) objective function on stomach content weight proportions** (Van Kirk et al. 2010, Table 4, p. 1139):

$$\mathcal{L}_{\text{stom}} = \sum_{p=1}^{3} w_{\text{stom},p} \sum_{i,a,j,b,t\in p} \left[\left(\frac{\varphi_{i,a,j,b,t}}{\varphi_{j,b,t}} + 0.0001\right) - \left(\frac{\hat{\varphi}_{i,a,j,b,t}}{\hat{\varphi}_{j,b,t}} + 0.0001\right)\right]^2 \tag{2.38}$$

where stomach proportions are pooled into three 7-year blocks ($p \in \{1981\text{–}1987,\ 1988\text{–}1994,\ 1995\text{–}2001\}$) with weights set to the square root of predator stomach sample sizes ($w_{\text{stom}} = \{11, 26, 44\}$).

3. **Key Lessons on Parameter Estimability (Van Kirk et al. 2010, 1140–45)**:

   - **Temporal Pooling of Stomach Data is Essential for Convergence**: Attempting to fit unpooled annual stomach proportions failed to converge in ADMB, and pooling all 21 years into a single time-invariant average (`M1`) produced a singular Hessian unless stomach weights were inflated so high that catch and survey fits degraded. Dividing diets into 7-year regime blocks (`M2`–`M4`) with time-varying species preference $\rho_{i,j}$ (`M3`, AIC = 2121.68) and size preference η_j, σ_j (`M4`, AIC = 2050.65) achieved positive-definite Hessians and captured the tripling of arrowtooth flounder predation on age-1 pollock.
   - **Confounding of Survey Catchability (Q_i) and Residual Mortality (M_i^r) with $P_{i,a,t}$**: When Van Kirk et al. attempted to estimate survey catchability Q_i or residual natural mortality M_i^r freely alongside predation mortality $P_{i,a,t}$, ADMB pushed prey abundance upward, drove $Q_i \to 0$, or counter-intuitively *increased* M_i^r to absorb the extra older cohorts generated by high juvenile recruitment. Thus, Q_i and M_i^r had to be fixed externally.

2.4.3 Curti et al. (2013): Georges Bank `MSCAA`, Ecosystem Biomass Closure, and Monte Carlo Error Propagation

(Curti et al. 2013) developed an ADMB `MSCAA` model for Georges Bank (1978–2007) covering Atlantic cod (*Gadus morhua*), silver hake (*Merluccius bilinearis*), and Atlantic herring (*Clupea harengus*), introducing three important methodological refinements:

1. **Sparre (1980) Stationary Total Ecosystem Biomass Closure**: Instead of assuming a constant biomass of "other food" B_{other}, (Curti et al. 2013, Eqs. 6–10) followed (Sparre 1980) and (Tsou and Collie 2001) in assuming a stationary **total ecosystem biomass** EB $= 15 \times 10^6$ metric tons (and tested EB $= 7.34 \times 10^6$ t from (Rochet et al. 2011) in a sensitivity analysis), so that non-modeled "other food" varies dynamically each year as the complement of modeled fish biomass:

$$B_{\text{other},t} = \text{EB} - \sum_i \sum_a B_{i,a,t} \tag{2.39}$$

Suitabilities $v_{i,a,j,b,t} = \rho_{i,j} g_{i,a,j,b,t}$ (with $v_{\text{other}} = \rho_{\text{other}} g_{\text{other}} = 1$) are normalized across all prey categories for each predator (j, b):

$$\tilde{v}_{i,a,j,b,t} = \frac{v_{i,a,j,b,t}}{\sum_k \sum_c v_{k,c,j,b,t} + v_{\text{other}}}, \qquad \phi_{i,a,j,b,t} = \tilde{v}_{i,a,j,b,t} B_{i,a,t}, \qquad \phi_{j,b,t} = \tilde{v}_{\text{other}} B_{\text{other},t} + \sum_{k,c} \phi_{k,c,j,b,t} \tag{2.40}$$

and predation mortality is driven by age-specific consumption-to-biomass ratios $(CB)_{j,b}$:

$$M2_{i,a,t} \approx \frac{1}{N_{i,a,t} W_{i,a,t}} \sum_j \sum_b (CB)_{j,b} \, B_{j,b,t} \, \frac{\phi_{i,a,j,b,t}}{\phi_{j,b,t}} \tag{2.41}$$

> **!** Ecosystem Biomass Scaling Invariance
>
> (Curti et al. 2013, 478) demonstrated that halving assumed total ecosystem biomass EB from 15 Mt to 7.34 Mt changed estimated recruitment, abundance, and $M2_{i,a,t}$ by **less than** 3.5% (0.53%, 0.59%, and 3.49% maximum differences, respectively!), because the estimated vulnerability parameters $\rho_{i,j}$ (defined relative to $\rho_{\text{other}} = 1$) automatically scaled down by 51–53% to preserve the observed diet proportions $\phi_{i,a,j,b,t}/\phi_{j,b,t}$.

2. **Empirical Pre-Estimation of Size Preference $(\eta_{i,j}, \sigma_{i,j})$ vs. Internal Estimation of $\ln(\rho_{i,j})$**: Because NEFSC Food Web Dynamics stomach records routinely measure predator length and prey species mass, but only a subset (1,900 records) measured individual prey lengths, Curti et al. found that attempting to estimate $(\eta_{i,j}, \sigma_{i,j})$ inside the integrated model caused severe parameter confounding. They therefore computed $\eta_{i,j}$ and $\sigma_{i,j}$ externally from the 1,900 paired predator-prey weight records (Table 5, p. 477) and estimated only the log species-preference coefficients $\ln(\rho_{i,j})$ inside ADMB against 5-year-binned multinomial food-habits proportions weighted by McAllister–Ianelli effective sample sizes (McAllister and Ianelli 1997).

3. **Regularizing Penalty Functions (Curti et al. 2013, Table 3, p. 474)**: To prevent biologically unrealistic boundary solutions during optimization and Monte Carlo runs, Curti et al. added three smooth quadratic penalties to the negative log-likelihood:

   - **Initial Cohort Decay Penalty (Yp_i)**: Penalizes deviations of first-year abundances $N_{i,a,1}$ from exponential decay $\text{Yr1}_{i,a,1} = N_{i,a-1,1} e^{-\bar{Z}_{i,a-1}}$:

$$Yp_i = \sum_{a>1} \left(N_{i,a,1} - \text{Yr1}_{i,a,1} \right)^2 \tag{2.42}$$

- **Log-Recruitment CV Threshold Penalty (Rp_i)**: Prevents the coefficient of variation of log-recruitment from exceeding species thresholds $\mathrm{Rth}_i \in \{1.0, 0.7, 0.4\}$:

$$Rp_i = 0.01 \left[\mathrm{CV}(\ln N_{i,1,t}) - \mathrm{Rth}_i\right]^2 \cdot \mathbb{I}\big(\mathrm{CV}(\ln N_{i,1,t}) > \mathrm{Rth}_i\big) \tag{2.43}$$

- **Minimum Age-Specific Biomass Threshold Penalty (Bp_i)**: Prevents any cohort biomass $B_{i,a,t}$ from collapsing below $\mathrm{Bth}_i = 10$ kg during trial steps:

$$Bp_i = 0.01 \sum_{a,t} \big(B_{i,a,t} - \mathrm{Bth}_i\big)^2 \cdot \mathbb{I}\big(B_{i,a,t} < \mathrm{Bth}_i\big) \tag{2.44}$$

4. **Monte Carlo Error-Propagation Findings (Curti et al. 2013, 474–82, Figs. 4–7)**: Running 400–500 Monte Carlo replicates across progressively higher observation error levels in commercial catch ($\sigma_{\mathrm{TC}} \in \{0.1, 0.2, 0.4, 0.6, 0.8\}$, $N_{\mathrm{eff,CP}} \in \{1000, 500, 75, 40, 20\}$), survey indices ($\sigma_{\mathrm{TS}} \in \{0.2, 0.4, 0.6, 0.8\}$, $N_{\mathrm{eff,SP}} \in \{1000, 200, 100, 60\}$), and predator diets ($N_{\mathrm{eff,FH}} \in \{2500, 500, 100, 50, 10\}$) revealed a clear separation of information sources:

   - **Fishing mortality** $F_{i,t}$ is most sensitive to measurement error in **commercial catches**.
   - **Recruitment** $N_{i,1,t}$ **and predation mortality** $M2_{i,a,t}$ are most sensitive to measurement error in **trawl survey indices** (breaking down when survey lognormal $\sigma_{\mathrm{TS}} \geq 0.6$).
   - **Robustness to Noisy Diet Data**: Across all parameters, increasing noise in predator stomach contents (even down to an effective sample size of $N_{\mathrm{eff,FH}} = 10$ per 5-year block) had the **smallest impact** on recruitment, F, and $M2$ precision, because the temporal trajectory of $M2_{i,a,t}$ is primarily driven by predator and prey abundance trajectories anchored by the catch and survey datasets!

2.4.4 Alternative Functional Responses in `MSCAA` and `Hydra`

Whereas SMS uses the fixed-ration Andersen–Ursin equation (which corresponds to a Holling Type II functional response in the limit where total consumption rate $R_{A,a}$ is held constant by adjusting "other food"), (Kinzey and Punt 2009) and Gaichas et al. (`Hydra`) formulated `MSCAA` to allow explicit comparison of **alternative functional response forms** for the per-predator consumption rate $I_{i,a,j,b,t}$ of prey (i,a) by predator (j,b):

Let $\tilde{B}_{i,a,j,b,t} = S_{i,a,j,b}\,\bar{N}_{i,a,t}\,w_{i,a,t}$ be the suitable prey biomass of (i,a) for predator (j,b), and let $\Phi_{j,b,t} = \sum_{k,c} \tilde{B}_{k,c,j,b,t} + S_{\mathrm{OF},j,b} B_{\mathrm{OF}}$ be total suitable food.

2.5 Type I (Linear)

Linear functional response with constant search / clearance rate $\alpha_{j,b}$:

$$M2_{i,a,t} = \sum_{j,b} \alpha_{j,b} \, S_{i,a,j,b} \, \bar{N}_{j,b,t} \tag{2.45}$$

2.6 Type II (Hyperbolic / SMS)

Hyperbolic satiation with half-saturation constant $H_{j,b}$:

$$M2_{i,a,t} = \sum_{j,b} \frac{R_{j,b,t} \, \bar{N}_{j,b,t} \, S_{i,a,j,b}}{\Phi_{j,b,t} + H_{j,b}} \tag{2.46}$$

(Note: when $H_{j,b} = 0$ and $S_{OF,j,b} B_{OF} > 0$, this reduces identically to the SMS equation).

2.7 Type III (Sigmoidal Switching)

Sigmoidal prey-switching with exponent $\theta_j > 1$:

$$M2_{i,a,t} = \sum_{j,b} \frac{R_{j,b,t} \, \bar{N}_{j,b,t} \, S_{i,a,j,b} \, \left(\bar{N}_{i,a,t} w_{i,a,t}\right)^{\theta_j - 1}}{\sum_{k,c} S_{k,c,j,b} \left(\bar{N}_{k,c,t} w_{k,c,t}\right)^{\theta_j} + \left(S_{\mathrm{OF},j,b} B_{\mathrm{OF}}\right)^{\theta_j}} \tag{2.47}$$

When $\theta_j = 2$, rare prey experience disproportionately low predation mortality ($M2_{i,a,t} \to 0$ as $B_{i,a,t} \to 0$), providing an automatic stabilizing refuge at low stock size.

2.8 Predator Interference

Beddington–DeAngelis / Crowley–Martin pre-emption with predator interference parameter γ_j:

$$M2_{i,a,t} = \sum_{j,b} \frac{R_{j,b,t} \, \bar{N}_{j,b,t} \, S_{i,a,j,b}}{\Phi_{j,b,t} + H_{j,b} \left(1 + \gamma_j \bar{N}_{j,b,t}\right)} \tag{2.48}$$

This dampens top-down control when predator abundance spikes, mimicking the behavioral foraging-arena limitation of Ecosim (Walters et al. 1997).

> Simulation-Estimation Takeaways: Trijoulet et al. (2019)
>
> Using a self-test and cross-test simulation framework in the TMB-based multispecies model `ms-keyrun` / `MSCAA`, (Trijoulet et al. 2019) demonstrated that:

- Ignoring trophic interactions (fitting single-species SCAA with constant M to data generated from a multispecies food web) produces severe, systematic biases in SSB ($> 30\%$), recruitment ($> 50\%$), and F_{MSY} reference points for both forage fish and cannibalistic predators.
- Even when diet composition data are noisy (effective sample size $N_{\text{stom}} = 20$–50), `MSCAA` models accurately recover the temporal trajectory of $M2_{i,a,t}$ and eliminate the retrospective bias seen in constant-M single-species assessments.

2.9 Climate-Enhanced Multispecies Stock Assessment (`CEATTLE` / `Rceattle`)

Developed by Kirstin Holsman, Jim Ianelli, Kerim Aydin, André Punt, Grant Adams, and colleagues at NOAA AFSC and the University of Washington (Holsman et al. 2016; Adams et al. 2022; ICES 2021, 2023), **CEATTLE** (*Climate-Enhanced, Age-based model with Temperature-specific Trophic Linkages and Energetics*) and its open-source TMB/R package `Rceattle` represent the state-of-the-art in integrating **temperature-dependent growth, Wisconsin bioenergetics, and multispecies harvest control rules** into an age-structured statistical stock assessment.

Originally built in ADMB for the Eastern Bering Sea complex—walleye pollock (*Gadus chalcogrammus*, ages 1–12), Pacific cod (*Gadus macrocephalus*, ages 1–12), and arrowtooth flounder (*Atheresthes stomias*, ages 1–21, sex-structured)—and subsequently ported to TMB (`Rceattle`) and expanded to the Gulf of Alaska (adding Pacific halibut *Hippoglossus stenolepis*; (Adams et al. 2022)), California Current, and Baltic Sea, `CEATTLE` is used operationally by the North Pacific Fishery Management Council (NPFMC).

2.9.1 Temperature-Dependent Generalized von Bertalanffy Growth (Holsman et al. 2016)

Unlike standard age-structured models that treat weight-at-age $W_{ij,y}$ solely as an empirical look-up table, (Holsman et al. 2016, Eqs. 1–2 & Table 1 Eqs. T1.10a–c, pp. 12–14, 50) derived a **temperature-dependent generalized von Bertalanffy growth function (VBGF)** based on (Essington et al. 2001) and (Holsman and Aydin 2015), fitted to $21,388$ pollock, $14,362$ Pacific cod, and 772 arrowtooth flounder otolith weight-at-age records:

$$W_{ij,y} = W_{\infty,iy}\left(1 - \exp\!\left(-K_i\left(1 - d_{i,y}\right)\left(j - t_{0,i}\right)\right)\right)^{\frac{1}{1-d_{i,y}}} e^{\varepsilon}, \qquad \varepsilon \sim \mathcal{N}\left(0, \sigma^2_{d,i}\right) \tag{2.49}$$

where:

- $W_{\infty,iy} = \left(\frac{H_i}{K_i}\right)^{\frac{1}{1-d_{i,y}}}$ is the time-varying asymptotic body mass of species i in year y, determined by the bioenergetic assimilation constant H_i ($\text{kg} \cdot \text{kg}^{-d} \cdot \text{yr}^{-1}$; e.g., 16.34 for pollock, 9.30 for cod) and catabolic energy loss constant K_i ($\text{kg} \cdot \text{kg}^{-1} \cdot \text{yr}^{-1}$; 0.22 for pollock, 0.45 for cod; Table 3, p. 54),
- $d_{i,y}$ is the **realized allometric slope of consumption**, which varies annually as an exponential-linear function of bottom temperature T_y (°C):

$$d_{i,y} = \exp\left(\alpha_{d,i,y} + \alpha 0_{d,i} + \beta_{d,i} T_y\right) \tag{2.50}$$

  allowing both the growth rate and asymptotic weight $W_{\infty,iy}$ to respond dynamically to historical and IPCC-projected bottom temperatures!
- For sexually dimorphic species (arrowtooth flounder, where males have higher natural mortality $M_{\text{male}} = 0.35$ vs. $M_{\text{fem}} = 0.20$ and smaller asymptotic size), sex-specific weights $W^{\text{fem}}_{ij,y}, W^{\text{male}}_{ij,y}$ and residual mortalities are combined using the surviving female cohort proportion (Holsman et al. 2016, Eqs. T1.18–T1.21, p. 51):

$$\omega_{ij} = \frac{e^{-jM_{\text{fem}}}}{e^{-jM_{\text{fem}}} + e^{-jM_{\text{male}}}}, \qquad W_{ij,y} = \omega_{ij} W^{\text{fem}}_{ij,y} + (1 - \omega_{ij}) W^{\text{male}}_{ij,y} \tag{2.51}$$

2.9.2 Wisconsin Bioenergetics Consumption and Thermal Scaling ($f(T_y)_p$)

In `CEATTLE`, the individual specific annual ration $\delta_{pa,y}$ ($\text{kg prey} \cdot \text{kg predator}^{-1} \cdot \text{yr}^{-1}$) of predator species p, age a in year y is modeled via the Wisconsin Bioenergetics allometric equation (Holsman et al. 2016, Table 2, Eqs. T2.4–T2.5, p. 52; Adams et al. 2022, Eqs. 10–11):

$$\delta_{pa,y} = \widehat{\varphi}_p \, \alpha_\delta \, W_{pa,y}^{(1+\beta_\delta)} \, f(T_y)_p \tag{2.52}$$

where:

- $\widehat{\varphi}_p$ is the annual relative foraging rate ($\text{d} \cdot \text{yr}^{-1}$ scalar),
- α_δ and $\beta_\delta < 0$ are the intercept and allometric slope of maximum specific consumption ($\beta_\delta = -0.460$ for pollock, -0.122 for cod, -0.245 for arrowtooth flounder; Table 3, p. 53),
- $f(T_y)_p$ is the (Kitchell et al. 1977) **dome-shaped temperature scaling algorithm**:

$$f(T_y)_p = V^X \exp(X(1 - V)) \tag{2.53}$$

  with:

$$V = \frac{T^{\text{cm}}_p - T_y}{T^{\text{cm}}_p - T^{\text{co}}_p}, \qquad X = \frac{Z^2 \left(1 + (1 + 40/Y)^{0.5}\right)^2}{400} \tag{2.54}$$

$$Z = \ln(Q^c_p)\left(T^{\text{cm}}_p - T^{\text{co}}_p\right), \qquad Y = \ln(Q^c_p)\left(T^{\text{cm}}_p - T^{\text{co}}_p + 2\right) \tag{2.55}$$

  where T^{co}_p is the physiological optimum consumption temperature (10.0°C for pollock, 13.7°C for cod, 19.6°C for arrowtooth), T^{cm}_p is the maximum lethal physiological temperature (15.0°C, 21.0°C, 34.13°C), and Q^c_p is the Q_{10} thermal slope $(2.60, 2.41, 2.18)$. (`Rceattle` also supports the (Thornton and Lessem 1978) two-logistic thermal curve and energy-density caloric scaling $\mathcal{E}_{\text{pred}} / \sum \widehat{W} \mathcal{E}_{\text{prey}}$).

2.9.3 Predation Mortality ($M2_{ij,y}$) and Analytical/Iterative Suitability ($\bar{S}_{paij}$)

Following (Jurado-Molina et al. 2005), `CEATTLE` expresses predation mortality $M2_{ij,y}$ on prey species i, age j in year y as a Holling Type II functional response where suitability $\bar{S}_{paij} \in [0, 1]$ of modeled prey (i, j) and suitability $1 - \sum_{ij} \bar{S}_{paij}$ of "other prey" B_p^{other} sum to unity (Holsman et al. 2016, Table 2, Eqs. T2.1–T2.3, p. 52):

$$M2_{ij,y} = \sum_p \sum_a \left(\frac{N_{pa,y}\, \delta_{pa,y}\, \bar{S}_{paij}}{\sum_{i,j} \left(\bar{S}_{paij} B_{ij,y} \right) + B_p^{\text{other}} \left(1 - \sum_{i,j} \bar{S}_{paij} \right)} \right) \tag{2.56}$$

Instead of adding thousands of noisy diet cells to the likelihood, (Holsman et al. 2016) and (Adams et al. 2022) calculate $\bar{S}_{paij}$ analytically from observed gravimetric stomach proportions $U_{paij,y}$ (converted from length-based stomach samples to age-based proportions using annual empirical age-length transition matrices ALK_y) and model-estimated biomasses $B_{ij,y}$:

$$\bar{S}_{paij} = \frac{1}{n_y} \sum_y \left(\frac{\dfrac{\bar{U}_{paij}}{B_{ij,y}}}{\sum_{i,j} \left(\dfrac{\bar{U}_{paij}}{B_{ij,y}} \right) + \dfrac{1 - \sum_{i,j} \bar{U}_{paij}}{B_p^{\text{other}}}} \right) \tag{2.57}$$

In `Rceattle` (Adams et al. 2022, sec. 2.4, p. 4), the single-species assessment models are first optimized (tuning McAllister–Ianelli effective sample sizes for age/length compositions; (McAllister and Ianelli 1997)), and then the multispecies model is iteratively re-optimized, updating $\bar{S}_{paij}$ at each outer step until convergence (which occurs in **3 iterations**).

2.9.4 Multispecies Biological Reference Points (MBRPs) and Harvest Control Rules (Holsman et al. 2016)

A major contribution of (Holsman et al. 2016, sec. 2.3 & Table 5, pp. 15–20, 57) is the rigorous mathematical formalization of **12 Multispecies Biological Reference Point (MBRP) harvest scenarios** under NPFMC Tier 3 control rules (where target Acceptable Biological Catch proxy biomass is $\overline{\text{SSB}}_{F,i} = 0.40\, \overline{\text{SSB}}_{0,i}$ and the minimum MSY biomass limit is $B_{\text{MSY},i} = 0.35\, \overline{\text{SSB}}_{0,i}$):

1. **Non-Uniqueness of "Unfished Biomass" ($\overline{\text{SSB}}_{0,i}$) and Fishing-Order Dependence (Scenarios 1.1–1.4)**:
    - Under **Scenario 1.1**, $\overline{\text{SSB}}_{0,i}$ is defined by setting $F_i = 0$ for **all species simultaneously**:

$$\overline{\text{SSB}}_{F,i} = 0.40\, \overline{\text{SSB}}_{0,i} \Big|_{F_1 = F_2 = F_3 = 0} \tag{2.58}$$

- Under **Scenarios 1.3 & 1.4**, $\overline{\mathrm{SSB}}'_{0,1}$ for prey (pollock) is defined **conditionally on predators (cod and arrowtooth) being fished at their target** $F^*_{\mathbf{ABC},2}, F^*_{\mathbf{ABC},3}$:

$$\overline{\mathrm{SSB}}'_{0,1} = \left(\overline{\mathrm{SSB}}_{0,1} \mid F_1 = 0,\, \overline{\mathrm{SSB}}_{F,2},\, \overline{\mathrm{SSB}}_{F,3}\right) \tag{2.59}$$

- *Mathematical Insight*: When predators are unfished ($F_2 = F_3 = 0$), predator biomass expands and inflicts massive $M2_{1,a}$ on juvenile pollock (plus intense pollock cannibalism, which triggers **over-compensatory damped oscillations** in projected SSB; Fig. 10, p. 60). Consequently, simultaneous unfished pollock biomass $\overline{\mathrm{SSB}}_{0,1}$ can actually be **lower** than harvested pollock biomass when predators are fished!

2. **Spawning Biomass-Per-Recruit Approach (Scenario 2.1, Eqs. T5.2.1a–b)**: Solves for $F_{x,i}$ such that $\mathrm{SPR}_{F,i} = 0.40\left(\mathrm{SPR}_{F,i} \mid F_{x,i} = 0\right)$, where:

$$\mathrm{SPR}_{F,i} = \frac{1}{R_{i,1}} \sum_{j=1}^{A_i} \rho_{ij}\, \bar{W}_{ij}\, N_{ij} \exp\Big(-\left(F_{x,i} s^f_{ij} + M1_{ij} + \overline{M2}_{ij}\right)\Big) \tag{2.60}$$

3. **System-Wide Aggregate Biomass Proxies (Scenarios 3.1–3.4, Eqs. T5.3.1–T5.3.4)**: Finds a system-wide multiplier F_{ALL} such that total community female spawning biomass satisfies $\sum_i \overline{\mathrm{SSB}}_{F,i} = 0.40 \sum_i \overline{\mathrm{SSB}}_{0,i}$, scaling species-specific fishing mortality either by total natural mortality $F_{x,i} = F_{\mathrm{ALL}}(\bar{M}_i / \sum_k \bar{M}_k)$ or by unfished biomass $F_{x,i} = F_{\mathrm{ALL}}(\overline{\mathrm{SSB}}_{0,i} / \sum_k \overline{\mathrm{SSB}}_{0,k})$, with (Scenarios 3.2, 3.4) or without (Scenarios 3.1, 3.3) the conservation constraint $\mathrm{SSB}_{F,i,y} \geq 0.35\, \mathrm{SSB}_{0,i}$.
4. **Simplex Optimization of Individual vs. Aggregate MSY (Scenarios 4.1–4.3, Eqs. T5.4.1–T5.4.3)**: Uses 8 iterations of Nelder–Mead simplex optimization to find $\mathbf{F}_x = (F_{x,1}, F_{x,2}, F_{x,3})$ maximizing either individual species yield $\max(\overline{\mathrm{ABC}}_{x,i} \mid F_{x,i})$ (Scenario 4.1) or total combined community yield $\max\left(\sum_i \overline{\mathrm{ABC}}_{x,i} \mid \mathbf{F}_x\right)$ without (Scenario 4.2) or with (Scenario 4.3) the minimum biomass threshold $\mathrm{SSB}_{F,i,y} \geq 0.35\, \mathrm{SSB}_{0,i}$:

    - *Predator Eradication Paradox*: Unconstrained maximization of aggregate yield (Scenario 4.2) achieved 3.2 million t by **fishing Pacific cod and arrowtooth flounder to extinction** in order to eliminate predation on pollock (since annual predation on juvenile pollock is 2–5× the annual fishery harvest!). Imposing the biodiversity constraint $\mathrm{SSB}_{F,i} \geq 0.35\, \mathrm{SSB}_{0,i}$ (Scenario 4.3) prevented predator collapse and demonstrated that **MBRPs are not inherently more or less conservative than single-species BRPs**; rather, their conservatism depends on whether single-species or aggregate targets are used and whether minimum biomass thresholds are enforced.

2.9.5 Gulf of Alaska `Rceattle` Ensemble and Hybrid "Predators-as-Fleets" Scaling (Adams et al. 2022)

In adapting `Rceattle` (implemented in TMB with recruitment and initial cohort deviations treated as Laplace-integrated random effects) to the Gulf of Alaska for pollock, Pacific cod, arrowtooth flounder, and Pacific halibut (*Hippoglossus stenolepis*), (Adams et al. 2022) addressed a universal practical problem in multispecies assessment: **how to include an important apex predator (Pacific halibut) whose stock assessment is conducted externally on a coastwide spatial scale** (by the International Pacific Halibut Commission, IPHC, using a 4-model Stock Synthesis ensemble).

(Adams et al. 2022) evaluated 16 model configurations across 3 time-series durations (1977–2018, 1993–2018, 1996–2018) grouped into 4 ensemble averages:

1. **Ensemble Avg 1 (Single-Species)** vs. **Ensemble Avg 2 (3-Species MS without Halibut)**: Including predation by arrowtooth flounder (the dominant predator in the GOA), cod, and pollock increased average estimated pollock total biomass by 2.03–2.21× and pollock SSB by 1.70–1.84× relative to single-species models.
2. **Ensemble Avg 3 (Fixed External Predator Abundance)**: Inputs sex-specific mid-year halibut numbers-at-age directly from each of the 4 IPHC Stock Synthesis ensemble models (scaled by the proportion of age-5+ halibut survey biomass in IPHC Regulatory Areas 3A & 3B) without fitting halibut data in the `Rceattle` likelihood.
3. **Ensemble Avg 4 (Relative Predator Survey Index with Estimated Scaling $\tilde{q}_i$)**: Inputs relative halibut abundance-at-age from the IPHC setline survey multiplied by a freely estimated scaling parameter $\tilde{q}_{\text{halibut}}$.

    - *Critical Identifiability Warning* (Adams et al. 2022, sec. 3.1 & 4.2, pp. 6, 11): When $\tilde{q}_{\text{halibut}}$ was freely estimated inside the multispecies likelihood (Models 11 & 16), the optimizer drove $\tilde{q}_{\text{halibut}} \to 0$ (0.0021–0.0025 with large standard errors), **shrinking halibut predation mortality to zero** and reproducing the 3-species no-halibut model! Even though Models 11 & 16 achieved a slightly lower AIC on the pollock/cod/arrowtooth survey and catch data by removing the external constraint of halibut predation, doing so was biologically spurious—a textbook demonstration of why predator abundance indices in "predators-as-fleets" extensions cannot have unanchored scaling parameters $\tilde{q}_i$ without informative priors on predator scale.

3 Age-Length and Length-Structured Assessment Models (Gadget/gadget3, SS3, MULTIFAN-CL, LeMans/LeMaRns, CSMS/CSM)

3.1 Why Track Length in Addition to (or Instead of) Age?

In marine ecosystems, nearly every ecological and fishery interaction—gape-limited predation, trawl mesh selection, gillnet dome-shaped selectivity, metabolic rate, and fecundity—is a direct physical function of **body size (length L or weight W)** rather than chronological age a (Stefánsson and Pálsson 1998; Punt et al. 2016). Purely age-structured models (`MSVPA`, `SMS`, `CEATTLE`) suffer from three structural limitations when size-dependent processes are strong:

1. **Rosa Lee's Phenomenon (Size-Selective Mortality Within an Age Class)**: When fishing or predation selectively removes the fastest-growing (or slowest-growing) individuals within an age cohort, the surviving individuals of age a have a distorted length and weight distribution that cannot be represented by a fixed empirical weight-at-age $w_{s,a,t}$ (Methot and Wetzel 2013; Begley and Howell 2004).
2. **Consumption-Driven Growth Plasticity**: When forage fish collapse, predators grow more slowly in length and lose body condition (weight-at-length). A joint age-length-weight model can track both stunted length growth and starvation-induced weight loss dynamically (Stefánsson and Pálsson 1998).
3. **Direct Observation of Length Frequencies**: Measuring fish lengths at sea or port is fast and cheap, whereas reading otolith ages is expensive, prone to reader bias, and unavailable for crustaceans (shrimp, Nephrops, crabs), cephalopods, and many tropical or data-limited stocks (Fournier et al. 1998; Punt et al. 2016).

3.2 Gadget (Gadget2 and gadget3): Statistical Age-Length-Structured Multispecies Framework

3.2.1 Lineage and Architecture

Gadget (*Globally Applicable Area Disaggregated General Ecosystem Toolbox*) evolved from two Nordic multispecies research models of the late 1980s and 1990s (Stefánsson and Pálsson

1998; Begley and Howell 2004), synthesized under the EU `dst2` project (Stefánsson 2003, 2004; Begley 2004; Begley and Howell 2004):

- **MULTSPEC** (Bogstad et al. 1997, IMR Norway): An area-, age-, and length-structured model of Barents Sea cod, capelin, herring, harp seals, and minke whales.
- **BORMICON** (*Boreal Migration and Consumption model*; (Stefánsson and Pálsson 1997), MRI Iceland): A statistical multi-area age-length assessment model of Icelandic cod, capelin, and northern shrimp (*Pandalus borealis*) (Taylor and Stefánsson 2004; Taylor et al. 2004, 2007).

Originally written in C++ (`Gadget2`, (Begley 2004)) and optimized via Hooke & Jeeves pattern search, Simulated Annealing, and BFGS, with applications ranging from Icelandic cod–capelin–shrimp interactions (Taylor and Stefánsson 2004; Taylor et al. 2004, 2007) and Mediterranean hake (Bartolino et al. 2011) to Eastern Baltic cod ontogenetic diet shifts (Kulatska et al. 2019), the framework has recently been re-engineered as **`gadget3` (`g3`)** (Lentz and Elvarsson 2024; ICES 2021), which compiles modular R model definitions (`g3a_grow_*`, `g3a_predate_*`, `g3l_distribution_*`) directly into **TMB C++ templates with exact automatic differentiation**.

3.2.2 State-Space Tensor ($N_{s,a,l,r,t}$ and $W_{s,a,l,r,t}$)

Gadget tracks sub-stocks s (which may represent species, sex, or immature vs. mature life stages, e.g., immature hake vs. mature hake in (Bartolino et al. 2011)) across discrete time steps t (typically monthly or quarterly, $\Delta t = 1/12$ or $1/4$ yr), spatial areas r, discrete ages $a \in \{a_{\min}, \dots, a_{\max}\}$, and discrete length bins $l \in \{1, \dots, n_L\}$ with bin boundaries $[L_{\min,l}, L_{\max,l})$ and midpoint L_l.

Crucially, in every (s, a, l, r, t) cell, Gadget tracks **two coupled state variables** (Begley and Howell 2004, 4; Lentz and Elvarsson 2024):

1. $N_{s,a,l,r,t}$: The **number of individuals** in cell (s, a, l, r) at time t.
2. $W_{s,a,l,r,t}$: The **mean body weight** of individuals in cell (s, a, l, r) at time t (which can either follow a fixed length-weight curve $W_l = \alpha_s L_l^{\beta_s}$ or vary dynamically in response to food intake and growth).

Within each time step t, biological and fishery processes are executed sequentially in a strict order:

$$\text{1. Migration } (\mathbf{M}) \longrightarrow \text{2. Predation \& Fishing } (\mathbf{P}) \longrightarrow \text{3. Somatic Growth \& Maturation } (\mathbf{G}) \longrightarrow \text{4. Ageing} \tag{3.1}$$

3.2.3 Somatic Growth and the Beta-Binomial Length-Transition Distribution

In a discrete length-bin model, somatic growth during time step Δt must specify:

1. The **mean length increment** ΔL_l and **mean weight increment** ΔW_l of fish currently in length bin l, and
2. The **probability distribution** $p(\Delta j \mid l)$ that a fish in length bin l jumps up by $\Delta j \in \{0, 1, \dots, \Delta j_{\max}\}$ length bins during Δt.

3.2.3.1 A. Mean Length & Weight Increments (`g3a_grow_*`)

3.3 von Bertalanffy Growth (Consumption-Independent)

Standard empirical growth functions (`g3a_grow_lengthvbsimple` / `g3a_grow_weightsimple`):

$$\Delta L_l = \left(L_{\infty,s} - L_l\right)\left(1 - \exp(-\kappa_s \Delta t)\right) \tag{3.2}$$

$$\Delta W_l = \alpha_s \left(L_l + \Delta L_l\right)^{\beta_s} - \alpha_s L_l^{\beta_s} \tag{3.3}$$

3.4 Bioenergetic Growth (Consumption-Dependent)

Let $\psi_{s,l,r,t} \in [0, 1]$ be the predator's realized **feeding level** in length bin l during time step t (ratio of actual biomass consumed to maximum possible consumption $M_{s,l,t}$; see Section 3.4.1), and let $T_{r,t}$ be ambient temperature:

- **MULTSPEC Length & Weight Increment (`g3a_grow_length_multspec` / `g3a_grow_weight_multspec`)**:

$$\Delta L_l = \Delta t \cdot p_{0,s}\, L_l^{p_{1,s}}\, \psi_{s,l,r,t}\left(p_{2,s} T_{r,t} + p_{3,s}\right) \tag{3.4}$$

$$\Delta W_l = \Delta t \cdot q_{0,s}\, W_{s,a,l}^{q_{1,s}}\left(\psi_{s,l,r,t} - q_{2,s}\right)\left(q_{3,s} T_{r,t} + q_{4,s}\right) \tag{3.5}$$

  where $q_{2,s}$ is the maintenance feeding level below which fish **lose weight** ($\Delta W_l < 0$, while $\Delta L_l \geq 0$ since fish skeletons do not shrink!).

- **Jones (1976) Bioenergetic Weight Increment (`g3a_grow_weight_weightjones`, (Jones 1976))**:

$$\Delta W_l = \frac{\Delta t\left(\frac{C_{s,l,t}}{\Delta t} - r_{0,s} W_{s,a,l}^{r_{1,s}} \exp(r_{2,s} T_{r,t})\right)}{r_{3,s}} \tag{3.6}$$

  coupled to length growth via a condition-factor feedback function $f(\delta W_l)$ that suppresses length growth ΔL_l whenever actual body weight $W_{s,a,l}$ falls below reference weight $W_{\text{ref},l} = \alpha_s L_l^{\beta_s}$ (Begley and Howell 2004, 9; Lentz and Elvarsson 2024).

3.4.0.1 B. The Beta-Binomial Length-Bin Jump Distribution (`g3a_grow_impl_bbinom`)

Once the mean length increment ΔL_l is computed (and normalized by the bin width δL to give mean bin growth $\mu_l = \Delta L_l / \delta L$), Gadget distributes growing individuals across integer bin jumps $\Delta j \in \{0, 1, \ldots, n_{\max}\}$ using a **Beta-Binomial distribution** (Stefánsson and Pálsson 1998; Taylor et al. 2004, Eq. 5; Lentz and Elvarsson 2024):

Given:

- $n_{\max}$: Maximum allowed number of length-bin jumps in a single time step Δt,
- $\mu_l = \frac{\Delta L_l}{\delta L} \in [0, n_{\max}]$: Expected number of bin jumps,
- $\beta_s > 0$: Single free dispersion parameter controlling variance in individual growth rates,

define α_l via the mean constraint $\mu_l = \frac{n_{\max}\alpha_l}{\alpha_l + \beta_s}$:

$$\alpha_l = \frac{\beta_s \, \mu_l}{n_{\max} - \mu_l} \tag{3.7}$$

Then the probability $G_{l,l+\Delta j} = \Pr(\text{Jump} = \Delta j \mid l)$ of growing by $\Delta j \in \{0, 1, \ldots, n_{\max}\}$ bins in Δt is:

$$G_{l,l+\Delta j} = \binom{n_{\max}}{\Delta j} \frac{\mathrm{B}(\Delta j + \alpha_l,\ n_{\max} - \Delta j + \beta_s)}{\mathrm{B}(\alpha_l, \beta_s)} = \frac{\Gamma(n_{\max}+1)}{\Gamma(\Delta j + 1)\Gamma(n_{\max} - \Delta j + 1)} \frac{\Gamma(\alpha_l + \beta_s)}{\Gamma(n_{\max} + \alpha_l + \beta_s)} \frac{\Gamma(\Delta j + \alpha_l}{\Gamma} \tag{3.8}$$

In `gadget3` (Lentz and Elvarsson 2024), this is evaluated efficiently via an exact forward recurrence relation in Δj so that no `lgamma` calls are needed inside the inner loop, and both cohort numbers $N_{s,a,l'}$ and cell mean weights $W_{s,a,l'}$ are updated cleanly via matrix multiplication:

$$N_{s,a,l',t+\Delta t} = \sum_{\Delta j=0}^{n_{\max}} G_{l'-\Delta j,l'} \, N_{s,a,l'-\Delta j,t}, \qquad (NW)_{s,a,l',t+\Delta t} = \sum_{\Delta j=0}^{n_{\max}} G_{l'-\Delta j,l'} \, N_{s,a,l'-\Delta j,t} \left(W_{s,a,l'-\Delta j,t} + \Delta W_{l'} \right. \tag{3.9}$$

> [!TIP] **Why the Beta-Binomial Growth Implementation is a Major Lesson for `mizer`**: With **only one extra parameter (β_s)** beyond the deterministic growth rate ΔL_l, `g3a_grow_impl_bbinom` generates realistic, strictly non-negative, bounded discrete-bin dispersion around the mean growth trajectory, completely eliminating the numerical vs. biological diffusion problem of deterministic first-order PDE schemes while remaining C^∞-differentiable in TMB!

3.4.1 Predation, Functional Response, and Overconsumption Protection (`g3a_predate`)

In Gadget, both **commercial/survey fishing fleets** and **biological predators** are unified under a single mathematical abstraction (`g3a_predate` in (Lentz and Elvarsson 2024)).

For a biological predator p in length group L preying on prey stock i in length group l (Begley and Howell 2004, 13–15; Taylor and Stefánsson 2004, Eqs. 7–11; Lentz and Elvarsson 2024):

1. **Predator-Prey Length Suitability ($S_{p,i}(L,l)$)**: Gadget offers several parametric suitability functions of predator length L and prey length l, most commonly:

   - **Andersen & Ursin Log-Ratio Suitability (`g3_suitability_andersen`)**:

   $$S_{p,i}(L,l) = p_0 + p_2 \exp\left(-\frac{\left(\ln \frac{L}{l} - p_1\right)^2}{p_3 \text{ (or } p_4)}\right) \tag{3.10}$$

   - **Richards / Asymmetric Logistic Suitability (`g3_suitability_richards`)**:

   $$S_{p,i}(L,l) = \left(\frac{p_3}{1+\exp(-p_0 - p_1 l - p_2 L)}\right)^{1/p_4} \tag{3.11}$$

2. **Maximum Possible Consumption ($M_{p,L,t}$, `g3a_predate_maxconsumption`)**: The maximum biomass a single predator of length L can consume in time step Δt at temperature T_t is:

   $$M_{p,L,t} = \Delta t \cdot m_{0,p}\, L^{m_{1,p}} \exp\left(m_{2,p} T_t - m_{3,p} T_t^3\right) \tag{3.12}$$

3. **Desired Consumption and Holling Type II/III Functional Response (`g3a_predate_catchability_predator`)**: Let $B_{i,l,t} = N_{i,l,t} W_{i,l,t}$ be the available prey biomass of (i,l), and let the suitability-weighted prey energy/biomass be:

   $$E_{p,L,i,l,t} = \left(S_{p,i}(L,l)\, \mathcal{E}_i\, N_{i,l,t}\, W_{i,l,t}\right)^{d_i} \tag{3.13}$$

   where $d_i = 1$ yields a **Holling Type II** functional response and $d_i > 1$ yields a **Holling Type III (prey-switching)** functional response. The predator's **feeding level** $\psi_{p,L,t} \in [0,1]$ is given by the half-saturation parameter H_p:

   $$\psi_{p,L,t} = \frac{\sum_{i,l} E_{p,L,i,l,t}}{\sum_{i,l} E_{p,L,i,l,t} + H_p\, \Delta t + E_{\text{other},p}} \tag{3.14}$$

   and the total desired consumption $C^{\text{des}}_{p,L,i,l,t}$ of prey (i,l) by all $N_{p,L,t}$ predators in length group L is:

   $$C^{\text{des}}_{p,L,i,l,t} = N_{p,L,t}\, M_{p,L,t}\, \psi_{p,L,t}\, \frac{E_{p,L,i,l,t}}{\sum_{k,m} E_{p,L,k,m,t} + E_{\text{other},p}} \tag{3.15}$$

4. **Differentiable Overconsumption Cap and Understocking Penalty**: During optimization, parameter trial values may cause total desired consumption across all predators and fleets $D_{i,l,t} = \sum_{p,L} C^{\text{des}}_{p,L,i,l,t} + \sum_f C^{\text{des}}_{f,i,l,t}$ to exceed the available prey biomass $B_{i,l,t} = N_{i,l,t} W_{i,l,t}$. In `gadget3` (Lentz and Elvarsson 2024), total consumption is smoothly capped at 95% of available biomass ($0.95 B_{i,l,t}$) using a differentiable logistic minimum function (`dif_pmin`), scaling each predator's actual consumption down proportionally:

$$C^{\text{act}}_{p,L,i,l,t} = C^{\text{des}}_{p,L,i,l,t} \cdot \text{dif_pmin}\left(1,\ \frac{0.95\, B_{i,l,t}}{D_{i,l,t}}\right) \tag{3.16}$$

Simultaneously, any "unmet demand" (understocking $U_{i,l,t} = \max(0, D_{i,l,t} - 0.95 B_{i,l,t})$) adds a smooth quadratic penalty $\ell_{\text{understock}} = w_U \sum_{i,l,t} U^p_{i,l,t}$ to the objective function, guiding the gradient optimizer away from biologically infeasible regions where prey stocks are driven negative.

3.4.2 Composite Likelihoods and Two-Stage Iterative Reweighting in `Gadget`

Gadget fits simultaneously to arbitrary combinations of raw, unaggregated observations (Begley and Howell 2004, 18–23; Lentz and Elvarsson 2024):

1. **Survey Abundance Indices (`g3l_distribution_surveyindices_log` / `linear`)**:

$$\ell_{\text{index}} = \sum_t \left(\ln I_t - \left(\alpha + \beta \ln \hat{I}_t\right)\right)^2 \tag{3.17}$$

where $\alpha = \ln q$ is log catchability and β is either fixed at 1 (linear proportionality) or estimated (hyperstability/hyperdepletion).

2. **Length-Frequency & Age-Length Compositions (`g3l_distribution_sumofsquares`, `multinomial`, `mvlogistic`)**:
   - *Sum of Squares of Proportions*: $\ell_{\text{SS}} = \sum_{t,a,l} \left(\pi_{t,a,l} - \hat{\pi}_{t,a,l}\right)^2$, where $\pi_{t,a,l} = N^{\text{obs}}_{t,a,l} / \sum_{a',l'} N^{\text{obs}}_{t,a',l'}$.
   - *Multinomial*: $\ell_{\text{mult}} = 2 \sum_t \left[\ln \Gamma(N_t + 1) - \sum_{a,l} \ln \Gamma(N^{\text{obs}}_{t,a,l} + 1) + \sum_{a,l} N^{\text{obs}}_{t,a,l} \ln \hat{\pi}_{t,a,l}\right]$.
3. **Stomach Content Proportions (`StomachContent` / `g3l_distribution_sumofsquares`)**: Compares observed prey species/length weight ratios $W^{\text{obs}}_{p,L,i,l,t}$ in predator (p, L) stomachs against model-predicted consumption fractions:

$$\hat{W}_{p,L,i,l,t} = \frac{C^{\text{act}}_{p,L,i,l,t}}{\sum_{k,m} C^{\text{act}}_{p,L,k,m,t}} \tag{3.18}$$

3.4.2.1 Stefánsson's (1998) Two-Stage Iterative Likelihood Reweighting

Because a typical Gadget model combines 10–30 disparate likelihood components ℓ_c ($c = 1, \ldots, C$) with different units and sample sizes, (Stefánsson and Pálsson 1998), (Stefánsson 2003), and (Taylor et al. 2004) developed an objective two-stage reweighting procedure (`gadget.iterative`):

1. **Stage 1 (Independent Component Minima)**: For each likelihood group g, run an optimization assigning a very high weight to group g to determine the minimum achievable residual sum of squares ℓ_g^* (representing the intrinsic observation variance $\hat{\sigma}_g^2 = \ell_g^*/\mathrm{df}_g$ of dataset g without model-conflict bias).
2. **Stage 2 (Inverse-Variance Weighting)**: Set the weight w_g of each likelihood component equal to its inverse estimated variance:

$$w_g = \frac{\mathrm{df}_g}{\ell_g^*} \tag{3.19}$$

   so that in the final joint optimization $\mathcal{L}_{\mathrm{tot}} = \sum_g w_g \ell_g(\theta)$, every well-specified likelihood component has an expected weighted score of $\approx \mathrm{df}_g$ (chi-square degrees of freedom).

3.4.3 Case Study: Baltic Sea Multispecies `Gadget` Model and Ontogenetic Diet Shifts (Kulatska et al. 2019)

(Kulatska et al. 2019) constructed a quarterly (1974–2013) age-length multispecies `Gadget` model (using `Rgadget` and the Marine Fisheries Database `MFDB`) for the Baltic Sea (ICES Subdivisions 25–32), providing a template for how benthic and pelagic prey functional groups are parameterized in a length-structured assessment:

1. **Dynamic Fish Stocks vs. Constant-Biomass Invertebrate Pools**: Eastern Baltic cod (*Gadus morhua*, split into juvenile substock ages 0–1, 4–35 cm and adult substock ages 2–12+, 10–120 cm at 1 cm resolution) is modeled as the active predator on two dynamic pelagic clupeid stocks—Central Baltic herring (*Clupea harengus*, ages 0–10+, 3.5–32 cm) and Baltic sprat (*Sprattus sprattus*, ages 0–8+, 3–17 cm)—plus three constant-biomass benthic/invertebrate prey pools: the benthic isopod *Saduria entomon* (0.5–9 cm), pelagic/hyperbenthic mysids (*Mysis mixta*, 0.5–3 cm), and "other food".
2. **Empirical Identification of Ontogenetic Predator Size Classes via Bray–Curtis Clustering**: Rather than choosing arbitrary predator length bins for the stomach likelihood (`StomachContent`), (Kulatska et al. 2019, sec. 2.3.1, p. 341) applied hierarchical cluster analysis with **Bray–Curtis dissimilarity** (BC ≥ 0.2 threshold) to 1974–2013 EU/ICES Baltic cod stomach records (Quarters 1 and 4, using only

digestion stages 1–2 and excluding stage 0 intact prey to eliminate trawl-net feeding!), identifying **five ontogenetic cod diet regimes**:

$$L_{\text{cod}} \in \{[16, 25),\ [25, 35),\ [35, 60),\ [60, 80),\ [80, 112]\ \text{cm}\} \tag{3.20}$$

reflecting the ontogenetic transition from mysids and *Saduria* (< 25 cm) to sprat (25–60 cm) and large herring (> 60 cm).

3. **Prey-Specific Suitability Functions ($S(L, l)$) and Evacuation Ration (M_L)**:

   - **Size-Structured Prey (Herring, Sprat, and *Saduria entomon*)**: Modeled via the symmetric Andersen–Ursin logarithmic length-ratio suitability curve (Kulatska et al. 2019, Eq. 4, p. 340):

   $$S_{\text{prey}}(L, l) = p_0 + p_2 \exp\left(-\frac{\left(\ln \frac{L}{l} - p_1\right)^2}{p_3}\right) \tag{3.21}$$

   yielding optimal prey lengths $l^* = Le^{-p_1}$ equal to 19% **of cod length for sprat** and 28% **of cod length for herring**.
   - **Mysids (*Mysis mixta*)**: Because mysids span a narrow size range (0.5–3 cm) eaten primarily by juvenile cod, suitability is parameterized as a decreasing exponential function of predator length L alone:

   $$S_{\text{mysid}}(L) = \frac{\gamma}{1 + \exp(-\alpha - \beta L)} \tag{3.22}$$

   - **Maximum Consumption Rate (M_L)**: Estimated externally from observed stomach content weights S_k (including empty stomachs) via the (Jones 1978) evacuation rate model, yielding the quarterly allometric ration:

   $$M_L = \Delta t_{\text{days}} \cdot m_0\, L^{m_1}, \qquad m_0 = 3.167 \times 10^{-4}\ \text{kg} \cdot \text{d}^{-1} \cdot \text{cm}^{-m_1}, \quad m_1 = 2.66 \tag{3.23}$$

4. **Diagnosing Regime Shifts and Spatial Mismatch (Kulatska et al. 2019, 343–47)**: Optimized across 21 likelihood components using Simulated Annealing → Hooke & Jeeves → BFGS with (Stefánsson 2003) / (Taylor et al. 2007) iterative inverse-variance reweighting, the model accurately captured both ontogenetic diet shifts and interdecadal clupeid switches during 1974–1988 (Bray–Curtis dissimilarity 0.08–0.28), but exhibited higher diet residuals in 2007–2013 (BC = 0.24–0.52). Kulatska et al. showed that this residual pattern was diagnostic of two unmodeled physical processes in the recent Baltic regime: (a) expanding bottom hypoxia decimating benthic *Saduria entomon* biomass, and (b) **spatial predator–prey mismatch** (Eastern Baltic cod contracting southwestward into ICES SD 25 while sprat and herring shifted northeastward into SD 28–32).

3.5 Single-Species Length-Based & Integrated Assessments as a Methodological Bridge

Building on classical cohort and yield theory (Beverton and Holt 1956, 1957; Cadima 2003; Lart 2022), single-species integrated assessment models (`MULTIFAN-CL` (Fournier et al. 1998), `Stock Synthesis SS3` (Methot and Wetzel 2013), Reviewed by (Maunder and Punt 2013)), size-transition matrix assessments (Punt et al. 2016), and non-equilibrium length-based mixed-effects models (`LIME` (Rudd and Thorson 2018)) solved several fundamental statistical problems in fitting length-composition data that are directly applicable to size-spectrum models like `mizer`.

3.5.1 `MULTIFAN-CL` and Fournier's Robustified Length-Frequency Likelihood

Developed by David Fournier, John Hampton, and John Sibert (Fournier et al. 1998) for Western and Central Pacific tuna assessments, **MULTIFAN-CL** was the first large-scale ADMB integrated model to estimate age-structured dynamics directly from length-frequency data.

1. **Length-at-Age Mixture Model**: At any time t, the lengths of fish in age class $a \in \{1, \dots, A\}$ are assumed normally distributed with mean $\mu_{a,t}$ (following a von Bertalanffy curve with seasonal oscillations and cohort-specific growth deviations) and standard deviation σ_a (scaling linearly with mean length):

$$\sigma_a = \sigma_1 \exp\left(\lambda_\sigma \left(\frac{\mu_a - \mu_1}{\mu_A - \mu_1}\right)\right) \tag{3.24}$$

   Crucially, fishery selectivity $s_{f,a}$ is parameterized as a smooth cubic-spline function of **mean length** μ_a rather than age (`length-dependent selectivity`), reducing the number of selectivity parameters and linking cohorts across growth variations.

2. **Fournier's Robustified Normal Likelihood for Length Compositions**: Standard multinomial likelihoods $\sum_l \tilde{P}_l \ln P_l$ are notoriously sensitive to **outliers in the tails** (a single large fish caught in a bin where $\hat{P}_l = 10^{-6}$ creates a massive log-likelihood penalty) and **overdispersion from schooling** (fish of similar sizes school together, so haul length samples have much smaller effective sample size τ_t than the number of fish measured). (Fournier et al. 1998, Eq. 3.6) derived the **robustified normal likelihood** for observed length proportions $\tilde{p}_{f,t,l}$ vs. predicted proportions $\hat{p}_{f,t,l}$ across L length bins:

$$\ln L_{\text{MFCL}} = -\frac{1}{2} \sum_{f,t} \sum_{l=1}^{L} \ln\left(2\pi \left(\xi_{f,t,l} + \frac{0.1}{L}\right)\right) + \sum_{f,t} \sum_{l=1}^{L} \ln\left[\exp\left(-\frac{\left(\tilde{p}_{f,t,l} - \hat{p}_{f,t,l}\right)^2}{2\left(\xi_{f,t,l} + \frac{0.1}{L}\right)/\tau_{f,t}}\right) + 0.01\right] \tag{3.25}$$

   where $\xi_{f,t,l} = (1 - \tilde{p}_{f,t,l})\tilde{p}_{f,t,l}$, $\tau_{f,t} = \min(N_{\text{fish},f,t}, 1000)$ is the effective sample size, $+\frac{0.1}{L}$ prevents the variance from vanishing in bins where $\tilde{p}_{f,t,l} \approx 0$ so empty tail bins do

not dominate the variance weight, and +0.01 inside the logarithm **caps the maximum penalty of any single outlier bin at** $\ln(0.01) \approx -4.605$ (equivalent to a 3σ normal residual!).

3.5.2 `Stock Synthesis` (SS3): Age-Length Kernels and Growth Morphs/Platoons

In **Stock Synthesis** (Methot and Wetzel 2013, 86–90), population numbers $N_{t,g,a}$ are tracked by age a and growth morph/platoon g, and mapped at every time step to discrete population length bins $l \in \{1, \dots, n_L\}$ via an **age-length transition matrix** $\varphi_{l,g,a}$:

$$\varphi_{l,g,a} = \Phi\left(\frac{L_{\max,l} - \tilde{L}_{g,a}}{\sigma_{g,a}}\right) - \Phi\left(\frac{L_{\min,l} - \tilde{L}_{g,a}}{\sigma_{g,a}}\right) \tag{3.26}$$

where $\Phi(\cdot)$ is the standard normal CDF, $\tilde{L}_{g,a}$ is the mean length at exact mid-season age for morph g, and $\sigma_{g,a}$ is its standard deviation.

- **Combined Age- and Length-Specific Selectivity**: For fleet f with age selectivity $S^{(A)}_{f,a}$ and length selectivity $S^{(L)}_{f,l}$, the effective survival and catch-at-age are integrated over the length distribution within each age class (Methot and Wetzel 2013, Eq. A.1.24):
$$F_{t,f,g,a} = E_{t,f}\, S^{(A)}_{f,a} \sum_{l=1}^{n_L} \varphi_{l,g,a}\, S^{(L)}_{f,l} \tag{3.27}$$
- **Capturing Rosa Lee's Phenomenon via Platoons**: To prevent the assumption that the normal length distribution $\varphi_{l,g,a}$ resets every year regardless of size-selective fishing on the right tail of the cohort, SS3 splits each recruiting cohort into 3 or 5 **growth platoons** g (slow, medium-slow, average, medium-fast, fast growers with initial Gaussian weights). Because fast-growing platoons enter the length-selective fishery earlier, they are depleted faster ($F_{t,f,\text{fast},a} > F_{t,f,\text{slow},a}$), naturally shifting the surviving cohort's mean length downward!

3.5.3 Pure Size-Structured Assessments and Size-Transition Matrices ($\mathbf{X}_{l,l'}$)

In their comprehensive review of size-structured stock assessments (used worldwide for lobsters, crabs, abalone, scallops, prawns, and tropical finfish), (Punt et al. 2016) formalized the general discrete-time, discrete-size population equation:

$$\mathbf{N}_{t+1} = \mathbf{X}_t\, \mathbf{S}_t\, \mathbf{N}_t + \mathbf{R}_{t+1} \tag{3.28}$$

where:

- $\mathbf{N}_t = (N_{1,t}, N_{2,t}, \dots, N_{n_L,t})^\top$ is the vector of numbers across n_L size bins,

- $\mathbf{S}_t = \mathrm{diag}\big(\exp(-Z_{1,t}), \ldots, \exp(-Z_{n_L,t})\big)$ is the diagonal survival matrix during time step t,
- $\mathbf{X}_t$ is the lower-triangular (since animals do not shrink in length) $n_L \times n_L$ **Size-Transition Matrix**, where $X_{l',l}$ is the probability that an individual in size bin l grows into size bin $l' \geq l$ at the end of the time step ($\sum_{l' \geq l} X_{l',l} = 1$).

3.5.3.1 Parameterizing the Size-Transition Matrix $\mathbf{X}_{l',l}$ (Punt et al. 2016, Eqs. 2–5)

Instead of estimating $n_L(n_L - 1)/2$ free transition probabilities, size-structured assessments model the continuous growth increment $\Delta L = L_{t+1} - L_l \geq 0$ of an individual at midpoint L_l using a **Gamma distribution** (or truncated Normal / Log-normal distribution) with mean increment $\bar{\Delta}_l = (L_\infty - L_l)(1 - e^{-k\Delta t})$ and variance $\sigma^2_{\Delta,l} = \beta_\Delta \bar{\Delta}_l$ (so shape $\alpha_l = \bar{\Delta}_l / \beta_\Delta$ and scale β_Δ):

$$X_{l',l} = \int_{L_{\min,l'} - L_l}^{L_{\max,l'} - L_l} \frac{\Delta^{\alpha_l - 1} \exp(-\Delta/\beta_\Delta)}{\Gamma(\alpha_l)\, \beta_\Delta^{\alpha_l}}\, d\Delta, \qquad l' \geq l \tag{3.29}$$

[!IMPORTANT] **Critical Insight from Punt et al. (2016) for Fitting `mizer` to Length Data**: (Punt et al. 2016, 50–53) prove via simulation that in any size-structured assessment model, **asymptotic size L_∞ (or growth rate $g(w)$), natural mortality $M(w)$, and dome-shaped vs. asymptotic gear selectivity $S_f(w)$ are strongly confounded** when fitted to commercial length compositions alone:

- Absence of large fish in the catch can be explained equally well by: (a) slower growth / lower L_∞, (b) higher predation/natural mortality M, (c) higher fishing mortality F, or (d) dome-shaped selectivity $S_f(w)$.
- **How to break the confounding**: (1) Use scientific trawl survey length distributions with known asymptotic selectivity, (2) include **mean weight-at-age / conditional age-at-length** data where available, or (3) include **stomach content / predator-prey size data** which independently anchors $M2(w)$ and growth!

3.5.4 `LIME`: Length-Based Integrated Mixed Effects for Non-Equilibrium Length Data (Rudd & Thorson 2018)

Traditional length-only assessment methods for data-limited stocks—such as Length-Based Spawning Potential Ratio (`LB-SPR`, (Hordyk et al. 2015)) and mean-length mortality estimators (Beverton and Holt 1956; Gedamke and Hoenig 2006)—assume **strict steady-state equilibrium** (constant recruitment $R_t = R_0$ and constant fishing mortality $F_t = F$). In reality, recruitment pulses and time-varying F_t distort single-year length compositions, causing severe bias in equilibrium length estimators.

To relax these equilibrium assumptions while requiring as little as **1 year of length-composition data** plus known basic life-history parameters ($L_\infty, k, t_0, M, L_m^{50}, \alpha, \beta, \mathrm{CV}_L$), Merrill Rudd and James Thorson (Rudd and Thorson 2018) developed **LIME** (*Length-based Integrated Mixed Effects*), implemented in **R and TMB**:

1. **Age-Length Transition Matrix ($\varphi_{a,l}$) and Length-Based Selectivity (S_l)**: Given von Bertalanffy mean length-at-age $L_a = L_\infty\left(1 - e^{-k(a-t_0)}\right)$ and length standard deviation $\sigma_{L,a} = L_a \cdot \mathrm{CV}_L$ (typically $\mathrm{CV}_L = 0.10$), the probability $\varphi_{a,l}$ of an individual of age a falling into length bin l with bounds $[l_{\text{low}}, l_{\text{high}})$ is (Rudd and Thorson 2018, Table 1, Eq. 5, p. 29):

$$\varphi_{a,l} = \begin{cases} \Phi\left(\dfrac{l_{\text{high}} - L_a}{\sigma_{L,a}}\right) & l = 1 \\ \Phi\left(\dfrac{l_{\text{high}} - L_a}{\sigma_{L,a}}\right) - \Phi\left(\dfrac{l_{\text{low}} - L_a}{\sigma_{L,a}}\right) & 1 < l < n_l \\ 1 - \Phi\left(\dfrac{l_{\text{low}} - L_a}{\sigma_{L,a}}\right) & l = n_l \end{cases} \tag{3.30}$$

   Gear selectivity is parameterized as a two-parameter logistic function of length x_l (with parameters L_s^{50} and L_s^{95}) and mapped to age a by integrating across the age-length matrix:

$$S_l = \frac{1}{1 + \exp\left(-\dfrac{\ln(19)\,(x_l - L_s^{50})}{L_s^{95} - L_s^{50}}\right)}, \qquad S_a = \sum_{l=1}^{n_l} \varphi_{a,l}\, S_l \tag{3.31}$$

2. **Mixed-Effects State-Space Dynamics (Random-Walk F_t and Laplace-Integrated Recruitment R_t)**:

   - **Autocorrelated Recruitment Random Effects**: Annual recruitment R_t is modeled as a lognormal random effect (with optional AR(1) autocorrelation ρ_R and TMB epsilon bias-correction; (Rudd and Thorson 2018, Table 2, Eqs. 7, 17–19, p. 30)):

$$R_t \sim \text{Lognormal}\left(\ln R_0 - \frac{\sigma_R^2}{2},\ \sigma_R^2\right) \tag{3.32}$$

     where recruitment standard deviation σ_R is estimated with a hierarchical lognormal prior centered on the meta-analytic fish average ($\sigma_R \sim \text{Lognormal}(\ln(0.737),\ 0.353^2)$; (Thorson et al. 2014)).
   - **Random-Walk Penalty on Fishing Mortality F_t**: Annual fishing mortality F_t is estimated as a fixed effect constrained by a first-difference random-walk penalty in log-space ($\sigma_F = 0.20$):

$$\ln F_t \sim \mathcal{N}(\ln F_{t-1},\ \sigma_F^2) \tag{3.33}$$

   - **Predicted Catch Length Proportions ($\hat{p}_{t,l}$)**: Predicted catch-at-age $\hat{C}_{t,a} = \frac{F_t S_a}{M + F_t S_a} N_{t,a}\left(1 - e^{-M - F_t S_a}\right)$ is projected into length bins via $\hat{C}_{t,l} = \sum_a \hat{C}_{t,a}\varphi_{a,l}$, giving predicted length proportions $\hat{p}_{t,l} = \hat{C}_{t,l} / \sum_{l'} \hat{C}_{t,l'}$.

3. **Self-Weighting Dirichlet-Multinomial Length Likelihood and Derived Reference Points**: LIME fits observed catch length counts $\tilde{\pi}_{t,l}$ using the **Dirichlet-Multinomial (DM) likelihood** of (Thorson et al. 2017) (see Section 5.6.2), estimating the DM linear overdispersion parameter θ (`log_theta`) simultaneously with $F_t, L_s^{50}, L_s^{95}, \sigma_R$, and integrating over the latent recruitment vector $\mathbf{r} = \{R_t\}$ via TMB's Laplace approximation (Rudd and Thorson 2018, Table 2, Eqs. 16, 22). From the estimated parameters, LIME computes annual Spawning Potential Ratio:

$$\mathrm{SPR}_t = \frac{\mathrm{SBPR}_t}{\mathrm{SBPR}_0} = \frac{\sum_a \mathrm{Mat}_a \, W_a \, N_{F_t,a}}{\sum_a \mathrm{Mat}_a \, W_a \, N_{F=0,a}} \tag{3.34}$$

Notice that because $\varphi_{a,l}$ allows cohorts of different ages to overlap in length, LIME can separate a recruitment pulse $\epsilon_{R,t}$ (which creates a mode in juvenile length bins that propagates across years at rate $g(L)$) from an increase in fishing mortality F_t (which truncates the right tail of the length distribution across all adult cohorts simultaneously)—a key mechanism for dynamic length-based calibration in `mizer`!

3.6 Discrete Length-Structured Multispecies Models (`LeMans`, `LeMaRns`, `CSMS`, `CSM`)

3.6.1 `LeMans` and `LeMaRns` (Hall et al. 2006; Rochet et al. 2011; Thorpe et al. 2015; Spence et al. 2020, 2021)

Originally created by Stephen Hall, Collie, Jennings, Bravington, and Pope (Hall et al. 2006) for Georges Bank, extended to evaluate selective fishing across size-based models (Rochet et al. 2011), and adapted to the North Sea by (Thorpe et al. 2015) and Michael Spence et al. as the R/C++ package **`LeMaRns`** (Spence et al. 2020; Spence, Dolder, et al. 2021; Spence, Thorpe, et al. 2021), **LeMans** (*Length-based Multispecies analysis by numerical simulation*) is a discrete-time, discrete-length-bin multispecies model with fixed von Bertalanffy growth and one-way predation mortality feedback.

3.6.1.1 Mathematical Equations of `LeMaRns` (Spence et al. 2020, 3–7)

Let $N_{i,j,t}$ be the number of individuals of species $i \in \{1, \ldots, n_s\}$ in length class $j \in \{1, \ldots, n_l\}$ with bounds $[L_j, L_{j+1})$ and midpoint $\bar{L}_j = (L_j + L_{j+1})/2$, body mass $W_{i,j} = \alpha_i \bar{L}_j^{\beta_i}$, and maturity ogive $m_{i,j} = \left(1 + \exp(-\kappa_{m,i}(\bar{L}_j - L_{\mathrm{mat},i}))\right)^{-1}$.

Step 1: Reproduction & Recruitment: Spawning stock biomass is $\mathrm{SSB}_{i,t} = \sum_{j=1}^{n_l} m_{i,j} N_{i,j,t} W_{i,j}$. At the first time step of each year, recruits $R_{i,t}$ enter the smallest

length bin $j = 1$ via a Hockey-Stick, Ricker, Beverton-Holt, or linear stock-recruitment relationship (with optional lognormal process error in (Spence, Dolder, et al. 2021)):

$$N^{(1)}_{i,1,t} = N_{i,1,t} + R_{i,t}, \qquad N^{(1)}_{i,j,t} = N_{i,j,t} \quad (j > 1) \tag{3.35}$$

Step 2: Mortality ($M1$, Fishing F, and Predation $M2$): Total mortality in time step Δt is $Z_{i,j,t} = \Delta t \left(M1_{i,j} + F_{i,j,t} + M2_{i,j,t}\right)$, so surviving numbers are:

$$N^{(2)}_{i,j,t} = N^{(1)}_{i,j,t} \exp(-Z_{i,j,t}) \tag{3.36}$$

where **predation mortality** $M2_{i,j,t}$ on prey (i, j) by all predators (m, k) is governed by a Holling Type II (or Type III) functional response:

$$M2_{i,j,t} = \sum_{m=1}^{n_s} \sum_{k=1}^{n_l} I_{m,k}\, N^{(1)}_{m,k,t}\, \frac{\theta_{m,k,i,j}}{\sum_{a=1}^{n_s} \sum_{b=1}^{n_l} \theta_{m,k,a,b}\, N^{(1)}_{a,b,t} W_{a,b} + W_{\text{other}}} \tag{3.37}$$

Here:

- $I_{m,k} = W_{m,k} \left(L_{\infty,m}/\bar{L}_k\right)^{0.4} \cdot \text{ration_scalar}_m$ is the maximum per-capita consumption rate of predator (m, k),
- $\theta_{m,k,i,j} = \tau_{m,i} \cdot \psi_{m,k,i,j} \cdot o_{m,i}$ is the suitability product of taxonomic preference $\tau_{m,i} \in \{0, 1\}$, spatial overlap $o_{m,i}$, and Ursin lognormal weight-ratio preference:

$$\psi_{m,k,i,j} = \exp\left(-\frac{\left(\log_{10} \frac{W_{m,k}}{W_{i,j}} - \eta_m\right)^2}{2\sigma_m^2} \right) \tag{3.38}$$

Step 3: Deterministic Fractional Somatic Growth: Under deterministic von Bertalanffy growth, the time required for an individual of species i to grow from the lower boundary L_j to the upper boundary L_{j+1} of length bin j is:

$$\delta t_{i,j} = \frac{1}{k_i} \ln\left(\frac{L_{\infty,i} - L_j}{L_{\infty,i} - L_{j+1}} \right) \tag{3.39}$$

Therefore, in a discrete time step Δt (chosen small enough that $\Delta t \leq \min_{i,j} \delta t_{i,j}$, e.g., $\Delta t = 0.1$ yr), a fixed fraction $\phi_{i,j}$ of individuals in length bin j advance to length bin $j+1$:

$$\phi_{i,j} = \frac{\Delta t}{\delta t_{i,j}} = \frac{\Delta t \cdot k_i}{\ln\left(\frac{L_{\infty,i} - L_j}{L_{\infty,i} - L_{j+1}}\right)} \tag{3.40}$$

giving the state update:

$$N_{i,j,t+1} = (1 - \phi_{i,j}) N^{(2)}_{i,j,t} + \phi_{i,j-1} N^{(2)}_{i,j-1,t} \tag{3.41}$$

⚠ Equivalence with mizer Finite-Difference Discretization

Mathematically, the `LeMaRns` state update (Eq. 3.41) is identical to the **explicit first-order upwind finite-difference scheme** of the McKendrick–von Foerster PDE in `mizer`! Consequently, it exhibits the exact same numerical Courant-Friedrichs-Lewy [CFL] artificial cohort diffusion $\approx \frac{1}{2} g \delta L (1 - \phi)$ analyzed by (Datta et al. 2010).

3.6.2 `CSMS` and `CSM` (Pope et al. 2006, 2019): Moment-Based Reduction of Length Spectra

John Pope and colleagues (Pope et al. 2006, 2019; Gislason et al. 2010) developed two remarkably parsimonious length-based multispecies models to explore multispecies MSY and community structure:

1. **CSMS (Charmingly Simple Multispecies Size-based model; (Pope et al. 2006))**: Parameterizes an entire 13-species North Sea fish community using only **15 life-history scaling parameters** (linking von Bertalanffy $k_i = c_k L_{\infty,i}^{-0.31}$, maturity $L_{\mathrm{mat},i} = c_m L_{\infty,i}$, Ricker recruitment parameters, and Andersen-Ursin predation solely to each species' asymptotic length $L_{\infty,i}$), reproducing the community size spectrum slope and steeper slopes under fishing.
2. **CSM (Charmingly Simple Model; (Pope et al. 2019))**: Instead of tracking 50 discrete length bins per cohort, CSM notes that the log-length distribution of a cohort is approximately Gaussian and therefore **tracks only the 0th through 4th moments** $\left(M_0 = N_{s,a},\ M_1 = \mathbb{E}[\ln L],\ M_2 = \mathrm{Var}(\ln L),\ \ldots\right)$ of each cohort's size distribution! Because the product of a Gaussian cohort log-length density and a Gaussian Ursin log-size preference kernel has an **exact closed-form Gaussian integral**, CSM computes size-based predation mortality $M2_{s,a}$ analytically without any discrete length bins, achieving a 50× speedup over bin-based models.

4 Continuous Size-Spectrum (`mizer`) and State-Space Food-Web Models (`EwE`, `Rpath`, `EcoState`)

4.1 Continuous Multispecies Size-Spectrum Models (`mizer`)

4.1.1 Theoretical Lineage and Core Philosophy

Size-spectrum theory originated from the empirical discovery by (Sheldon et al. 1972) and (Sheldon et al. 1977) that pelagic marine biomass is approximately constant across logarithmic body-mass bins ($B(w)dw \propto w^0 d(\ln w) \implies N(w) \propto w^{-2}$), and from the continuous mass-balance and pelagic ecosystem derivations of (Platt and Denman 1977), (Silvert and Platt 1978), (Benoît and Rochet 2004), (Andersen and Beyer 2006), and (Cuesta et al. 2018).

In a landmark synthesis, (Hartvig et al. 2011) and (Scott et al. 2014) resolved species identity within the continuous size spectrum to create the **`mizer`** framework (*Multi-species size-spectrum ecological modelling in R*; applied to North Sea reference points by (Blanchard et al. 2014) and reviewed comprehensively by (Andersen et al. 2016), (Andersen 2019), (Andersen 2020), and (Pennino et al. 2025)).

Unlike age-structured models where somatic growth and natural mortality are external inputs, in `mizer` **both individual somatic growth $g_i(w,t)$ and predation mortality $\mu_{p,i}(w,t)$ emerge dynamically from individual-level foraging and strict conservation of mass/energy**.

4.1.2 Complete Mathematical Formulation of Standard `mizer`

Let $N_i(w,t)$ (numbers $\cdot\, \mathrm{g}^{-1} \cdot \mathrm{m}^{-3}$ or total numbers per weight interval) denote the number density of species $i \in \{1, \dots, S\}$ at body mass $w \in [w_0, w_{\max,i}]$ and time t, and let $N_R(w,t)$ denote the background plankton/benthic resource spectrum for $w \in [w_{\text{cut}}, w_{\max,R}]$.

4.1.2.1 1. Governing McKendrick–von Foerster Conservation PDE

The time evolution of $N_i(w,t)$ is governed by the first-order hyperbolic continuity (advection-reaction) equation (Hartvig et al. 2011, Eq. 1; Scott et al. 2014, Eq. 3.1):

$$\frac{\partial N_i(w,t)}{\partial t} + \frac{\partial \left(g_i(w,t) N_i(w,t)\right)}{\partial w} = -\mu_i(w,t) N_i(w,t) \tag{4.1}$$

coupled to the **boundary condition (recruitment flux)** at egg weight w_0:

$$g_i(w_0,t) N_i(w_0,t) = R_{\mathrm{dd},i}(t) \tag{4.2}$$

4.1.2.2 2. Volumetric Search Rate, Predation Kernel, and Encounter Rate ($E_i(w,t)$)

A predator of species i and mass w searches a water volume at allometric rate $\gamma_i(w) = \gamma_i w^q$ ($q \approx 0.8$). Its size preference for prey of mass w_p is described by a **predation kernel** $\phi_i(w/w_p)$, most commonly the Andersen–Ursin lognormal kernel (Scott et al. 2014, Eq. 3.4):

$$\phi_i\left(\frac{w}{w_p}\right) = \exp\left(-\frac{\left(\ln\left(\frac{w}{\beta_i w_p}\right)\right)^2}{2\sigma_i^2}\right) \tag{4.3}$$

(or a power-law / box / double-logistic kernel). Given the species interaction (spatial overlap / vulnerability) matrix $\theta_{ij} \in [0,1]$, the **total biomass of suitable prey encountered per unit time** $E_i(w,t)$ by predator (i,w) is:

$$E_i(w,t) = \gamma_i(w) \int \phi_i\left(\frac{w}{w_p}\right) \left[\theta_{iR} N_R(w_p,t) + \sum_{j=1}^{S} \theta_{ij} N_j(w_p,t)\right] w_p \, dw_p \tag{4.4}$$

4.1.2.3 3. Feeding Level ($f_i(w,t)$) and Holling Type II Satiation

Given physiological maximum intake rate $I_{\max,i}(w) = h_i w^n$ ($n \approx 0.67$–0.75), the dimensionless **feeding level** $f_i(w,t) \in [0,1)$ follows a Holling Type II functional response:

$$f_i(w,t) = \frac{E_i(w,t)}{E_i(w,t) + h_i w^n} \tag{4.5}$$

Thus actual biomass consumption per predator per unit time is $C_i(w,t) = f_i(w,t) h_i w^n$.

4.1.2.4 4. Bioenergetic Allocation, Somatic Growth ($g_i(w,t)$), and Reproduction ($R_{p,i}(t)$)

Assimilated energy is a fraction $\alpha_i \in (0,1)$ of ingested food ($\alpha_i \approx 0.6$), from which standard metabolic maintenance $k_{s,i}w^p$ ($p \approx 0.7$–1.0) is subtracted to give net available energy $E_{\text{avail},i}(w,t)$:

$$E_{\text{avail},i}(w,t) = \alpha_i \, f_i(w,t) \, h_i w^n - k_{s,i} w^p \tag{4.6}$$

A size-dependent maturation ogive $\psi_i(w)$ (transitioning from 0 to 1 around maturation weight $w_{\text{mat},i}$ and scaling with $\left(w/w_{\text{max},i}\right)^{1-n}$ or $1-(w/w_{\text{max},i})^{p-n}$ so growth ceases at $w_{\text{max},i}$) partitions net positive energy between **somatic growth** $g_i(w,t)$ and **gonadal reproduction** $E_{r,i}(w,t)$:

$$g_i(w,t) = \max\Big(0,\ E_{\text{avail},i}(w,t)\Big)\Big(1-\psi_i(w)\Big) \tag{4.7}$$

$$\psi_i(w) = \left[1+\left(\frac{w}{w_{\text{mat},i}}\right)^{-U}\right]^{-1}\left(\frac{w}{w_{\text{max},i}}\right)^{1-n} \tag{4.8}$$

The **density-independent physiological egg production rate** $R_{\text{di},i}(t)$ ($\text{eggs}\cdot\text{yr}^{-1}$) is obtained by integrating reproductive energy over all sizes, multiplying by reproductive efficiency $\epsilon_{\text{rep},i}$ and female fraction $1/2$, and dividing by egg weight w_0:

$$R_{\text{di},i}(t) = \frac{\epsilon_{\text{rep},i}}{2w_0}\int_{w_0}^{w_{\text{max},i}} N_i(w,t)\max\Big(0,\ E_{\text{avail},i}(w,t)\Big)\psi_i(w)\,dw \tag{4.9}$$

In standard `mizer`, external density dependence at the larval stage (e.g., limited larval habitat or unmodeled larval mortality) is represented by passing $R_{\text{di},i}(t)$ through a **Beverton–Holt stock-recruitment function** with maximum recruitment cap $R_{\text{max},i}$:

$$R_{\text{dd},i}(t) = \frac{R_{\text{max},i}\,R_{\text{di},i}(t)}{R_{\text{di},i}(t)+R_{\text{max},i}} \tag{4.10}$$

4.1.2.5 5. Total Mortality ($\mu_i(w,t)$): Background, Predation, Starvation, and Fishing

Total instantaneous mortality $\mu_i(w,t)$ is the sum of four mechanistic components:

$$\mu_i(w,t) = \mu_{0,i}(w) + \mu_{p,i}(w,t) + \mu_{s,i}(w,t) + F_i(w,t) \tag{4.11}$$

1. **Background / Senescence Mortality**: $\mu_{0,i}(w) = \mu_{0,i} w_{\text{max},i}^{n-1}$ (allometrically scaled with $w_{\text{max},i}$).

2. **Endogenous Predation Mortality ($\mu_{p,i}(w_p,t) \equiv M2$)**: By strict mass balance, every gram of prey (j, w_p) eaten by predators (i, w) must appear as mortality on (j, w_p) (Scott et al. 2014, Eq. 3.11):

$$\mu_{p,j}(w_p,t) = \sum_{i=1}^{S} \theta_{ij} \int_{w_0}^{w_{\max,i}} \phi_i\left(\frac{w}{w_p}\right) \Big(1 - f_i(w,t)\Big) \gamma_i(w)\, N_i(w,t)\, dw \tag{4.12}$$

(Notice the identity $\frac{f_i(w) h_i w^n}{E_i(w)} = 1 - f_i(w)$: when a predator is satiated [$f_i(w) \to 1$], its per-unit-prey clearance rate $(1 - f_i)\gamma_i(w)$ drops toward zero!)

3. **Starvation Mortality ($\mu_{s,i}(w,t)$)**: Whenever available assimilated energy falls below metabolic maintenance ($E_{\text{avail},i}(w,t) < 0$), individuals suffer starvation mortality proportional to the energy deficit divided by lipid energy stores $\xi_{\text{store}} w$ (Scott et al. 2014, Eq. 3.12):

$$\mu_{s,i}(w,t) = \begin{cases} \dfrac{-E_{\text{avail},i}(w,t)}{\xi_{\text{store}} w} & \text{if } E_{\text{avail},i}(w,t) < 0 \\ 0 & \text{otherwise} \end{cases} \tag{4.13}$$

4. **Fishing Mortality ($F_i(w,t)$)**: Summed over fishing gears/fleets f with size selectivity $S_{f,i}(w)$, catchability $Q_{f,i}$, and fishing effort $E_f(t)$:

$$F_i(w,t) = \sum_f S_{f,i}(w)\, Q_{f,i}\, E_f(t) \tag{4.14}$$

generating total fishery yield in weight $Y_{f,i}(t) = \int_{w_0}^{w_{\max,i}} F_{f,i}(w,t)\, w\, N_i(w,t)\, dw$.

4.1.2.6 6. Resource Spectrum Dynamics ($N_R(w,t)$)

The background plankton/benthos spectrum $N_R(w,t)$ evolves via semi-chemostat (or logistic) renewal at rate $r_R(w) = r_0 w^{n-1}$ toward carrying capacity $\kappa_R(w) = \kappa w^{-\lambda}$ ($\lambda \approx 2 + q - n \approx 2.05$):

$$\frac{\partial N_R(w,t)}{\partial t} = r_R(w)\Big(\kappa_R(w) - N_R(w,t)\Big) - \mu_{p,R}(w,t) N_R(w,t) \tag{4.15}$$

4.1.2.7 7. Semi-Implicit Upwind Finite-Difference Discretization in `mizer`

On a logarithmically spaced weight grid $w_j = w_0 \cdot 10^{(j-1)\Delta x}$ with bin widths $\Delta w_j = w_{j+1} - w_j$, `mizer` solves the McKendrick–von Foerster PDE over time step Δt by evaluating encounter E_i, feeding level f_i, growth $g_{i,j}^t$, and mortality $\mu_{i,j}^t$ explicitly at time t (using Fast Fourier Transform [FFT] convolution on the log-weight grid for the integrals E_i and $\mu_{p,i}$ when kernels are shift-invariant in $\ln(w/w_p)$), and then advancing $N_{i,j}^{t+\Delta t}$ via a **first-order upwind, backward-Euler (semi-implicit) scheme** (Scott et al. 2014, Appendix A):

$$\frac{N_{i,j}^{t+\Delta t} - N_{i,j}^t}{\Delta t} + \frac{g_{i,j}^t N_{i,j}^{t+\Delta t} - g_{i,j-1}^t N_{i,j-1}^{t+\Delta t}}{\Delta w_j} = -\mu_{i,j}^t N_{i,j}^{t+\Delta t} \tag{4.16}$$

Solving for $N_{i,j}^{t+\Delta t}$ gives a lower-bidiagonal linear system solved in $O(n_w)$ forward substitution:

$$N_{i,j}^{t+\Delta t} = \frac{N_{i,j}^t + \frac{\Delta t}{\Delta w_j} g_{i,j-1}^t N_{i,j-1}^{t+\Delta t}}{1 + \Delta t \left(\frac{g_{i,j}^t}{\Delta w_j} + \mu_{i,j}^t \right)} \tag{4.17}$$

4.2 The Jump-Growth Fokker–Planck PDE (Datta, Delius & Law 2010, 2011)

4.2.1 Why the First-Order McKendrick–von Foerster PDE Fails for Dynamic Cohort Fitting

In the standard McKendrick–von Foerster PDE $\partial_t N + \partial_w(gN) = -\mu N$, somatic growth is treated as a purely deterministic continuous drift $dw/dt = g(w,t)$. Consequently:

1. **No Intrinsic Cohort Broadening**: All individuals born at the same instant t_0 at egg weight w_0 travel along the exact same characteristic trajectory $w(t)$ with zero variance in size-at-age! If recruitment is pulsed seasonally (or fluctuates interannually as in real stock assessments), a narrow delta-function pulse at w_0 remains an unrealistically sharp spike (broadened only by **numerical discretization error** $\propto \Delta w_j$ of the upwind scheme!).
2. **Grid-Dependent Numerical Diffusion**: In `mizer`'s first-order upwind scheme, changing the number of weight bins n_w or time step Δt alters the numerical diffusivity $D_{\text{num}} \approx \frac{1}{2} g \Delta w$, which artificially changes how recruitment pulses smear across length compositions!
3. **Deterministic Instability of the Community Size Spectrum**: Without growth dispersion or an external stock-recruitment clamp, the continuous McKendrick–von Foerster PDE exhibits traveling-wave instabilities (cohort cycles) for realistic predator-prey mass ratios ($\beta \sim 100$; (Datta et al. 2011)).

4.2.2 Derivation of the Jump-Growth Equation (Datta, Delius & Law 2010)

(Datta et al. 2010) resolved this fundamental limitation from first principles by observing that fish do not grow by absorbing a continuous fluid: they grow by **discrete stochastic jumps** $\Delta w = \alpha w_p$ each time they capture and assimilate a discrete prey organism of body mass w_p.

Starting from the exact Markov **Master Equation** for discrete prey-capture events in continuous time (Datta et al. 2010, Eq. 5):

$$\frac{\partial N_i(w,t)}{\partial t} = \int dw_p \Big[\text{Transition in from } (w - \alpha_i w_p) - \text{Transition out from } w\Big] - \mu_i(w,t) N_i(w,t) \tag{4.18}$$

and performing a second-order **Kramers–Moyal Taylor expansion** in the prey mass increment $\alpha_i w_p \ll w$, (Datta et al. 2010) derived the **Jump-Growth Fokker–Planck (Advection–Diffusion–Reaction) PDE**:

$$\frac{\partial N_i(w,t)}{\partial t} + \frac{\partial \left(g_i(w,t) N_i(w,t)\right)}{\partial w} = \frac{1}{2} \frac{\partial^2 \left(d_i(w,t) N_i(w,t)\right)}{\partial w^2} - \mu_i(w,t) N_i(w,t) \tag{4.19}$$

where:

- The **first-order drift (mean somatic growth rate)** $g_i(w,t)$ is the first moment of assimilated prey mass:

$$g_i(w,t) = \alpha_i \Big(1 - \psi_i(w)\Big)\Big(1 - f_i(w,t)\Big)\gamma_i(w) \int \phi_i\left(\frac{w}{w_p}\right) N_c(w_p,t)\, \mathbf{w_p}\, dw_p - k_{s,i} w^p \Big(1 - \psi_i(w)\Big) \tag{4.20}$$

- The **second-order macro-diffusion (growth variance) coefficient** $d_i(w,t)$ is the **second moment** of assimilated prey mass (Datta et al. 2010, Eq. 9):

$$d_i(w,t) = \alpha_i^2 \Big(1 - \psi_i(w)\Big)^2 \Big(1 - f_i(w,t)\Big)\gamma_i(w) \int \phi_i\left(\frac{w}{w_p}\right) N_c(w_p,t)\, \mathbf{w_p^2}\, dw_p \tag{4.21}$$

  [!IMPORTANT] **Mathematical & Practical Significance of the Jump-Growth Diffusion Term** $d_i(w,t)$:

  1. **Zero Free Parameters**: In the pure mechanistic derivation, $d_i(w,t)$ requires **no additional free parameters** (or a single scaling multiplier if accounting for behavioral bout-feeding / schooling variation, analogous to Gadget's Beta-Binomial β_s).
  2. **Physical Cohort Dispersion**: A recruiting year-class pulse at w_0 naturally broadens into a realistic bell-shaped size-at-age distribution as it advects to larger sizes, enabling direct likelihood fitting to time-series length-frequency distributions!
  3. **Grid-Independent Convergence**: When solved on a sufficiently fine grid (or via Crank–Nicolson / implicit central differences for the $\partial_w^2(d_i N_i)$ term), the physical diffusivity $d_i(w,t)$ dominates numerical grid diffusivity, making model predictions invariant to grid resolution.

4.3 Population Regulation Without Ad-Hoc Stock–Recruitment Relationships (Canales, Delius & Law 2020)

In standard fisheries models (including standard `mizer`, `SMS`, `Gadget`, and `LeMaRns`), population abundance is stabilized externally by imposing an ad-hoc **Beverton–Holt or Ricker stock-recruitment relationship (SRR)** at the boundary w_0, with one or two free species-specific parameters ($R_{\max,i}$ or steepness h_i) that have little direct physiological basis and strongly dictate MSY predictions.

(Canales et al. 2020) demonstrated that in a size-spectrum model with linear density-independent reproduction ($R_{\mathrm{dd},i} = R_{\mathrm{di},i}$, i.e., $R_{\max,i} \to \infty$), **three endogenous biological mechanisms** naturally regulate population abundance and coexistence without any ad-hoc SRR:

1. **Density-Dependent Somatic Growth via Prey Competition**: When a species' abundance increases, it depletes its prey size range, lowering its feeding level $f_i(w,t)$ and slowing juvenile growth $g_i(w,t)$. Because juveniles spend longer in small vulnerable size classes ($w < w_{\mathrm{mat},i}$) while experiencing predation mortality $\mu_{p,i}(w,t)$, cumulative survival to maturity $\exp\!\Big(-\int_{w_0}^{w_{\mathrm{mat}}} \frac{\mu_i(w)}{g_i(w)} dw\Big)$ drops sharply!
2. **Early-Life Predation and Cannibalism**: Extending the predation kernel to planktonic/larval stages and accounting for intra- and inter-specific predation on larvae creates a self-regulating mortality bottleneck.
3. **Reproductive Allometry ($m_i > n_i$)**: When reproductive investment scales steeper with body mass than metabolic intake (matching empirical BOFFFF—Big Old Fat Fecund Female—observations), truncating the adult size structure via fishing directly reduces per-biomass egg output without needing an SRR curve.

4.4 Mass-Balance, Foraging Arena, and State-Space Food-Web Models (`Ecopath`, `Ecosim`, `Rpath`, `EcoState`)

While size-spectrum models resolve continuous body mass, the most widely applied whole-ecosystem modeling framework in marine science is **Ecopath with Ecosim (EwE)**—originating with (Polovina 1984) and (Christensen and Pauly 1992), extended dynamically and spatially via Foraging Arena Theory in `Ecosim` and `Ecospace` by (Walters et al. 1997), (Walters et al. 1999), (Christensen and Walters 2004), (Christensen et al. 2005), and (Christensen et al. 2024), critically evaluated by (Plagányi and Butterworth 2004), and automated via stepwise fitting by (Scott et al. 2016)—along with its reproducible R implementation **`Rpath`** (Lucey et al. 2020) and its recent TMB state-space stock assessment reformulation **`EcoState`** (Thorson et al. 2024).

4.4.1 `Ecopath`: Static Mass Balance at Reference Equilibrium

`Ecopath` partitions the ecosystem into $i \in \{1, \dots, n\}$ functional groups (which may be single species, multi-species guilds, or juvenile/adult life stanzas). At a reference equilibrium period, production of group i equals the sum of all mortality and export fluxes (Christensen and Walters 2004, Eq. 1; Lucey et al. 2020, Eq. 1):

$$B_i \left(\frac{P}{B}\right)_i \cdot EE_i = Y_i + E_i + BA_i + \sum_{j=1}^{n} B_j \left(\frac{Q}{B}\right)_j DC_{ij} \tag{4.22}$$

where:

- B_i is biomass, $(P/B)_i = Z_i$ is the production-to-biomass ratio (total mortality rate under equilibrium), and $(Q/B)_j$ is the consumption-to-biomass ratio of predator j,
- $EE_i \in [0, 1]$ is **Ecotrophic Efficiency** (the fraction of production explained within the modeled system, so $M0_i = (P/B)_i(1 - EE_i)$ is residual unexplained mortality),
- Y_i is fishery catch, E_i is net migration, BA_i is biomass accumulation, and
- DC_{ij} is the **diet composition matrix** (fraction by weight of predator j's diet consisting of prey i, with $\sum_i DC_{ij} = 1$). For each group i, three of the four core parameters $\{B_i, (P/B)_i, (Q/B)_i, EE_i\}$ are supplied along with DC_{ij} and Y_i, and the linear system is solved for the missing parameter (typically EE_i or B_i).

4.4.2 `Ecosim` and Walters' Foraging Arena Theory

When `Ecopath` is projected dynamically through time as a system of coupled ODEs (`Ecosim`; (Walters et al. 1997)):

$$\frac{dB_i}{dt} = g_i \sum_j Q_{ji}(t) - \sum_k Q_{ik}(t) + I_i - (M0_i + F_i(t) + e_i)\, B_i(t) \tag{4.23}$$

where $g_i = (P/Q)_i$ is net growth efficiency, a critical question arises: how should the consumption flux $Q_{ij}(t)$ of prey i by predator j depend on $B_i(t)$ and $B_j(t)$?

> **!** Theoretical Necessity of Foraging Arena Refugia
>
> (Walters et al. 1997) showed that if standard mass-action Lotka–Volterra ($Q_{ij} = a_{ij} B_i B_j$) or standard Holling Type II is used in a complex 30-species food web, **the ecosystem rapidly collapses via Gause competitive exclusion or violent predator-prey limit cycles**. Foraging Arena Theory resolves this by creating behavioral or spatial refugia for prey.

To resolve this, (Walters et al. 1997) formulated **Foraging Arena Theory**: at any given moment, total prey biomass B_i is partitioned with respect to predator j into an **invulnerable**

(refuge/sheltered/un-overlapping) state $B_i - V_{ij}$ and a small **vulnerable foraging arena state** V_{ij}, with behavioural/spatial exchange rate v_{ij} between them:

$$\frac{dV_{ij}}{dt} = v_{ij}\left(B_i - V_{ij}\right) - v'_{ij}V_{ij} - a_{ij}V_{ij}B_j \tag{4.24}$$

Assuming the behavioral exchange V_{ij} equilibrates rapidly relative to annual population changes ($dV_{ij}/dt \approx 0$, with $v'_{ij} = v_{ij}$), solving for V_{ij} yields the **Foraging Arena Functional Response** (Walters et al. 1997, Eq. 4; Christensen and Walters 2004, Eq. 5; Thorson et al. 2024, Eq. 4):

$$Q_{ij}(t) = a_{ij}V_{ij}(t)B_j(t) = \frac{a_{ij}\, v_{ij}\, B_i(t)\, B_j(t)}{2v_{ij} + a_{ij}B_j(t)} \tag{4.25}$$

Calibrating to the Ecopath baseline $Q_{ij,0} = B_{j,0}(Q/B)_{j,0}DC_{ij,0}$ at reference biomasses $(B_{i,0}, B_{j,0})$ eliminates a_{ij} in favor of a single dimensionless **vulnerability multiplier** $X_{ij} = \frac{v_{ij}B_{i,0}}{Q_{ij,0}} \in (1, \infty)$ (default $X_{ij} = 2$):

$$Q_{ij}(t) = Q_{ij,0}\,\frac{X_{ij}}{X_{ij} - 1 + \frac{B_j(t)}{B_{j,0}}}\,\frac{B_i(t)}{B_{i,0}} \tag{4.26}$$

- **Low Vulnerability ($X_{ij} \to 1$, "Bottom-Up / Donor Control")**: As $X_{ij} \to 1$, $Q_{ij}(t) \to Q_{ij,0}\frac{B_i(t)}{B_{i,0}}$ is **independent of predator biomass** $B_j(t)$! Doubling predator abundance B_j simply means twice as many predators compete for the same trickle of prey entering the foraging arena, halving per-capita predator intake and preventing top-down prey extermination.
- **High Vulnerability ($X_{ij} \to \infty$, "Top-Down / Lotka–Volterra Control")**: As $X_{ij} \to \infty$, $Q_{ij}(t) \to Q_{ij,0}\frac{B_i(t)}{B_{i,0}}\frac{B_j(t)}{B_{j,0}}$ recovers mass-action top-down predation.

(For critique of EwE's delay-difference multi-stanza assumptions and vulnerability overfitting when many X_{ij} are tuned to short time series, see (Plagányi and Butterworth 2004) and the stepwise AIC fitting protocol of (Scott et al. 2016)).

4.4.3 `EcoState` (Thorson et al. 2024): State-Space Mass-Balance Assessment in TMB

Historically, EwE was fitted by shooting (deterministic trajectory matching) without process errors or formal state-space marginal likelihoods. In 2024, James Thorson, Kirstin Holsman, Gaichas, Aydin, and Punt introduced `EcoState` (Thorson et al. 2024), which embeds the Ecopath + Ecosim ODEs directly inside a **Template Model Builder (TMB) state-space stock assessment model**:

1. **Continuous-Time ODE with Annual Process Errors** $\epsilon_{i,t}$: Within each year $t \in \{t_0, \dots, T\}$, biomass $B_i(s)$ evolves over $s \in [t, t+1]$ via the Ecosim Foraging Arena ODE augmented with an estimated annual unexplained process-error rate $\epsilon_{i,t}$ (Thorson et al. 2024, Eq. 5):

$$\frac{dB_i(s)}{ds} = g_i \sum_j Q_{ji}(s) - \sum_k Q_{ik}(s) - \Big(M0_i + F_{i,t} - \epsilon_{i,t}\Big) B_i(s) \tag{4.27}$$

   integrated numerically in TMB using a differentiable Runge–Kutta / Adams–Bashforth ODE solver (`odeint`), where $\epsilon_{i,t} \sim \mathcal{N}(0, \sigma^2_{\text{proc},i})$ are random effects marginalized via the **Laplace approximation**.
2. **Simultaneous Joint Likelihood Across All Data Streams**: `EcoState` estimates Ecopath baseline parameters ($\ln B_{i,0}, EE_i, DC_{ij,0}$), Ecosim vulnerabilities $\ln(X_{ij} - 1)$, fishing mortalities $\ln F_{i,t}$, and process error variances $\sigma_{\text{proc},i}$ simultaneously by fitting to:

   - Survey biomass indices: $\ln I_{i,t} \sim \mathcal{N}\big(\ln(q_i \hat{B}_{i,t}),\, \sigma^2_{I,i,t}\big)$,
   - Fishery catches: $\ln C^{\text{obs}}_{i,t} \sim \mathcal{N}\big(\ln \hat{C}_{i,t},\, \sigma^2_{C,i}\big)$,
   - Time-varying stomach diet proportions $\mathbf{W}^{\text{obs}}_{j,t}$: modeled via a **Dirichlet likelihood** on predicted Ecosim consumption shares $\hat{p}_{ij,t} = \frac{\hat{Q}_{ij,t}}{\sum_k \hat{Q}_{kj,t}}$!

5 Stomach Content Data, Gastric Evacuation Physics, and Diet Likelihood Formulations

5.1 Why Stomach Content Data Cannot Be Treated as Direct Diet Proportions

A common pitfall when calibrating ecological models is to treat the wet weight $S_{i,k}$ of prey i observed inside stomach k as a direct measure of the predator's consumption rate C_i. In reality, what is observed inside a sampled fish stomach is the **standing stock of partially digested prey**, which is related to the **ingestion flux (consumption rate)** only through the physics of **gastric evacuation** and **spatial sampling design** (Rindorf and Lewy 2004; Andersen and Beyer 2005; Buckland et al. 2017; Jacobsen et al. 2023):

$$\text{Ingestion Rate } C_i(t) \xrightarrow{\text{Digestion / Evacuation Rate } \frac{dS_i}{dt}} \text{True Stomach Content } S_i(t) \xrightarrow{\text{Trawl Survey \& Lab Protocol}} \text{Observe} \tag{5.1}$$

5.2 Biophysical Modelling of Gastric Evacuation (Andersen & Beyer 2005)

5.2.1 The Square-Root Cylinder Evacuation Equation

Through extensive laboratory feeding experiments on gadoids and other predatory fishes, (Andersen and Beyer 2005) established the mechanistic **cylinder model of gastric evacuation**.

Because digestive enzymes and hydrochloric acid are secreted by the stomach mucosa and act only on the **outer surface area** of the food bolus in the stomach (modeled as a cylinder of fixed length $l_s \propto L_{\text{pred}}$ and radius r, where volume/mass is $S \propto l_s r^2$ and surface area is $A_s \propto l_s r \propto S^{0.5}$), the instantaneous gastric evacuation rate (GER) of total stomach mass $S(t)$ (in $\text{g} \cdot \text{h}^{-1}$) follows a power law with exponent $\alpha = 0.5$ (Andersen and Beyer 2005, Eqs. 1–3):

$$\frac{dS(t)}{dt} = -\rho\, S(t)^{\alpha}, \qquad \alpha \approx 0.5 \tag{5.2}$$

where the rate parameter ρ scales with predator total length L (cm), ambient water temperature T (°C), and meal energy density E (kJ · g^{-1} wet weight):

$$\rho(L, T, E) = \rho_0\, L^{\lambda}\, \exp(\delta T)\, E^{-\mu} \tag{5.3}$$

Typical empirical values for North Sea and Baltic piscivores (cod, whiting, saithe; (Andersen and Beyer 2005, Table 1)) are:

- Surface-to-volume exponent: $\alpha = 0.50$,
- Predator length scaling: $\lambda \approx 1.30$–1.44 (since $\lambda + 3\alpha \approx 2.8$–$2.95$, matching the allometric scaling of maximum consumption $W^{0.75\text{–}0.8}$),
- Temperature coefficient: $\delta \approx 0.08$–$0.10\ ^{\circ}\mathrm{C}^{-1}$,
- Prey energy-density resistance exponent: $\mu \approx 0.85$–0.90 (lipid-rich, high-energy prey such as herring or sprat [$E \approx 6$–9 kJ · g^{-1}] are evacuated significantly more slowly in wet weight per hour than lean fish or crustaceans [$E \approx 3.5$–4.5 kJ · g^{-1}], so **rate of energy delivery** $E \cdot |dS/dt| \propto E^{1-\mu} \approx E^{0.12}$ is nearly invariant across prey types!).

5.2.2 Evacuation of Individual Prey Items in Mixed Meals

When a predator stomach contains K distinct prey items/categories with individual wet masses $S_1, S_2, \ldots, S_K$ and total stomach mass $S_{\text{tot}} = \sum_{k=1}^{K} S_k$, the evacuation rate of prey category i is proportional to its mass share S_i/S_{tot} in the cylinder multiplied by the total evacuation rate (Andersen and Beyer 2005, Eq. 6; Rindorf and Lewy 2004, Eq. 2):

$$\mathrm{GER}_i = -\frac{dS_i}{dt} = \rho_i \left(\frac{S_i}{S_{\text{tot}}}\right) S_{\text{tot}}^{\alpha} = \rho_i\, S_i\, S_{\text{tot}}^{\alpha-1} \tag{5.4}$$

where $\rho_i = \rho_0 L^{\lambda} e^{\delta T} E_i^{-\mu}$ accounts for the specific energy density and cuticle/exoskeleton resistance of prey i.

5.3 The Nonlinear Stomach-Pooling Bias (Rindorf & Lewy 2004)

In over a century of stomach sampling (including the 1981 and 1991 ICES North Sea Stomach Sampling Projects), stomachs of predators of the same species and length class caught in the same trawl haul were **physically pooled in a bucket aboard ship** to save laboratory processing time, and only the **mean stomach content weight** $\bar{S}_i = \frac{1}{n}\sum_{k=1}^{n} S_{i,k}$ across the n sampled predators (including empty stomachs) was recorded (Rindorf and Lewy 2004).

5.3.1 Proof of Jensen's Inequality Bias

Over a stationary feeding period of duration D, the true mean population consumption rate $\bar{C}_i$ per predator is the expectation of the individual evacuation rate across all individual stomachs $k \in \{1, \dots, n\}$:

$$\bar{C}_{\text{true},i} = D \cdot \mathbb{E}\left[\rho_i \, S_{i,k} \, S_{\text{tot},k}^{\alpha-1}\right] = \frac{D}{n} \sum_{k=1}^{n} \rho_i \, S_{i,k} \, S_{\text{tot},k}^{\alpha-1} \tag{5.5}$$

However, when stomachs are pooled prior to calculation, only the sample means $\bar{S}_i = \frac{1}{n}\sum_{k=1}^{n} S_{i,k}$ and $\bar{S}_{\text{tot}} = \frac{1}{n}\sum_{k=1}^{n} S_{\text{tot},k}$ are known, and the naive pooled estimator is:

$$\hat{C}_{\text{pooled},i} = D \, \rho_i \, \bar{S}_i \, \bar{S}_{\text{tot}}^{\alpha-1} \tag{5.6}$$

For total consumption ($S_i = S_{\text{tot}}$), $\bar{C}_{\text{true}} = D\rho \, \mathbb{E}[S_{\text{tot}}^{\alpha}]$ whereas $\hat{C}_{\text{pooled}} = D\rho \, (\mathbb{E}[S_{\text{tot}}])^{\alpha}$. Because $f(S) = S^{\alpha}$ is a **strictly concave function** for $0 < \alpha < 1$ ($\alpha \approx 0.5$), **Jensen's Inequality** guarantees that:

$$\mathbb{E}[S_{\text{tot}}^{\alpha}] < \Big(\mathbb{E}[S_{\text{tot}}]\Big)^{\alpha} \implies \hat{C}_{\text{pooled}} > \bar{C}_{\text{true}} \tag{5.7}$$

⚠ Jensen's Inequality Pooling Biases (Rindorf & Lewy 2004)

(Rindorf and Lewy 2004) demonstrated analytically and via bootstrap simulations on individual cod stomachs that:

1. **Total Consumption Overestimation**: Because individual fish stomachs are highly skewed (many empty or near-empty stomachs, and a few stomachs with recently ingested large fish), applying $\bar{S}_{\text{tot}}^{0.5}$ to pooled haul averages **overestimates total food consumption by** 25% **to** $>$ 100%!
2. **Prey-Composition Bias Toward Large-Meal Piscivory**: When a predator occasionally eats a large fish prey item A (high $S_{\text{tot},k}$, so $S_{\text{tot},k}^{\alpha-1} = S_{\text{tot},k}^{-0.5}$ is small) and frequently eats small benthic invertebrate meals B (low $S_{\text{tot},k}$, so $S_{\text{tot},k}^{-0.5}$ is large), the naive pooled formula $\hat{C}_{\text{pooled},i} \propto \bar{S}_i \bar{S}_{\text{tot}}^{-0.5}$ applies the **same** average total stomach weight $\bar{S}_{\text{tot}}^{-0.5}$ to both prey types. This **overestimates the diet share of large fish prey** and **underestimates the diet share of small invertebrate prey** (Rindorf and Lewy 2004, 2491–94)!

5.3.2 Second-Order Taylor Bias Correction (Rindorf & Lewy 2004)

When individual stomachs cannot be analyzed separately, (Rindorf and Lewy 2004, Eqs. 11–14) derived a second-order Taylor series correction for total consumption among non-empty stomachs (where p_+ is the proportion of non-empty stomachs in the sample and CV_+ is the coefficient of variation of stomach content weight among feeding fish):

$$\bar{C}_{\text{corrected}} \approx D\,\rho\,p_+^{1-\alpha}\,\bar{S}_{\text{tot}}^{\alpha}\left[1+\frac{\alpha(\alpha-1)}{2}\mathrm{CV}_+^2\right] \tag{5.8}$$

For $\alpha = 0.5$, $\frac{\alpha(\alpha-1)}{2} = -\frac{1}{8} = -0.125$, reducing the pooling bias by more than an order of magnitude when p_+ (fraction of non-empty stomachs) is recorded.

5.4 Prey Condition, Differential Digestion, and Dietary Metrics (Hyslop 1980; Buckland et al. 2017)

In empirical stomach content analysis, prey are quantified using one or more of four standard metrics (Hyslop 1980; Buckland et al. 2017, Table 1):

1. **Gravimetric** ($\%W$): Wet or dry mass of prey category i as a percentage of total stomach content mass.
2. **Volumetric** ($\%V$): Displaced volume, grid area, or visual points of prey i as a percentage of total volume.
3. **Numerical** ($\%N$): Count of prey individuals of category i as a percentage of total counted prey items.
4. **Frequency of Occurrence** ($\%F_i$): Percentage of sampled non-empty stomachs in which prey category i is present (binary presence/absence per stomach).

5.4.1 Biases Revealed by Prey Condition Grading (Buckland et al. 2017)

By grading stomach contents across **7 digestion states** (from State 1 = intact freshly ingested prey to State 6 = isolated hard parts/otoliths/chaetae and State 7 = unidentifiable organic mush and mucus; (Buckland et al. 2017, Table A1)), Buckland et al. demonstrated that:

- **Unidentifiable Matter & Mucus Adhesion**: Unidentifiable organic tissue and stomach mucus were present in 33–77% of stomachs and physically adhered to small prey in $> 61\%$ of stomachs, making direct separation for $\%W$ and $\%V$ imprecise.

- **Differential Retention of Hard Parts**: Soft-bodied prey (polychaetes, small teleost larvae, nematodes) digest rapidly into unidentifiable mush, whereas crustacean carapaces, bivalve shells, and large teleost otoliths persist in stomachs for days. Raw $\%W$ of partially digested remains can either severely underestimate teleost importance (if only a tiny otolith remains) or overestimate shelled prey (if indigestible shells are weighed).
- **Prey Weight Reconstruction vs. Digested Stomach Weight**:
  - For size-preference estimation (e.g., fitting the `mizer` or `SMS` predation kernel $\phi(w_{\text{pred}}/w_{\text{prey}})$), one requires the **fresh ingested body weight** $w_{\mathbf{prey}}$ of the prey at the moment of capture, which must be reconstructed from intact prey lengths or otolith/carapace allometric regressions ($w_{\text{prey}} = aL^b_{\text{otolith}}$).
  - Conversely, for evacuation-rate calculation (`SMS` / `FishStomachs`), one requires the **actual digested wet mass** S_i **currently in the stomach** to plug into $\rho_i S_i S_{\text{tot}}^{\alpha-1}$! Recording *both* digested mass S_i and reconstructed ingested size/mass w_{prey} (as in the ICES and DAPSTOM databases) is therefore essential.
- **Robustness of Hierarchical** $\%F$: When comparing across studies with variable digestion states, `%F` (reported in a nested taxonomic hierarchy from Class → Order → Family → Genus → Species; (Buckland et al. 2017, Table B1)) is invariant to mucus/evaporation weight distortion and partial fragmentation.

5.5 Stomach Content Databases and Spatiotemporal Raising Pipelines (Jacobsen et al. 2023)

5.5.1 Major Open-Access Stomach Content Databases

Two primary databases underpin North Atlantic and European multispecies stock assessments (ICES 2010; Jacobsen et al. 2023), complemented by global predator–prey body-size compilations (Barnes et al. 2008, 2010):

1. **ICES Stomach Content Database** (`stomachdata.ices.dk`; (ICES 2010)): Contains $> 800,000$ prey records from the North Sea (1981 & 1991 "Years of the Stomach" plus IBTS collections), Baltic Sea (cod and flounder stomachs 1963–2021+), and Barents/Celtic Seas. Organized into four normalized relational tables linked by `tblUploadID` / `tblHaulID` / `tblPredatorID` (Jacobsen et al. 2023, 13–17):
   - `File_Information (FI)`: Upload metadata, country, institute, evacuation parameters.
   - `Haul_Information (HH)`: ICES survey, gear, ship, haul duration, latitude/longitude, ICES statistical rectangle, bottom depth, bottom temperature T, year, month, time of day.

- `Predator_Information (PI)`: Predator species (`AphiaID`), predator length class L, individual or pooled sample flag, number of feeding stomachs n_+, number of regurgitated stomachs n_{reg}, number of empty stomachs n_0, mean predator weight.
- `Prey_Information (PP)`: Prey species (`AphiaID`), digestion stage, prey length class l, prey count N_{prey}, total observed wet weight S_i in stomach, and gravimetric method.

2. **CEFAS `DAPSTOM` Database** (Pinnegar 2015; Jacobsen et al. 2023, 10–12; Morgan 2022): Contains $> 225,000$ stomach records from 1836 to 2016 across Northwest European shelf seas, providing individual predator-prey length and mass ratios (PPMR) across a 180-year historical baseline.

5.5.2 Design-Based Stratified Raising (`FishStomachs` R Package)

Because research vessels sample a fixed number of stomachs per length class per haul (e.g., at most 5 cod per 5-cm length bin per haul) rather than sampling proportionally to true predator density in the sea, simple arithmetic averages across all sampled stomachs are **spatially biased** toward areas with low predator density.

The ICES standard raising pipeline—implemented in Morten Vinther's **`FishStomachs`** R package (Jacobsen et al. 2023, 34–38)—raises individual/haul stomach observations to the population level in three hierarchical steps:

1. **Step 1 (Regurgitation Correction & Evacuation Conversion at Haul Level h)**: Within each haul h and predator length class L, regurgitated stomachs ($n_{\text{reg},h,L}$) are assumed to have had the same mean diet as feeding stomachs ($n_{+,h,L}$), so observed weights are scaled by $\frac{n_{+,h,L}+n_{\text{reg},h,L}}{n_{+,h,L}(n_{+,h,L}+n_{\text{reg},h,L}+n_{0,h,L})}$ and converted from stomach mass $S_{i,h,L}$ to evacuated ration rate $R_{i,h,L}$ using the (Andersen and Beyer 2005) cylinder equation and (Rindorf and Lewy 2004) pooling bias correction.
2. **Step 2 (Haul-to-Rectangle Raising via √Stomachs or CPUE)**: Within each ICES statistical rectangle r, haul rations $R_{i,h,L}$ are weighted by $\sqrt{n_{\text{stom},h,L}}$ (or haul predator CPUE) to obtain the rectangle mean ration $R_{i,r,L} = \frac{\sum_{h\in r} w_{h,L} R_{i,h,L}}{\sum_{h\in r} w_{h,L}}$.
3. **Step 3 (Rectangle-to-Stock Raising via Predator Survey Abundance $I_{r,L}$)**: Across all rectangles $r \in \mathcal{R}$ in the stock assessment area, rectangle rations are weighted by the **survey abundance index $I_{r,L}$ of predators of length L in rectangle r**:

$$\bar{R}_{\text{pop},i,L} = \frac{\sum_{r\in\mathcal{R}} I_{r,L}\, R_{i,r,L}}{\sum_{r\in\mathcal{R}} I_{r,L}}, \qquad \hat{W}_{\text{pop},i,L} = \frac{\bar{R}_{\text{pop},i,L}}{\sum_k \bar{R}_{\text{pop},k,L}} \tag{5.9}$$

5.5.3 Model-Based Spatiotemporal Standardization (`sdmTMB`)

When stomach sampling is patchy in space and time (missing ICES rectangles in certain quarters/years), design-based raising suffers from missing-stratum bias. (Jacobsen et al. 2023, 26–33) developed a **model-based spatiotemporal standardization workflow** using **`sdmTMB`** (Anderson et al. 2022):

1. Define the **Feeding Ratio (FR)** of prey i in individual predator k of length L_k at spatial coordinates $\mathbf{s}_k$ and year-quarter t as:

$$\mathrm{FR}_{i,k} = \frac{S_{i,k}}{W_{\mathrm{pred},k}} \tag{5.10}$$

2. Fit a spatial/spatiotemporal Generalized Linear Mixed Model (GLMM) with a **Tweedie** distribution (power variance $V(\mu) = \phi\mu^p$ with $1 < p < 2$, which naturally models continuous positive ratios with a point mass at exact zero $\mathrm{FR}_{i,k} = 0$!) and log link:

$$\mathbb{E}[\mathrm{FR}_{i,k}] = \mu_{i,k}, \qquad \ln \mu_{i,k} = \mathbf{X}_k\beta + s(L_k) + \omega_i(\mathbf{s}_k) + \epsilon_i(\mathbf{s}_k, t) \tag{5.11}$$

   where $s(L_k)$ is a smooth spline of predator length, $\omega_i(\mathbf{s}) \sim \mathrm{MVN}(\mathbf{0}, \Sigma_\omega)$ is a time-invariant spatial Gaussian Markov Random Field (GMRF) approximated via SPDE on a finite-element mesh, and $\epsilon_i(\mathbf{s}, t)$ is a spatiotemporal random field.
3. Predict $\hat{\mu}_i(\mathbf{s}_g, L, t)$ across a fine spatial prediction grid $g \in \mathcal{G}$, weight by spatially predicted predator density $\hat{D}_{\mathrm{pred}}(\mathbf{s}_g, L, t)$ (also estimated via `sdmTMB` on trawl survey catches), and normalize across prey types i to obtain spatially unbiased population diet proportions and their standard errors!

5.6 Statistical Observation Models and Likelihoods for Stomach Content Data

Once population-level diet proportions $\mathbf{W}_U = (W_{1,U}, \ldots, W_{K,U})^\top$ (satisfying $\sum_{k=1}^{K} W_{k,U} = 1$) or individual predator-prey mass ratios (PPMR) are prepared for predator stratum $U = (A, L, y, q)$, how should they enter the stock assessment objective function?

5.6.1 Summary Comparison of Compositional Diet Likelihoods

The four principal likelihood formulations for fitting multispecies stomach composition data are summarized in Table 5.1:

Table 5.1: Statistical Comparison of Compositional Diet and Stomach Content Likelihood Formulations

Likelihood Distribution	Data Support	Log-Likelihood $\ln L(\mathbf{W}_U \mid \hat{\mathbf{p}}_U, \theta)$	Variance Structure $\mathrm{Var}(W_k)$	Handling of Zeros ($W_k = 0$)	Used In
1. Multinomial	Proportions $\mathbf{W}_U \in [0,1]^K$ scaled by sample size N_U	$N_U \sum_{k=1}^{K} W_{k,U} \ln \hat{p}_{k,U} + \text{const}$	$\frac{\hat{p}_k(1-\hat{p}_k)}{N_U}$ (Fixed by N_U; no free dispersion parameter)	**Natural**: $W_{k,U} = 0$ simply contributes $0 \cdot \ln \hat{p}_{k,U} = 0$!	`MSCAA`, `CEATTLE`, `Gadget`, `SMS` (Option A)
2. Dirichlet	Strictly positive simplex $W_{k,U} \in (0,1)$, $\sum_k W_{k,U} = 1$	$\ln \Gamma(v_U) - \sum_{k=1}^{K} \ln \Gamma(v_U \hat{p}_{k,U}) + \sum_{k=1}^{K} (v_U \hat{p}_{k,U} - 1) \ln W_{k,U}$	$\frac{\hat{p}_k(1-\hat{p}_k)}{v_U+1}$ (Estimable precision $v_U = N_U/\theta - 1$)	**Undefined at** $W_k = 0$ ($\ln 0 = -\infty$); requires pooling zero bins or tail compression	`SMS` (Default), `EcoState`, `MSCAA`
3. Dirichlet-Multinomial (DM)	Count vector $N_U \mathbf{W}_U$ with overdispersion $\beta = \theta N_U$	$\ln \Gamma(N_U + 1) + \ln \Gamma(\beta) - \ln \Gamma(N_U + \beta) + \sum_{k=1}^{K} \left[\ln \frac{\Gamma(N_U W_k + \beta \hat{p}_k)}{\Gamma(N_U W_k + 1) \Gamma(\beta \hat{p}_k)} \right]$	$\frac{\hat{p}_k(1-\hat{p}_k)}{N_U} \left(\frac{N_U+\beta}{1+\beta} \right)$ (Linear or saturating parameterization (Thorson et al. 2017))	**Natural**: Exact handling of $W_{k,U} = 0$ AND estimable overdispersion θ!	`SS3`, `LIME`, `WHAM`, modern `Rceattle`
4. Additive Logistic-Normal (Aitchison)	Log-ratios $y_k = \ln(W_k/W_K) \in \mathbb{R}^{K-1}$	$-\frac{1}{2} \ln \lvert 2\pi \Sigma_U \rvert - \frac{1}{2} (\mathbf{y}_U - \hat{\mu}_U)^\top \Sigma_U^{-1} (\mathbf{y}_U - \hat{\mu}_U) - \sum_{k=1}^{K} \ln W_{k,U}$	Arbitrary $(K-1) \times (K-1)$ covariance Σ_U (allows positive/negative correlations!)	Requires zero replacement or censored Tobit normal integration	`SAM` (compositions), `gadget3` (`mvlogistic`)

5.6.2 Mathematical Derivation of the Compound Dirichlet-Multinomial Likelihood (Thorson et al. 2017)

Because compositional observations in both age/length sampling and stomach content analysis exhibit strong **overdispersion** relative to a pure multinomial draw of nominal sample size n, (Thorson et al. 2017) derived a model-based compound **Dirichlet-Multinomial (DM)**

likelihood that estimates effective sample size n_{eff} simultaneously with all stock assessment parameters.

1. **Hierarchical Compound Integral Formulation**: Let $\mathbf{p} = (p_1, \ldots, p_A)$ denote the assessment model's expected proportion vector across A categories ($\sum_{a=1}^{A} p_a = 1$), and suppose that due to **process error** (unmodeled spatial/temporal variation in cohort availability or predator foraging) or **clustered sampling** (schooling of fish within trawl hauls), the true local proportion vector $\mu = (\mu_1, \ldots, \mu_A)$ encountered during sampling is drawn from a Dirichlet distribution with expectation $\mathbf{p}$ and concentration sum $\beta = \sum_{a=1}^{A} \beta p_a > 0$, followed by a multinomial sample of size n yielding observed proportions $\tilde{\pi} = (\tilde{\pi}_1, \ldots, \tilde{\pi}_A)$ (Thorson et al. 2017, Eqs. 2–5, p. 86):

$$\Pr(\tilde{\pi} \mid \mathbf{p}, n, \beta) = \int \text{Multinomial}(n\tilde{\pi} \mid \mu, n)\, \text{Dirichlet}(\mu \mid \beta\mathbf{p})\, d\mu \tag{5.12}$$

   Evaluating this Dirichlet integral analytically yields the closed-form **Dirichlet-Multinomial log-likelihood**:

$$\ln f_{\text{DM}}(\tilde{\pi} \mid \mathbf{p}, n, \beta) = \ln\Gamma(n+1) - \sum_{a=1}^{A} \ln\Gamma(n\tilde{\pi}_a + 1) + \ln\Gamma(\beta) - \ln\Gamma(n+\beta) + \sum_{a=1}^{A} \Big[\ln\Gamma(n\tilde{\pi}_a + \beta p_a) - l \tag{5.13}$$

   Notice that when $\tilde{\pi}_a = 0$, the term inside the final sum is $\ln\Gamma(\beta p_a) - \ln\Gamma(\beta p_a) = 0$, so **exact observed zeros require no ad hoc constant addition or bin pooling**!

2. **Two Parameterizations of Effective Sample Size (n_{eff})**: The marginal variance of an observed proportion $\tilde{\pi}_a$ under the DM distribution is $\text{Var}(\tilde{\pi}_a) = \frac{p_a(1-p_a)}{n}\left(\frac{n+\beta}{1+\beta}\right)$. Equating this to the equivalent multinomial variance $\frac{p_a(1-p_a)}{n_{\text{eff}}}$ yields $n_{\text{eff}} = \frac{n+n\beta}{n+\beta}$. (Thorson et al. 2017, Eqs. 9–12, p. 86) introduced two distinct parameterizations of how n_{eff} scales with input sample size n_y across years:

   - **Parameterization #1 (Linear Scaling, $\beta_y = \theta n_y$, Default)**:

$$n_{\text{eff},y} = \frac{1+\theta n_y}{1+\theta} = \frac{1}{1+\theta} + n_y\left(\frac{\theta}{1+\theta}\right) \tag{5.14}$$

     As $\theta \to \infty$, $n_{\text{eff},y} \to n_y$ (recovering the unweighted multinomial); for small θ and large n_y, $\frac{\theta}{1+\theta} \in (0,1)$ acts directly as a **multiplicative data-weighting factor** λ (analogous to McAllister–Ianelli or Francis `TA1.8` reweighting), suitable when overdispersion arises from **clustered/schooling sampling**.
   - **Parameterization #2 (Saturating Scaling, $\beta_y = \beta$ constant)**:

$$n_{\text{eff},y} = \frac{n_y + n_y\beta}{n_y + \beta} \xrightarrow{n_y \gg \beta} 1 + \beta \tag{5.15}$$

     Here $n_{\text{eff},y}$ increases linearly with n_y at small sample sizes ($n_y \ll \beta$) but **asymptotes at an upper ceiling** $1 + \beta$ as $n_y \to \infty$, which is theoretically exact

when overdispersion arises from **process error / model misspecification** in selectivity or availability (because measuring infinitely many fish in year y cannot reduce year-specific process error $\mathrm{Var}(\mu_a) = \frac{p_a(1-p_a)}{1+\beta}$!).

[!TIP] **Choosing Between Dirichlet, Dirichlet-Multinomial, and Logistic-Normal for Multispecies Assessments**: While the **Dirichlet** distribution (used in `SMS` and `EcoState`) works well for aggregated species-level diet shares where all modeled prey are present ($W_{k,U} > 0$), when diet data are disaggregated simultaneously by **prey species AND prey size bins** (as required for `Gadget` or dynamic `mizer`), many (i, l) prey bins have observed $W_{k,U} = 0$. In that regime, the **Dirichlet-Multinomial** (Thorson et al. 2017) is strictly superior to the plain Dirichlet because $W_{k,U} = 0$ is in the natural support of the distribution while still estimating θ_{stom} via maximum likelihood. Note, however, that both Dirichlet and DM imply strictly negative correlations $\mathrm{Corr}(\tilde{\pi}_a, \tilde{\pi}_{a'}) < 0$ between bins; if positive correlations between adjacent length bins are strong, the **Logistic-Normal** or **Tweedie spatial GLMM** (`sdmTMB`) should be used.

5.7 Fitting Exponential Family & Gamma Distributions to Predator–Prey Mass Ratios (Morgan 2022)

For continuous size-spectrum models like `mizer`, stomach content data provide a second, uniquely powerful source of information beyond species diet shares: the **empirical distribution of individual Predator–Prey Mass Ratios (PPMR)**, $\mathrm{PPMR} = w_{\text{pred}}/w_{\text{prey}}$.

In his University of York dissertation analyzing $> 300,000$ stomach records from the combined CEFAS `DAPSTOM` and ICES databases, Owain Morgan (Morgan 2022) derived an elegant analytical link between **realized stomach log-PPMR distributions** and the **intrinsic `mizer` feeding kernel** $\varphi_{w'}(w)$ using the multiplicative closure properties of the exponential family.

5.7.1 Linking Realized Diet Size Distribution to the Feeding Kernel via the Sheldon Spectrum

Let w' denote the body mass of the predator and w denote the body mass of an individual prey item in its stomach, so the logarithmic predator-prey mass ratio is:

$$x = \ln\left(\frac{w'}{w}\right) \iff w = w' e^{-x} \tag{5.16}$$

In `mizer`, the realized number density $I_{w'}(w)$ of prey of weight w consumed by a predator of weight w' is proportional to the product of the predator's intrinsic **feeding preference**

kernel $\varphi_{w'}(w) = \phi(w'/w) = \phi(e^x)$ and the **ambient prey abundance spectrum** $N(w)$ in the sea (Morgan 2022, Eq. 1):

$$I_{w'}(w) \propto \varphi_{w'}(w)\, N(w) \tag{5.17}$$

By the Sheldon spectrum power law (Morgan 2022, Eq. 2), background marine abundance density scales as $N(w) = n_0 w^{-\lambda}$ (with $\lambda \approx 2$ for number density per unit weight, or $wN(w) \propto w^{1-\lambda} = w^{-1}$ per logarithmic weight bin $dx = -dw/w$). Substituting $w = w'e^{-x}$ into $N(w)$ gives:

$$N(w'e^{-x}) = n_0 (w')^{-\lambda} \exp(\lambda x) \tag{5.18}$$

Therefore, in terms of the log-PPMR variable $x = \ln(w'/w)$, the **realized stomach log-PPMR probability density** $f_{\text{real}}(x)$ and the **intrinsic feeding preference kernel** $\phi(x)$ are related by an **exponential tilt factor** $e^{\tilde{\lambda}x}$ (where $\tilde{\lambda} = \lambda - 1 \approx 1$ for number counts in log-bins, or $\tilde{\lambda} = \lambda - 2 \approx 0$ for biomass shares in log-bins):

$$f_{\text{real}}(x) \propto \phi(x)\, \exp(\tilde{\lambda}x) \iff \phi(x) \propto f_{\text{real}}(x)\, \exp(-\tilde{\lambda}x) \tag{5.19}$$

5.7.2 Exact Multiplicative Closure of Exponential Family & Gamma Distributions (Morgan 2022, Section 4.6)

(Morgan 2022, 7–9) showed that if the realized log-PPMR $X = \ln(w'/w) > 0$ follows any distribution in the natural **Exponential Family** with sufficient statistic vector $T(x)$ containing x:

$$f_{\text{real}}(x \mid \eta) = h(x) \exp(\eta^\top T(x) - A(\eta)) \tag{5.20}$$

then multiplying by the Sheldon abundance correction $\exp(-\tilde{\lambda}x)$ simply shifts the canonical parameter corresponding to x by $-\tilde{\lambda}$, keeping the intrinsic feeding kernel $\phi(x)$ within the **exact same parametric distribution family**!

Specifically:

1. **Normal Distribution on $x = \ln(w'/w)$ (Lognormal on w'/w, as in standard `mizer`)**: If $f_{\text{real}}(x) = \mathcal{N}(x \mid \mu_{\text{real}}, \sigma^2)$, then multiplying by $e^{-\tilde{\lambda}x}$ yields another Normal distribution with the **same variance** σ^2 and a shifted mean:

$$\phi(x) = \mathcal{N}\big(x \,\big|\, \mu_{\text{pref}} = \mu_{\text{real}} - \tilde{\lambda}\sigma^2,\ \sigma^2\big) \tag{5.21}$$

2. **Gamma Distribution on $x = \ln(w'/w)$ (Morgan 2022, Eqs. 19–24)**: Because empirical log-PPMR distributions in `DAPSTOM` and ICES stomachs are positively supported ($x = \ln(w'/w) > 0$, since predators rarely eat prey heavier than themselves) and right-skewed (sharp gape-limitation cutoff at small $x \approx 2$–3, long tail toward

small benthic/planktonic prey at large $x \approx 10$–15), (Morgan 2022) fitted a **Gamma distribution** to $x = \ln(w'/w)$:

$$f^{\Gamma}_{\alpha,\beta}(x) = \frac{\beta^{\alpha}}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}, \qquad x > 0 \tag{5.22}$$

When multiplied by $e^{-\tilde{\lambda}x}$, the Gamma density satisfies the exact algebraic identity (Morgan 2022, Eq. 24):

$$e^{-\tilde{\lambda}x} f^{\Gamma}_{\alpha,\beta}(x) = \left(\frac{\beta}{\beta + \tilde{\lambda}}\right)^{\alpha} f^{\Gamma}_{\alpha,\ \beta+\tilde{\lambda}}(x) \tag{5.23}$$

Thus, if the observed log-PPMR in stomachs is Gamma$(\hat{\alpha}, \hat{\beta})$ (with empirical estimates across North Sea cod, haddock, whiting, plaice, and sole clustering around $\hat{\alpha} \approx 6.8$–12.3, $\hat{\beta} \approx 1.15$–1.85; (Morgan 2022, Table 1)), the underlying `mizer` feeding kernel is analytically a **Gamma**$(\hat{\alpha},\ \hat{\beta} + \tilde{\lambda})$ distribution on $\ln(w'/w)$!

6 Statistical Estimation, Data Weighting, Uncertainty Quantification, and an Architectural Roadmap for Dynamic `mizer` Stock Assessment

6.1 Master Comparative Synthesis Across Multispecies Stock Assessment Model Families

A comprehensive architectural comparison across the eight multispecies stock assessment model families is presented in Table 6.1:

Table 6.1: Comprehensive Comparative Synthesis of Multispecies Stock Assessment Model Frameworks

Feature / Dimension	`MSVPA` / `MSFOR`	`SMS` / `smsR`	`MSCAA` / `Hydra`	`CEATTLE` / `Rceattle`	`Gadget2` / `gadget3`	`LeMans` / `LeMaRns`	`EwE` / `Rpath` / `EcoState`	`mizer` (Standard & Jump-Growth)
State Space	Discrete Age (s, a, y, q)	Discrete Age (s, a, y, q) (+ ALK for stomachs)	Discrete Age (s, a, y) (+ size bins in `Hydra`)	Discrete Age (s, a, y) (sex-structured)	Joint Age-Length (s, a, l, r, t) + Cell Mean Weight $W_{s,a,l,r,t}$	Discrete Length (s, l, t) (no age tracking)	Biomass Pools $B_i(t)$ + Multi-Stanza Age Groups	Continuous Body Mass Density $N_i(w, t)$

Feature / Dimension	`MSVPA` / `MSFOR`	`SMS` / `smsR`	`MSCAA` / `Hydra`	`CEATTLE` / `Rceattle`	`Gadget2` / `gadget3`	`LeMans` / `LeMaRns`	`EwE` / `Rpath` / `EcoState`	`mizer` (Standard & Jump-Growth)
Governing Equations	Backward VPA difference eqs.	Forward seasonal exponential survival	Forward SCAA difference eqs.	Forward SCAA difference eqs.	Sequential matrix operators (Migration, Predation, Beta-Binomial Growth)	Matrix projection with fractional bin growth $\phi_{i,j}$	Coupled Foraging Arena ODEs	McKendrick–von Foerster PDE (or Jump-Growth Fokker–Planck PDE)
Trophic Feedback	**One-Way** ($M2$ only; fixed $w_{a,t}$ and ration $R_{a,q}$)	**One-Way** ($M2$ only; fixed $w_{a,t}$ and evacuation ration $R_{a,q}$)	**One-Way** ($M2$ only; fixed $w_{a,t}$)	**One-Way** or **Bioenergetic** (T-dependent ration & $w_{a,t}$)	**Two-Way** ($M2$ + consumption-driven $\Delta L_l, \Delta W_l$)	**One-Way** ($M2$ only; fixed von Bertalanffy growth)	**Two-Way** (Prey $M2 \leftrightarrow$ Predator production $g_i \sum_j Q_{ji}$)	**Full Two-Way** (Encounter $\rightarrow$ $f_i(w) \rightarrow g_i(w), R_{p,i}, \mu_{s,i} \leftrightarrow \mu_{p,j}(w_p)$)
Predation Functional Response	Holling Type II (fixed total ration)	Holling Type II (fixed total ration)	Holling Type I, II, III, or Predator Interference	Holling Type II/III + Wisconsin Bioenergetics $f(T)$	Holling Type II or III (`g3a_predate_catchability_type`)	Holling Type II or III	**Foraging Arena** $Q_{ij} = \frac{a_{ij} v_{ij} B_i B_j}{2v_{ij} + a_{ij} B_j}$	Holling Type II (or Predator III) with search rate $\gamma_i w^q$

Feature / Dimension	`MSVPA` / `MSFOR`	`SMS` / `smsR`	`MSCAA` / `Hydra`	`CEATTLE` / `Rceattle`	`Gadget2` / `gadget3`	`LeMans` / `LeMaRns`	`EwE` / `Rpath` / `EcoState`	`mizer` (Standard & Jump-Growth)
Somatic Growth Dispersion	None (empirical $w_{a,y,q}$)	None (empirical $w_{a,y,q}$ + ALK)	None (empirical $w_{a,y}$)	Empirical or bioenergetic $w_{a,y}$	**Beta-Binomial** jump distribution (`g3a_grow_impl_bbinom`)	Deterministic upwind fraction $\phi_{i,j}$	Multi-stanza von Bertalanffy	Upwind numerical diffusion (Standard) or **Jump-Growth PDE** $d_i(w)$
Stomach / Diet Likelihood	Exact empirical iterative matching	**Dirichlet** (with $V_U = \theta/N_{\text{haul}}$) or Multinomial	Multinomial, Dirichlet, or Lognormal	Multinomial or Dirichlet	Weighted Sum of Squares or Multinomial on length-prey ratios	History Matching / SMC implausibility	Fixed Ecopath $DC_{ij,0}$ (`EwE`) or **Dirichlet** (`EcoState`)	Steady-state calibration or **Gamma/Normal log-PPMR** (Morgan 2022)
Catch & Survey Likelihoods	Exact catch (no error)	Lognormal $C_{s,a,y,q}$ & Lognormal CPUE $I_{f,s,a,y}$	Lognormal total catch/index + Multinomial age comps	Lognormal catch/index + Multinomial/DM age/length comps	Weighted composite likelihoods (`g3l_distribution_sp`)	History Matching + SMC against subset biomass	SSQ (`Ecosim`) or Lognormal state-space (`EcoState`)	Steady-state yield/biomass SSQ or GA/MCMC (Zhang et al. 2015)
Software / AD Engine	Fortran / R	**ADMB** (`SMS`) & **TMB** (`smsR`)	**ADMB / TMB**	**TMB** (`Rceattle`)	C++ (`Gadget2`) & **TMB/R** (`gadget3`)	R / Rcpp (`LeMaRns`)	C#/.NET (`EwE`), R (`Rpath`), **TMB** (`EcoState`)	R / Rcpp (`mizer`)

Feature / Dimension	`MSVPA` / `MSFOR`	`SMS` / `smsR`	`MSCAA` / `Hydra`	`CEATTLE` / `Rceattle`	`Gadget2` / `gadget3`	`LeMans` / `LeMaRns`	`EwE` / `Rpath` / `EcoState`	`mizer` (Standard & Jump-Growth)
Key Reference	(Helgason and Gislason 1979; Livingston 1983)	(Lewy and Vinther 2004)	(Kinzey and Punt 2009; Van Kirk et al. 2010; Curti et al. 2013; Trijoulet et al. 2019)	(Holsman et al. 2016; Adams et al. 2022)	(Stefánsson and Pálsson 1998; Begley and Howell 2004; Kulatska et al. 2019)	(Spence et al. 2020)	(Walters et al. 1997; Thorson et al. 2024)	(Scott et al. 2014; Pennino et al. 2025)

6.2 Data Weighting and Likelihood Conflict in Integrated & Multispecies Assessments (Francis 2011; Maunder & Punt 2013; Thorson et al. 2017)

When fitting an integrated single-species or multispecies model simultaneously to **abundance indices** ($I_{f,y}$), **catch biomass** ($Y_{f,y}$), **length/age compositions** ($\mathbf{P}_{f,y}$), and **stomach content compositions** ($\mathbf{W}_{A,L,y,q}$), the most critical statistical hazard is **likelihood conflict and compositional domination** (Francis 2011; Maunder and Punt 2013; Thorson et al. 2017).

6.2.1 Why Unadjusted Composition Likelihoods Destroy Abundance Fits ("Francis Principle 1")

Suppose a survey measures $10,000$ fish lengths across 50 length bins each year, while also providing 1 annual biomass index point I_y per year. If the nominal number of fish measured ($N_y = 10,000$) is plugged into a Multinomial likelihood $\sum_y N_y \sum_l P_{l,y} \ln \hat{P}_{l,y}$, the composition likelihood will have $1000\times$ larger gradient and curvature than the lognormal survey index likelihood $\frac{1}{2\sigma_I^2}(\ln I_y - \ln \hat{I}_y)^2$.

! Francis's First Principle of Data Weighting

Because fish school by size and models have minor selectivity misspecifications, positive correlations among bins within a year mean the true **effective sample size** $N_{\text{eff},y}$ is typically 20–200 (closer to the number of **trawl hauls**, not individual fish!). If nominal sample size $N_y \sim 10,000$ is not downweighted, the optimizer will distort recruitment R_y, fishing mortality F_y, and biomass trends $\hat{B}_y$ just to chase noise in length compositions, violating **Francis's First Principle** (Francis 2011, 1125):

> *"Do not let other data stop the model from fitting abundance data well."*

6.2.2 Three Standard Data-Weighting Algorithms for Compositional Data

Let $O_{l,y}$ and $E_{l,y}$ denote the observed and model-expected proportions in length (or age) bin $l \in \{1, \dots, n_L\}$ with midpoint x_l in year $y \in \{1, \dots, n_Y\}$, with initial nominal sample size N_y.

1. **McAllister & Ianelli (1997) Harmonic Mean Weighting**: Estimates the effective sample size $\hat{N}_y = \frac{\sum_l E_{l,y}(1-E_{l,y})}{\sum_l (O_{l,y}-E_{l,y})^2}$ for each year and sets the dataset multiplier w_{MI} using the harmonic mean across years (used in `MSCAA` (Curti et al. 2013) and `Rceattle` (Adams et al. 2022); see (McAllister and Ianelli 1997)).

   - *Critique (Francis 2011, 1128–31)*: Assumes residuals across bins l within the same year y are uncorrelated. When residuals are positively correlated across adjacent length bins (the norm when growth or selectivity varies), McAllister–Ianelli **dramatically overweights** the composition data!

2. **Francis (2011) Method `TA1.8` (Mean-Length / Mean-Age Residual Variance)**: Collapses each year's length composition into its **observed mean length** $\bar{O}_y = \sum_l x_l O_{l,y}$ and **expected mean length** $\bar{E}_y = \sum_l x_l E_{l,y}$, with multinomial within-year variance $v_y = \sum_l x_l^2 E_{l,y} - \bar{E}_y^2$. Because collapsing to the annual mean $\bar{O}_y$ absorbs all within-year inter-bin correlations, the standardized mean-length residuals $\frac{\bar{O}_y - \bar{E}_y}{\sqrt{v_y/(wN_y)}}$ must have unit variance, yielding the **Francis `TA1.8` weight multiplier** (Francis 2011, Appendix A, Table A1):

$$w_{\text{Francis}} = \left[\text{Var}_y \left(\frac{\bar{O}_y - \bar{E}_y}{\sqrt{v_y / N_y}} \right) \right]^{-1} \tag{6.1}$$

3. **Self-Weighting Likelihoods (Dirichlet-Multinomial / Logistic-Normal / SMS Dirichlet)**: Rather than outer fixed-point reweighting loops (`Francis TA1.8` or `Gadget` two-stage reweighting), modern TMB and SS3 models (`SMS`, `EcoState`, `LIME`

(Rudd and Thorson 2018), `SS3` (Thorson et al. 2017)) use a **Dirichlet** or **Dirichlet-Multinomial** likelihood with an estimable dispersion parameter θ, where effective sample size is either linear in N_y:

$$N_{\text{eff},y} = \frac{1 + \theta N_y}{1 + \theta} \tag{6.2}$$

or saturating at $1 + \beta$:

$$N_{\text{eff},y} = \frac{N_y + N_y \beta}{N_y + \beta} \tag{6.3}$$

In a 4,500-replicate simulation study and Pacific hake application, (Thorson et al. 2017) proved that simultaneous DM estimation eliminates the severe bias of unweighted multinomial likelihoods, matches the accuracy and precision of iterative McAllister–Ianelli tuning for M, $\ln R_0$, and spawning output, and propagates data-weighting uncertainty directly into the joint Hessian and MCMC posterior!

6.3 Parameter Estimation and Uncertainty Quantification Engines

6.3.1 Frequentist / Empirical Bayes State-Space Estimation in `TMB` (Kristensen et al. 2016)

Modern statistical stock assessment models (`SAM` (Nielsen and Berg 2014), `smsR`, `Rceattle`, `gadget3`, `EcoState` (Thorson et al. 2024)) are built on **Template Model Builder (`TMB`)** (Kristensen et al. 2016) or its pure-R interface `RTMB`.

Let $\theta \in \mathbb{R}^p$ denote the vector of **fixed effects** (life-history scaling parameters, selectivities, catchabilities, suitability parameters β_i, σ_i, and variance parameters $\sigma_{\text{proc}}, \sigma_{\text{obs}}$) and let $\mathbf{u} \in \mathbb{R}^m$ denote the vector of **latent random effects (process errors)** (e.g., annual recruitment deviations $\epsilon_{R,i,y}$, time-varying fishing mortality random walks $\ln F_{f,y}$, or biomass process errors $\epsilon_{B,i,y}$). Given joint negative log-likelihood $f(\theta, \mathbf{u}) = -\ln p(\mathbf{y}, \mathbf{u} \mid \theta)$, TMB approximates the marginal likelihood $L(\theta) = \int \exp(-f(\theta, \mathbf{u}))\, d\mathbf{u}$ via the **Laplace Approximation** (Kristensen et al. 2016, Eq. 2):

$$-\ln L^*(\theta) = f(\theta, \hat{\mathbf{u}}(\theta)) + \frac{1}{2} \ln \det \mathbf{H}(\theta) - \frac{m}{2} \ln(2\pi) \tag{6.4}$$

where:

- $\hat{\mathbf{u}}(\theta) = \arg\min_{\mathbf{u}} f(\theta, \mathbf{u})$ is the inner-loop empirical Bayes mode of the random effects, and
- $\mathbf{H}(\theta) = \left. \frac{\partial^2 f(\theta, \mathbf{u})}{\partial \mathbf{u} \partial \mathbf{u}^\top} \right|_{\mathbf{u} = \hat{\mathbf{u}}(\theta)}$ is the sparse $m \times m$ Hessian matrix of the random effects, factored via sparse Cholesky decomposition (`CHOLMOD`).

Using up to third-order automatic differentiation (`CppAD`), TMB computes the exact gradient $\nabla_\theta(-\ln L^*(\theta))$ of the Laplace-approximated marginal likelihood in $O(1)$ reverse-mode sweeps, allowing `nlminb` or `optim` to estimate hundreds of fixed effects and thousands of random effects in seconds to minutes, followed by joint precision matrix inversion (`sdreport`) for derived quantities ($\mathrm{SSB}_{i,y}, M2_{i,a,y}, F_{\mathrm{MSY},i}$).

6.3.2 Bayesian, Evolutionary, and Ensemble Approaches (Zhang et al. 2015, 2016; Spence et al. 2016, 2018, 2021)

When models are implemented as black-box simulation packages without automatic differentiation (or when likelihood surfaces are rugged and multimodal), researchers have developed four complementary calibration and uncertainty frameworks for `mizer` and `LeMaRns`:

6.3.2.1 Two-Step Genetic Algorithm + Parallel Tempering MCMC (Zhang et al. 2015, 2016)

Global Optimization Strategy for mizer Spin-Up

In calibrating a multispecies `mizer` model to data-poor and data-rich ecosystems (Zhang et al. 2015, 2016), Zhang et al. found that standard gradient descent or single-chain Metropolis–Hastings gets trapped in local modes where one or more species go extinct during spin-up. They introduced a two-stage global optimizer:

- *Stage 1*: A parallel **Real-Coded Genetic Algorithm (GA)** searches the high-dimensional parameter space ($\ln R_{\max,i}, \kappa, q_{f,i}, F_i$) using a survival penalty for extinct species to locate the global basin of attraction.
- *Stage 2*: **Parallel Tempering MCMC** (running multiple interacting Markov chains at temperatures $T_1 = 1 < T_2 < \cdots < T_M$) samples the posterior distribution across multimodal ridges.

6.3.2.2 History Matching + Sequential Monte Carlo (Spence et al. 2016, 2021)

To calibrate `mizer` (Spence et al. 2016) and `LeMaRns` (Spence, Thorpe, et al. 2021) to North Sea time series while explicitly accounting for **structural model discrepancy** η_{disc} (the difference between the true ocean and the best possible parameterization of the model), Spence et al. combined:

- **History Matching**: Defines an **implausibility measure** $I_k(\theta)$ for summary statistic k:

$$I_k(\theta) = \frac{\left|y_k^{\mathrm{obs}} - \mathbb{E}[f_k(\theta)]\right|}{\sqrt{\mathrm{Var}(y_k^{\mathrm{obs}}) + \mathrm{Var}(f_k(\theta)) + \mathrm{Var}(\eta_{\mathrm{disc},k})}} \tag{6.5}$$

and rules out any parameter vector θ where $\max_k I_k(\theta) > 3$ (by Pukelsheim's 3σ rule), rapidly shrinking the feasible space before running full time-series likelihood evaluations.

- **Sequential Monte Carlo (SMC)**: Propagates a particle cloud $\left\{\theta^{(p)}\right\}_{p=1}^{P}$ through sequential annealing or time-series filtering steps using Normalizing Flows / Metropolis-Hastings rejuvenation moves.

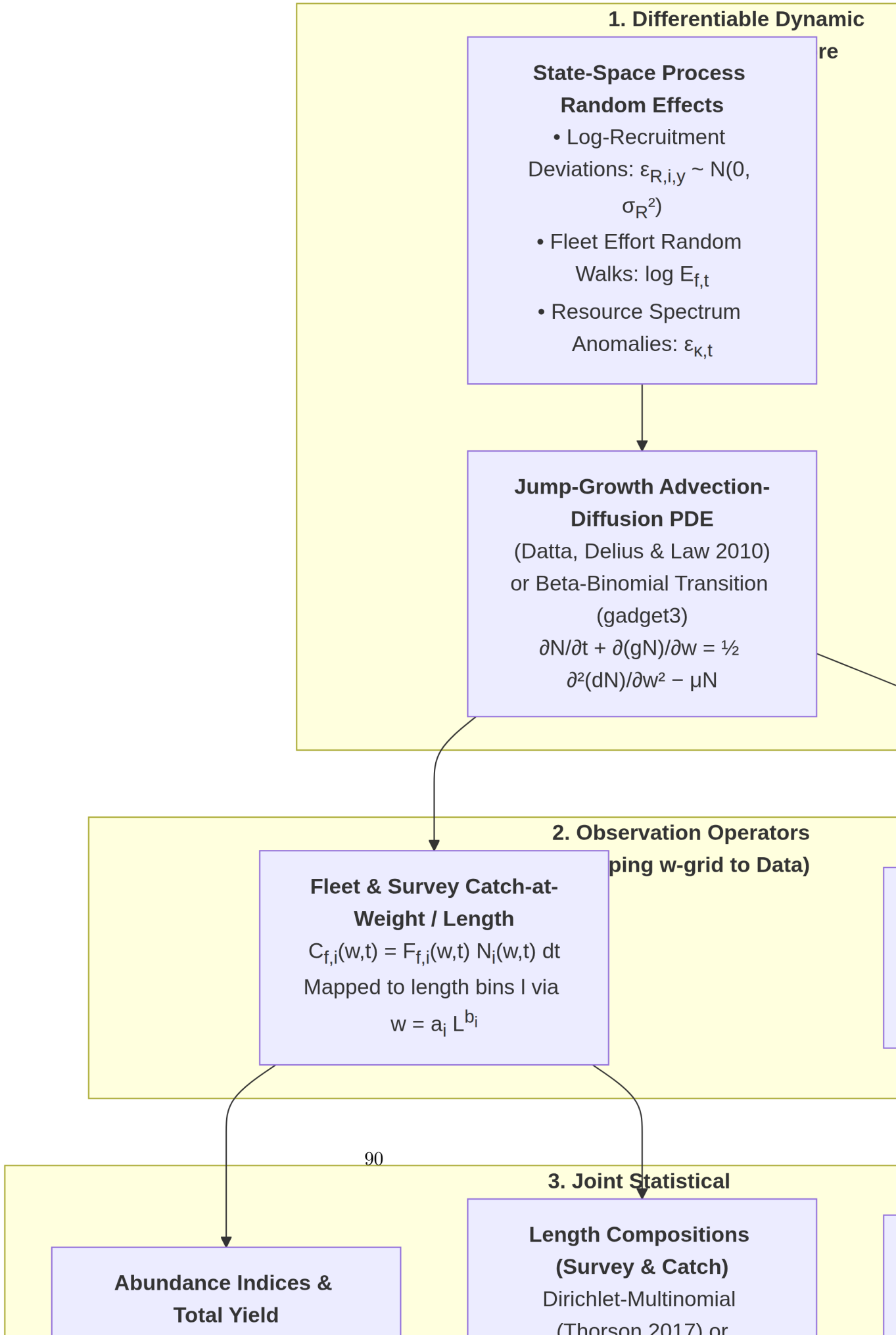
6.3.2.3 Bayesian Multi-Model Ensembles (Spence et al. 2018)

Instead of selecting a single "winner" model via AIC, (Spence et al. 2018) constructed a hierarchical Bayesian ensemble linking five distinct multispecies simulators (`EcoPath with Ecosim`, `LeMans`, `mizer`, `SMS`, and single-species assessments) as biased noisy realizations around a latent **true ecosystem trajectory** μ_t, decomposing structural discrepancy into shared vs. simulator-specific discrepancy processes.

6.4 Architectural Blueprint for Next-Generation Statistical Dynamic Stock Assessment in `mizer`

In their 10-year review of `mizer`, (Pennino et al. 2025, Section *"Challenges and Future Directions"*, p. 9) identify the central barrier preventing `mizer` from being used as a routine tactical stock assessment tool: > *"First, there is a lack of a standardized optimization routine for parameter estimation and uncertainty quantification… Second, `mizer` models are typically calibrated to a time-averaged steady state rather than fitted dynamically to time series of catches, survey indices, and size compositions."*

Drawing together the mathematical lessons from **`SMS`** (Chapter 2), **`Gadget/gadget3`**, **`MULTIFAN-CL/SS3`**, and (Punt et al. 2016) (Chapter 3), **`EcoState`** (Chapter 4), and **stomach content likelihoods** (Chapter 5), we can synthesize a concrete, five-pillar architectural roadmap for transforming `mizer` into a **differentiable, state-space statistical multispecies stock assessment model** (`mizer-TMB` / `mizer-RTMB`).


1. Differentiable Dynamic
re
State-Space Process Random Effects
• Log-Recruitment Deviations: $\varepsilon_{R,i,y} \sim N(0, \sigma_R^2)$
• Fleet Effort Random Walks: log $E_{f,t}$
• Resource Spectrum Anomalies: $\varepsilon_{k,t}$
Jump-Growth Advection-Diffusion PDE
(Datta, Delius & Law 2010) or Beta-Binomial Transition (gadget3)
$\partial N/\partial t + \partial(gN)/\partial w = ½ \partial^2(dN)/\partial w^2 - \mu N$
2. Observation Operators
ping w-grid to Data)
Fleet & Survey Catch-at-Weight / Length
$C_{f,i}(w,t) = F_{f,i}(w,t)\ N_i(w,t)\ dt$
Mapped to length bins l via $w = a_i\ L^{b_i}$
90
3. Joint Statistical
Abundance Indices & Total Yield
Length Compositions (Survey & Catch)
Dirichlet-Multinomial
(Thorson 2017) or

6.4.1 Pillar 1: Resolving Cohort Dispersion via Jump-Growth Diffusion (Datta, Delius & Law 2010)

When `mizer` is projected dynamically with interannual recruitment fluctuations $R_{\mathrm{dd},i}(t)\exp(\epsilon_{R,i,y})$, a strong year class enters at w_0 and propagates across the weight grid.

- **Problem**: Under the standard first-order upwind scheme, cohort width is governed by numerical grid viscosity $D_{\mathrm{num}} \approx \frac{1}{2} g_i \Delta w$, which fails to match the true biological variance of size-at-age in survey length-frequency distributions.
- **Solution**: Replace the first-order upwind step with the **semi-implicit Crank–Nicolson / Chang–Cooper finite-volume discretization of the Jump-Growth Fokker–Planck PDE** (Datta et al. 2010, Eq. 7):

$$\frac{\partial N_i}{\partial t} + \frac{\partial (g_i N_i)}{\partial w} - \frac{1}{2}\frac{\partial^2 \left(\psi_{\mathrm{disp},i}\, d_i(w,t)\, N_i\right)}{\partial w^2} = -\mu_i(w,t) N_i \tag{6.6}$$

  where $d_i(w,t)$ is the mechanistic second moment of prey mass intake, optionally scaled by a species-specific dispersion multiplier $\psi_{\mathrm{disp},i} \geq 1$ (analogous to Gadget's Beta-Binomial dispersion parameter β_s in `g3a_grow_impl_bbinom`; (Lentz and Elvarsson 2024)) to account for genetic growth variation and patchy foraging.
- **Computational Cost**: Discretizing the second derivative $\partial_w^2(d_i N_i)$ changes the implicit linear system per species per time step from **lower-bidiagonal** to **tridiagonal**, which is still solved in exact $O(n_w)$ **operations** via the Thomas algorithm (`solve_tridiag`) and is 100% differentiable in `TMB` / `RTMB`!

6.4.2 Pillar 2: Two-Stage Initialization (Differentiable Steady-State Spin-Up + Dynamic Projection)

In single-species assessments (`SS3`, `SAM`), the initial age vector N_{i,a,t_0} at the start of the historical time series t_0 is initialized at a fished equilibrium $N_{\mathrm{eq}}(F_{\mathrm{init},i})$ plus initial deviations. For dynamic `mizer`:

1. **Differentiable Steady-State Initialization at** $t = t_0$: Given trial parameters θ and baseline initial fishing effort E_{f,t_0}, compute the steady-state spectra $N_{i,0}(w)$ via a fixed number ($k = 15$–25) of damped fixed-point iterations of the analytical steady-state envelope $N_i(w) = N_i(w_0)\frac{g_i(w_0)}{g_i(w)}\exp\Big(-\int_{w_0}^{w}\frac{\mu_i(w')}{g_i(w')}dw'\Big)$ (with matching resource $N_{R,0}(w)$). Because a fixed number of unrolled iterations uses only elementary vector operations, `TMB` / `RTMB` differentiates smoothly through the steady-state initialization!

2. **Dynamic Time-Stepping ($t \in [t_0, T]$)**: Step forward in quarterly or monthly increments Δt (e.g., $\Delta t = 0.1$ or 0.25 yr) driven by time-varying fishing mortality $F_{f,i}(w,t)$, temperature anomalies $T(t)$ (scaling metabolic rates $k_{s,i}$, intake h_i, and search rates γ_i via Arrhenius activation $e^{-E_a/k_B T}$), and state-space recruitment deviations $\epsilon_{R,i,y}$.

6.4.3 Pillar 3: State-Space Process Random Effects (u)

Following `SAM` (Nielsen and Berg 2014) and `EcoState` (Thorson et al. 2024), introduce three compact sets of latent Gaussian random effects integrated out by TMB's Laplace approximation:

1. **Annual Recruitment Deviations**:

$$R_i(t) = R_{\mathrm{dd},i}(t) \exp\!\left(\epsilon_{R,i,y(t)} - \frac{1}{2}\sigma_{R,i}^2\right), \qquad \epsilon_{R,i,y} \sim \mathcal{N}\!\left(0, \sigma_{R,i}^2\right) \tag{6.7}$$

2. **Resource Spectrum Productivity / Carrying Capacity Anomalies** (capturing bottom-up climate/plankton regimes):

$$\kappa_R(w,t) = \kappa_0 \exp\!\left(\epsilon_{\kappa,y(t)}\right) w^{-\lambda}, \qquad \epsilon_{\kappa,y} \sim \mathrm{AR1}(\rho_\kappa, \sigma_\kappa^2) \tag{6.8}$$

3. **Random-Walk Fishing Effort / Mortality** (when commercial effort is noisy or unrecorded):

$$\ln E_{f,y+1} \sim \mathcal{N}\!\left(\ln E_{f,y},\ \sigma_{E,f}^2\right) \tag{6.9}$$

6.4.4 Pillar 4: Observation Operators and Joint Statistical Likelihood Function

To avoid the parameter confounding identified by (Punt et al. 2016), dynamic `mizer` should be fitted simultaneously to four orthogonal empirical data streams:

1. **Survey Biomass / Abundance Indices ($I_{f,i,y}$, Lognormal Likelihood)**: Given survey selectivity $S_{f,i}^{\mathrm{surv}}(w)$ and catchability $Q_{f,i}^{\mathrm{surv}}$:

$$\hat{I}_{f,i,y} = Q_{f,i}^{\mathrm{surv}} \int_{w_0}^{w_{\max,i}} S_{f,i}^{\mathrm{surv}}(w)\, w\, N_i(w,t_y)\, dw, \qquad \ln I_{f,i,y}^{\mathrm{obs}} \sim \mathcal{N}\!\left(\ln \hat{I}_{f,i,y},\ \sigma_{I,f,i,y}^2\right) \tag{6.10}$$

2. **Commercial Fishery Yields ($Y_{f,i,y}$, Lognormal Likelihood)**:

$$\hat{Y}_{f,i,y} = \int_{y}^{y+1} dt \int_{w_0}^{w_{\max,i}} F_{f,i}(w,t)\, w\, N_i(w,t)\, dw, \qquad \ln Y_{f,i,y}^{\mathrm{obs}} \sim \mathcal{N}\!\left(\ln \hat{Y}_{f,i,y},\ \sigma_{Y,f,i}^2\right) \tag{6.11}$$

3. **Survey & Commercial Length-Frequency Compositions ($\mathbf{P}^{\mathbf{obs}}_{f,i,y}$, Dirichlet-Multinomial or Fournier Robustified Normal Likelihood)**: For observed length bin l with boundaries $[L_l, L_{l+1})$ corresponding to body weights $[w_{i,l}, w_{i,l+1}) = [a_i L_l^{b_i}, a_i L_{l+1}^{b_i})$, the expected catch number in bin l is:

$$\hat{C}_{f,i,l,y} = \int_{w_{i,l}}^{w_{i,l+1}} S_{f,i}(w)\, N_i(w, t_y)\, dw, \qquad \hat{p}_{f,i,l,y} = \frac{\hat{C}_{f,i,l,y}}{\sum_m \hat{C}_{f,i,m,y}} \tag{6.12}$$

   Evaluated via the **self-weighting Dirichlet-Multinomial likelihood** (Thorson et al. 2017) with haul sample size $N_{\text{haul},f,y}$ and estimated overdispersion $\theta_{\text{comp},f,i}$ (or Fournier's robustified normal likelihood from `MULTIFAN-CL` (Fournier et al. 1998) to prevent tail outliers from distorting growth parameters).

4. **Stomach Content Diet Shares & Log-PPMR Distributions (Dirichlet-Multinomial + Gamma Likelihood)**: For predator (i, w') in year-quarter t, the model-predicted mass share of prey species j in prey weight bin $[w_{p,m}, w_{p,m+1})$ in the diet is:

$$\hat{W}_{i,w',j,m,t} = \frac{\gamma_i(w')\, \theta_{ij} \int_{w_{p,m}}^{w_{p,m+1}} \phi_i(w'/w_p)\, N_j(w_p, t)\, w_p\, dw_p}{E_i(w', t)} \tag{6.13}$$

   - **Taxonomic Diet Shares**: Fit evacuated, CPUE-weighted (or `sdmTMB`-standardized; (Jacobsen et al. 2023)) stomach weight proportions against $\hat{W}_{i,w',j,m,t}$ using the **SMS Dirichlet** (Lewy and Vinther 2004) or **Dirichlet-Multinomial** (Thorson et al. 2017) likelihood, which directly identifies the interaction matrix θ_{ij}!
   - **Predator–Prey Size Kernel Parameters (β_i, σ_i or Gamma α_i, β_i)**: Fit the observed log-PPMR distribution $x = \ln(w'/w_p)$ using the **Exponential Family / Gamma likelihood** of (Morgan 2022).

6.4.5 Pillar 5: Stabilizing Dynamic Multi-Decadal Fits via Prey Refuges & Overconsumption Caps

A major lesson from `Gadget` (Lentz and Elvarsson 2024), `MSCAA` (Curti et al. 2013), and `Ecosim/EcoState` (Walters et al. 1997; Thorson et al. 2024) is that during gradient optimization over a 40-year time series, trial parameter values can easily trigger runaway trophic cascades unless two stabilizing mechanisms are included:

1. **Holling Type III Exponent or Foraging Arena Vulnerability**: Replacing linear encounter $\theta_{ij} N_j(w_p)$ with a weakly sigmoid Type III encounter $\theta_{ij} N_j(w_p)^{1+\zeta}$ ($\zeta \approx 0.1$–0.3) or a Foraging Arena predator-interference denominator $\frac{\theta_{ij} v_{ij}}{2v_{ij} + \gamma_i N_i}$ provides a low-density prey refuge and prevents unrealistic species extinctions during parameter search.

2. **Smooth Differentiable Overconsumption Cap (`dif_pmin`)**: Borrowing `gadget3`'s smooth logistic minimum cap on total mortality per time step ensures that $N_i(w,t)$ remains strictly positive and gradients $\nabla_\theta \mathcal{L}$ remain finite across the entire parameter space.

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